Physical world and Measurements

Question1

Consider a modified Bernoulli equation.

$$\left(\mathrm{P}+\frac{A}{Bt^2}\right)+\rho g(h+Bt)+\frac{1}{2}\rho V^2=\text{ constant}$$

If t has the dimension of time then the dimensions of A and B are _____ , _____ respectively.

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Options:

A.

$\left[\mathrm{ML}^0\ \mathrm{T}^{-1}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}\right]$

B.

$\left[\mathrm{ML}^0\ \mathrm{T}^{-2}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}^{-1}\right]$

C.

$\left[\mathrm{ML}^0\ \mathrm{T}^{-2}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}^{-2}\right]$

D.

$\left[\mathrm{ML}^0\ \mathrm{T}^{-1}\right]$ and $\left[\mathrm{M}^0\mathrm{LT}^{-1}\right]$

Answer: D

Solution:

In the term $(h+Bt)$, Bt is added to h, where h is the height or length which has the dimension: $[h]=[L]$.

Therefore, using principle of homogeneity the product Bt must also have the dimension of length: $[\mathrm{L}]$.

Given that t is time whose dimension is $[\mathrm{T}]$:

$[\mathrm{B}][\mathrm{t}]=[\mathrm{L}]$

$\Rightarrow[\mathrm{B}][\mathrm{T}]=[\mathrm{L}]$

$\Rightarrow[\mathrm{B}]=\frac{[\mathrm{L}]}{[\mathrm{T}]}=\left[\mathrm{LT}^{-1}\right]$

Therefore, the dimensions of B is $\left[\mathrm{M}^0 \mathrm{LT}^{-1}\right]$.

In the term $\left(\mathrm{P}+\frac{\mathrm{A}}{\mathrm{Bt}^2}\right)$, $\frac{\mathrm{A}}{\mathrm{Bt}^2}$ is added to P (Pressure).

The dimension of pressure is $\mathrm{P}=\frac{\text{Force}}{\text{Area}}$:

$[\mathrm{P}]=\frac{\left[\mathrm{MLT}^{-2}\right]}{\left[\mathrm{L}^2\right]}=\left[\mathrm{ML}^{-1} \mathrm{~T}^{-2}\right]$

According to the principle of homogeneity, the dimension of $\frac{\mathrm{A}}{\mathrm{Bt}^2}$ must equal the dimension of P :

$\left[\frac{\mathrm{A}}{\mathrm{Bt}^2}\right]=\left[\mathrm{ML}^{-1} \mathrm{~T}^{-2}\right]$

$\Rightarrow \frac{[\mathrm{A}]}{\left[\mathrm{LT}^{-1}\right]\left[\mathrm{T}^2\right]}=\left[\mathrm{ML}^{-1} \mathrm{~T}^{-2}\right]$

$\Rightarrow[\mathrm{A}]=\left[\mathrm{ML}^{-1} \mathrm{~T}^{-2}\right] \times[\mathrm{LT}]$

$\Rightarrow[\mathrm{A}]=[\mathrm{M}]\left[\mathrm{L}^{-1+1}\right]\left[\mathrm{T}^{-2+1}\right]$

$\Rightarrow[\mathrm{A}]=\left[\mathrm{ML}^0 \mathrm{~T}^{-1}\right]$

Therefore, dimensions of A is $\left[\mathrm{ML}^0 \mathrm{~T}^{-1}\right]$ and the dimensions of B is $\left[\mathrm{M}^0 \mathrm{LT}^{-1}\right]$. Hence, the correct option is (D).

Question2

In an experiment the values of two spring constants were measured as $k_1=(10 \pm 0.2) \mathrm{N} / \mathrm{m}$ and $k_2=(20 \pm 0.3) \mathrm{N} / \mathrm{m}$. If these springs are connected in parallel, then the percentage error in equivalent spring constant is :

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Options:

A.

1.33%

B.

2.67%

C.

1.67%

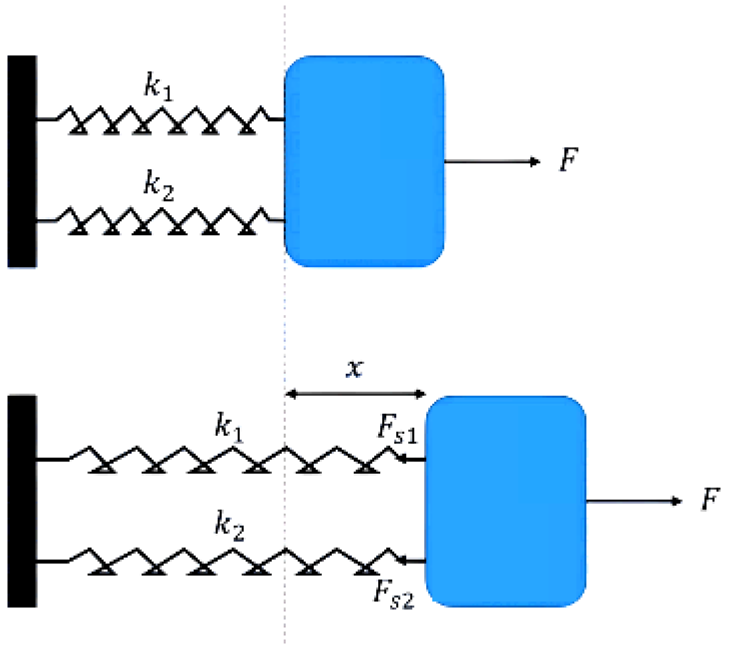
D.

2.33%

Answer: C

Solution:

When two springs are connected in parallel, they both undergo the same displacement (x) when a force is applied.



Force in spring 1 is $F_1 = k_1 x$

Force in spring 2 is $F_2 = k_2 x$

The total force F is the sum of individual forces :

$F = F_1 + F_2$

$\Rightarrow k_{eq} x = k_1 x + k_2 x$

$\Rightarrow k_{eq} = k_1 + k_2$

Substituting the given values, $k_1 = 10\,N/m, k_2 = 20\,N/m$

$k_{eq} = 10 + 20 = 30\,\mathrm{N/m}$

When two measured quantities are added, the absolute error in the result is the sum of the absolute errors of the individual quantities.

$\Delta k_{eq} = \Delta k_1 + \Delta k_2$

Substituting the given values, $\Delta k_1 = 0.2\,\mathrm{N/m}, \Delta k_2 = 0.3\,\mathrm{N/m}$

$\Delta k_{eq} = 0.2 + 0.3 = 0.5\,\mathrm{N/m}$

So, the percentage error in measurement of equivalent spring constant is,

Percentage Error $= \left(\frac{\Delta k_{eq}}{k_{eq}}\right) \times 100\% = \left(\frac{0.5}{30}\right) \times 100\% \approx 1.666\ldots\%$

Rounding to two decimal places, we get the percentage error in measurement of spring constant as 1.67%.

Therefore, the equivalent spring constant with its error is $(30 \pm 0.5)\mathrm{N/m}$, which corresponds to a percentage error of 1.67%.

Hence, the correct option is (C).

Question3

Keeping the significant figures in view, the sum of the physical quantities 52.01 m, 153.2 m and 0.123 m is :

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Options:

A.

205.3 m

B.

205 m

C.

205.333 m

D.

205.33 m

Answer: A

Solution:

In addition, or subtraction, the final result should be reported to the same number of decimal places as the measurement with the least number of decimal places. This ensures the result is no more precise than the least precise measurement.

The given measured values are,

- 52.01 m : 2 decimal places.
- 153.2 m : 1 decimal place.
- 0.123 m : 3 decimal places.

The measurement with the least number of decimal places is 153.2 m (1 decimal place). Therefore, our final answer must be rounded to one decimal place.

$52.01 + 153.2 + 0.123 = 205.333 \mathrm{~m}$

Now, we round 205.333 to one decimal place :

- The digit at the first decimal place is $3 \rightarrow 205.33$
- The digit following it is 3 (which is less than 5) → 205.3

Therefore, the rounded result is 205.3 m . Hence, the correct option is (A)

Question4

Match the LIST-I with LIST-II

List-I		List-II	
A.	Spring constant	I.	$\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~K}^{-1}$
B.	Thermal conductivity	II.	$\mathrm{ML}^0 \mathrm{~T}^{-2}$
C.	Boltzmann constant	III.	$\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-2}$
D.	Inductive reactance	IV.	$\mathrm{MLT}^{-3} \mathrm{~K}^{-1}$

Choose the correct answer from the options given below:

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Options:

A.

A-III, B-II, C-IV, D-I

B.

A-I, B-IV, C-II, D-III

C.

A-II, B-I, C-IV, D-III

D.

A-II, B-IV, C-I, D-III

Answer: D

Solution:

A. Spring constant (k)

According to Hooke's Law, the restoring force F of a spring is directly proportional to its displacement x :

$\mathrm{F}=-\mathrm{kx}$

Taking the magnitude to find dimensions :

$\mathrm{k}=\frac{\mathrm{F}}{\mathrm{x}} \Rightarrow[\mathrm{k}]=\left[\frac{\mathrm{F}}{\mathrm{x}}\right]=\frac{\left[\mathrm{MLT}^{-2}\right]}{\left[\mathrm{M}^0 \mathrm{LT}^0\right]}$

$[\mathrm{k}]=\frac{\left[\mathrm{MLT}^{-2}\right]}{[\mathrm{L}]}=\left[\mathrm{ML}^{1-1} \mathrm{~T}^{-2}\right]=\left[\mathrm{ML}^0 \mathrm{~T}^{-2}\right]$

$\mathrm{A} \rightarrow$ II.

B. Thermal conductivity (K)

From Law of heat conduction, the rate of heat flow $\left(\mathrm{H}=\frac{\mathrm{dQ}}{\mathrm{dt}}\right)$ is given by:

$H=\frac{K A \Delta T}{L}$

Where A is area, $\Delta \mathrm{T}$ is temperature difference, and L is length.

$\mathrm{K}=\frac{\mathrm{H} \cdot \mathrm{L}}{\mathrm{A} \cdot \Delta \mathrm{~T}}$

The dimensions of rate of heat flow, $[\mathrm{H}]=\frac{[\text { Energy }]}{[\text { Time }]}=\frac{[\text { Force } \times \text { distance }]}{[\text { Time }]}=\frac{\left[\mathrm{MLT}^{-2} \times \mathrm{L}\right]}{[\mathrm{T}]}=\left[\mathrm{ML}^2 \mathrm{~T}^{-3}\right]$

So the dimensions of thermal conductivity is,

$[\mathrm{K}]=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-3}\right] \cdot[\mathrm{L}]}{\left[\mathrm{L}^2\right] \cdot[\mathrm{K}]}=\frac{\left[\mathrm{ML}^3 \mathrm{~T}^{-3}\right]}{\left[\mathrm{L}^2 \mathrm{~K}\right]}$

$\Rightarrow[\mathrm{K}]=\left[\mathrm{ML}^{3-2} \mathrm{~T}^{-3} \mathrm{~K}^{-1}\right]=\left[\mathrm{MLT}^{-3} \mathrm{~K}^{-1}\right]$

$\mathrm{B} \rightarrow$ IV.

C. Boltzmann constant (k_{B})

The average kinetic energy (E) of a gas molecule is related to its absolute temperature (T) by :

$\mathrm{E}=\frac{3}{2} \mathrm{k}_{\mathrm{B}} \mathrm{T}$

The dimensionless constant is $\frac{3}{2}$.

$\left[\mathrm{k}_{\mathrm{B}}\right]=\left[\frac{\mathrm{E}}{\mathrm{T}}\right]$

We know Energy has the same dimensions as Work ($\mathrm{F} \times \mathrm{d}$):

$[\mathrm{E}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]$

So, the dimensions of Boltzmann constant is,

$\left[\mathrm{k}_{\mathrm{B}}\right]=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]}{[\mathrm{K}]}=\left[\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~K}^{-1}\right]$

$\mathrm{C} \rightarrow \mathrm{I}$.

D. Inductive reactance (X_{L})

Inductive reactance is the opposition to alternating current by an inductor. It acts exactly like resistance, so it must have the same dimensional formula as electrical resistance (R).

From Ohm's Law: $\mathrm{V}=\mathrm{IR}$, so $\mathrm{R}=\frac{\mathrm{V}}{\mathrm{I}}$.

Potential is Work done per unit charge (W/q):

$[\mathrm{V}]=\frac{[\text { Work }]}{[\text { Charge }]}=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]}{[\mathrm{A} \cdot \mathrm{~T}]}=\left[\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-1}\right]$

So, the dimensions of inductive reactance, i.e., resistance is,

$\left[\mathrm{X}_{\mathrm{L}}\right]=[\mathrm{R}]=\frac{[\mathrm{V}]}{[\mathrm{I}]}=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-1}\right]}{[\mathrm{A}]}$

$$\left[\mathrm{X}_{\mathrm{L}}\right]=\left[\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-2}\right]$$

$\mathrm{D} \rightarrow \mathrm{III}$.

Therefore, the correct match is, $\mathrm{A} \rightarrow \mathrm{II}, \mathrm{B} \rightarrow \mathrm{IV}, \mathrm{C} \rightarrow \mathrm{I}, \mathrm{D} \rightarrow \mathrm{III}$.

Hence, the correct option is (D).

Question5

If ϵ, E and t represent the free space permittivity, electric field and time respectively, then the unit of $\frac{\epsilon E}{t}$ will be :

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Options:

A.

Am

B.

$\mathrm{A/m}$

C.

$\mathrm{A/m^2}$

D.

$\mathrm{Am^2}$

Answer: C

Solution:

The SI unit of permittivity of free space (ϵ) is derived from Coulomb's law $\left(F=\frac{q_1 q_2}{4\pi\epsilon_0 r^2}\right)$. The SI unit is $C^2/N.\,\mathrm{m}^2$.

The electric field (E) is defined as force per unit charge, so its SI unit is Newtons per Coulomb (N/C).

The standard SI unit is the second (s).

The given expression is $\frac{\epsilon \mathrm{E}}{\mathrm{t}}$.

So, the SI unit of given expression is, $\frac{\mathrm{C}^2}{\mathrm{N}\cdot\,\mathrm{m}^2}\times\left(\frac{\mathrm{N}}{\mathrm{C}}\right)=\frac{\mathrm{C}}{\mathrm{s}}=\frac{\mathrm{s}}{\mathrm{s}\cdot\mathrm{m}^2}$

Electric current is defined as the rate of flow of electric charge over time. Therefore, $1\text{ Ampere }(\mathrm{A})=$

$1\text{ Coulomb per second }\left(\frac{\mathrm{C}}{\mathrm{s}}\right)$.

Unit of $\frac{\epsilon \mathrm{E}}{\mathrm{t}}=\left(\frac{\mathrm{C}}{\mathrm{s}}\right)\times\frac{1}{\mathrm{m}^2}=\mathrm{A}\times\frac{1}{\mathrm{m}^2}=\mathrm{A/m^2}$

Therefore, the SI unit of $\frac{\epsilon E}{t}$ is $\mathrm{A/m^2}$. Hence, the correct option is (C).

Question6

Four persons measure the length of a rod as $20.00\text{ cm}, 19.75\text{ cm}, 17.01\text{ cm}$ and 18.25 cm . The relative error in the measurement of average length of the rod is :

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Options:

A.

0.18

B.

0.24

C.

0.06

D.

0.08

Answer: C

Solution:

The average (mean) of the measured values is the true value for our error calculation.

$\bar{x} = \frac{x_1+x_2+x_3+x_4}{4}$

Substituting the given values :

$\bar{x} = \frac{20.00+19.75+17.01+18.25}{4} = \frac{75.01}{4}$

$\overline{\mathrm{x}} = 18.7525 \approx 18.75 \text{ cm}$

The absolute error is the magnitude of the difference between each individual measurement and the mean value.

$|\Delta \mathrm{x_i}| = |\mathrm{x_i} - \overline{\mathrm{x}}|$

$\Rightarrow |\Delta \mathrm{x_1}| = |20.00 - 18.75| = 1.25$

$\Rightarrow |\Delta \mathrm{x_2}| = |19.75 - 18.75| = 1.00$

$\Rightarrow |\Delta x_3| = |17.01 - 18.75| = |-1.74| = 1.74$

$\Rightarrow |\Delta \mathrm{x_4}| = |18.25 - 18.75| = |-0.50| = 0.50$

Then the average of these absolute errors.

$\Delta\overline{\mathrm{x}} = \frac{\sum|\Delta \mathrm{x_i}|}{4}$

$\Rightarrow \Delta \bar{x} = \frac{1.25+1.00+1.74+0.50}{4} = \frac{4.494}{4}$

$\Rightarrow \Delta \overline{\mathrm{x}} = 1.1225 \approx 1.12 \mathrm{~cm}$

The relative error is defined as the ratio of the mean absolute error to the mean value.

Relative Error $= \frac{\Delta \overline{\mathrm{x}}}{\overline{\mathrm{x}}}$

Relative Error $= \frac{1.1225}{18.7525}$

Relative Error ≈ 0.0598

Rounding to two decimal places,

Relative Error ≈ 0.06

Therefore, the relative error in the measurement is 0.06 . Hence, the correct option is (C).

Question7

In a screw gauge, the zero of the circular scale lies 3 divisions above the horizontal pitch line when their metallic studs are brought in contact. Using this instrument thickness of a sheet is measured. If pitch scale reading is 1 mm and the circular scale reading is 51 then the correct thickness of the sheet is ____ mm.

[Assume least count is 0.01 mm]

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Options:

A.

1.54

B.

1.50

C.

1.51

D.

1.48

Answer: A

Solution:

The circular scale lies 3 divisions above the horizontal pitch line when the studs are in contact.

When the zero mark is above the reference line, the screw has been turned past the zero point. This indicates a negative zero error. The error corresponds to 3 divisions.

Hence, the zero error is,

$\mathrm{E}_0=-(3 \times \text{ Least Count })=-(3 \times 0.01 \mathrm{~mm})=-0.03 \mathrm{~mm}$

The observed reading is the sum of the Pitch Scale Reading (PSR) and the Circular Scale Reading (CSR) multiplied by the Least Count (LC).

The given Pitch Scale Reading (PSR) is 1 mm and the Circular Scale Reading (CSR) is 51

So, the observed reading $=\mathrm{PSR}+(\mathrm{CSR} \times \mathrm{LC})$

Observed Reading $=1 \mathrm{~mm}+(51 \times 0.01 \mathrm{~mm})$

Observed Reading $=1 \mathrm{~mm}+0.51 \mathrm{~mm}$

Observed Reading $=1.51 \mathrm{~mm}$

To find the true thickness, we must subtract the zero error from the observed reading.

Correct Reading $=$ Observed Reading - Zero Error

Correct Reading $=1.51 \mathrm{~mm}-(-0.03 \mathrm{~mm})$

Correct Reading $=1.51 \mathrm{~mm}+0.03 \mathrm{~mm}$

Correct Reading $=1.54 \mathrm{~mm}$

Therefore, the correct thickness of the sheet is 1.54 mm.

Hence, the correct option is (A).

Question8

In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division $=0.05 \mathrm{~mm}$, then the least count of the vernier callipers is ____ mm.

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Options:

A.

0.002

B.

0.02

C.

0.05

D.

0.005

Answer: A

Solution:

Value of one Main Scale Division, $1 \mathrm{MSD}=0.05 \mathrm{~mm}$

Number of Vernier Scale Divisions $\mathrm{m}=50$

Number of Main Scale Divisions is $\mathrm{n}=48$

$50 \mathrm{VSD}=48 \mathrm{MSD}$

$\Rightarrow 1 \mathrm{VSD}=\frac{48}{50} \mathrm{MSD}$

The Least Count of a Vernier instrument is the smallest measurement that can be taken accurately.

Least Count (LC) of Vernier scale = 1 MSD -1 VSD

Using the least count formula,

$\mathrm{LC}=1 \mathrm{MSD}-1 \mathrm{VSD}$

$\Rightarrow \mathrm{LC}=1 \mathrm{MSD}-\left(\frac{48}{50}\right) \mathrm{MSD}$

$\Rightarrow \mathrm{LC}=\left(1-\frac{48}{50}\right) \mathrm{MSD}$

$\Rightarrow \mathrm{LC}=\left(\frac{50-48}{50}\right) \mathrm{MSD}$

$\Rightarrow \mathrm{LC}=\frac{2}{50} \mathrm{MSD}=\frac{2}{50} \times 0.05 \mathrm{~mm}=0.002 \mathrm{~mm}$

Therefore, the smallest measurement that this Vernier calliper can record is 0.002 mm .

Hence, the correct option is (A).

Question9

When both jaws of vernier callipers touch each other, zero mark of the vernier scale is right to zero mark of main scale, 4^{th} mark on vernier scale coincides with certain mark on the main scale. While measuring the length of a cylinder, observer observes 15 divisions on main scale and 5^{th} division of vernier scale coincides with a main scale division. Measured length of cylinder is ____ mm.

(Least count of Vernier calliper $= 0.1 \text{ mm}$)

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Options:

A.

15.5

B.

15.4

C.

15.9

D.

15.1

Answer: D

Solution:

When jaws touch, the vernier zero is to the right of the main scale zero, which indicates a positive zero error.

Coinciding vernier division (CVD) for error $= 4$.

Least Count $(\text{LC}) = 0.1$ mm.

So, zero error $= +(\text{CVD} \times \text{LC}) = +(4 \times 0.1) = +0.4$ mm

Main scale reading $(\text{MSR}) = 15$ divisions.

Assuming 1 division $= 1$ mm, then main scale reading is $\mathrm{MSR} = 15$ mm

Vernier Scale Reading (VSR) $=$ (Coinciding vernier division) $\times$ LC.

$\mathrm{VSR} = 5 \times 0.1 = 0.5$ mm

Total Observed Reading $= \mathrm{MSR} + \mathrm{VSR} = 15 + 0.5 = 15.5$ mm

True Length $=$ Observed Reading - Zero Error

True Length $= 15.5$ mm $- (+0.4$ mm$)$

True Length $= 15.1$ mm

Hence, the correct option is (D).

Question10

In an experiment, a set of reading are obtained as follows - 1.24 mm, 1.25 mm, 1.23 mm, 1.21 mm. The expected least count of the instrument used in recording these readings is _______ mm.

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Options:

A.

0.01

B.

0.05

C.

0.001

D.

0.1

Answer: A

Solution:

The least count of an instrument is the smallest measurement that can be taken accurately with that instrument. The least count is shown by the position of the last significant digit (the smallest decimal place) to which all readings are recorded.

The obtained readings are :

- 1. Reading 1 : 1.24 mm
- 2. Reading 2 : 1.25 mm
- 3. Reading 3 : 1.23 mm
- 4. Reading 4 : 1.21 mm

All the readings are recorded up to two decimal places.

The difference between consecutive possible readings

$$1.25 \text{ mm} - 1.24 \text{ mm} = 0.01 \text{ mm}$$
$$1.24 \text{ mm} - 1.23 \text{ mm} = 0.01 \text{ m}$$

If the least count were 0.1 mm (the smallest decimal place is 1), the readings would look like 1.2 mm or 1.3 mm .

If the least count were 0.001 mm (the smallest decimal place is 3), the readings would look like 1.240 mm .

If the least count were 0.05 mm , the readings would only end in .00 or .05 (e.g., 1.20, 1.25). Since we have 1.24 and 1.23, this is not possible.

Therefore, the least count of the instrument is 0.01 mm . Hence, the correct option is (A).

Question11

Match List - I with List - II.

List – I	List – II
A. Coefficient of viscosity	I. $[ML^{-1}T^{-2}]$
B. Surface tension	II. $[ML^{2}T^{-2}]$
C. Pressure	III. $[ML^{0}T^{-2}]$
D. Surface energy	IV. $[ML^{-1}T^{-1}]$

Choose the correct answer from the options given below :

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Options:

A.

A-I, B-II, C-IV, D-III

B.

A-IV, B-III, C-I, D-II

C.

A-IV, B-I, C-II, D-III

D.

A-I, B-III, C-II, D-IV

Answer: B

Solution:

From Newton's law of viscosity, the force is $\mathrm{F}=\eta \mathrm{A}\left(\frac{\Delta \mathrm{v}}{\Delta \mathrm{x}}\right)$, where A is area and $\frac{\Delta \mathrm{v}}{\Delta \mathrm{x}}$ is the velocity gradient.

$\eta=\frac{\mathrm{F}}{\mathrm{A} \cdot\left(\frac{\Delta \mathrm{v}}{\Delta \mathrm{x}}\right)}$

$\Rightarrow[\eta]=\frac{\left[\mathrm{MLT}^{-2}\right]}{\left[\mathrm{L}^2\right] \cdot\left(\frac{\left[\mathrm{LT}^{-1}\right]}{[\mathrm{L}]}\right)}=\frac{\left[\mathrm{MLT}^{-2}\right]}{\left[\mathrm{L}^2\right] \cdot\left[\mathrm{T}^{-1}\right]}=\left[\mathrm{ML}^{-1} \mathrm{~T}^{-1}\right]$

$\mathrm{A} \rightarrow \mathrm{IV}$

Surface tension is defined as the force acting per unit length.

$\mathrm{T}=\frac{\text { Force }}{\text { Length }}$

$[\mathrm{T}]=\frac{\left[\mathrm{MLT}^{-2}\right]}{[\mathrm{L}]}=\left[\mathrm{ML}^0 \mathrm{~T}^{-2}\right]$

$\mathrm{B} \rightarrow \mathrm{III}$

Pressure is defined as force acting per unit area.

$\mathrm{P}=\frac{\text { Force }}{\text { Area }}$

$[\mathrm{P}]=\frac{\left[\mathrm{MLT}^{-2}\right]}{\left[\mathrm{L}^2\right]}=\left[\mathrm{ML}^{-1} \mathrm{~T}^{-2}\right]$

$C \rightarrow I$

Surface energy $\left(\mathrm{U}_{\mathrm{s}}\right)$ is defined as the work done by surface tension in changing the surface area

$\mathrm{U}_{\mathrm{s}}=\mathrm{T} \cdot(\Delta \mathrm{~A})$

$\Rightarrow\left[\mathrm{U}_{\mathrm{S}}\right]=[\text { Surface Tension }] \cdot[\text { Area }]$

$$\Rightarrow [U_S] = \left[ML^0 T^{-2}\right] \cdot \left[\left[M^0 L^2 T^0\right]\right] = \left[\left[ML^2 T^{-2}\right]\right]$$

$D \rightarrow II$

List - I	List - II
A. Coefficient of viscosity	IV. $\left[ML^{-1} T^{-1}\right]$
B. Surface tension	III. $\left[ML^0 T^{-2}\right]$
C. Pressure	I. $\left[ML^{-1} T^{-2}\right]$
D. Surface energy	II. $\left[ML^2 T^{-2}\right]$

Hence, the correct option is (B).

Question12

The speed of a longitudinal wave in a metallic bar is 400 m/s. If the density and Young's modulus of the bar material are increased by 0.5% and 1%, respectively then the speed of the wave is changed approximately to ______ m/s.

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Options:

A.

398

B.

402

C.

401

D.

399

Answer: C

Solution:

The formula for the speed of a longitudinal wave (v) in a solid metallic bar is given as

$\mathrm{v}=\sqrt{\frac{\mathrm{Y}}{\rho}}$

Where, $\mathrm{Y}=$ Young's modulus and its ρ is its density

Taking the natural log on both sides of $\mathrm{v}=\left(\frac{\mathrm{Y}}{\rho}\right)^{1/2}$:

$\ln(\mathrm{v})=\frac{1}{2}\ln(\mathrm{Y})-\frac{1}{2}\ln(\rho)$

Differentiating both sides gives the fractional change:

$\frac{\Delta v}{v}=\frac{1}{2}\frac{\Delta Y}{Y}-\frac{1}{2}\frac{\Delta\rho}{\rho}$

Initial speed $=\mathrm{v}=400\ \mathrm{m/s}$

Increase in Young's modulus $=\frac{\Delta\mathrm{Y}}{\mathrm{Y}}\times 100=+1\%$

Increase in density $\frac{\Delta\rho}{\rho}\times 100=+0.5\%$

So, the percentage change in speed,

$\%$ change in $\mathrm{v}=\frac{1}{2}(1\%)-\frac{1}{2}(0.5\%)=0.5\%-0.25\%=0.25\%$

Since the result is positive, the speed increases by 0.25%.

So, new speed $(\mathrm{v}')=$ Initial Speed + Change in Speed

$$\begin{aligned}&\mathrm{v}'=\mathrm{v}+(0.25\%\text{ of v})\\ \Rightarrow\ &\mathrm{v}'=400+\left(\frac{0.25}{100}\times 400\right)\\ \Rightarrow\ &\mathrm{v}'=400+1=401\ \mathrm{m/s}\end{aligned}$$

Hence, the correct option is (C).

Question13

The time period of a simple harmonic oscillator is $T=2\pi\sqrt{\frac{k}{m}}$. Measured value of mass (m) of the object is 10 g with an accuracy of 10 mg and time for 50 oscillations of the spring is found to be 60 s using a watch of 2 s resolution. Percentage error in determination of spring constant (k) is ________%.

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Options:

A.

3.43

B.

7.60

C.

3.35

D.

6.76

Answer: D

Solution:

The formula for the time period of simple harmonic oscillator is

$T = 2\pi\sqrt{\frac{k}{m}}.$

$\Rightarrow T^2 = (2\pi)^2\left(\frac{k}{m}\right)$

$\Rightarrow k = \frac{T^2 \cdot m}{4\pi^2}$

Using propagation of error, when quantities are multiplied or divided, their relative errors (percentage errors) are added. The constants have no error.

$\frac{\Delta k}{k} \times 100 = \left(\frac{\Delta m}{m} + 2 \cdot \frac{\Delta T}{T}\right) \times 100$

The measured value of mass is, $m = 10$ g

Accuracy/Error in measurement of mass is, $\Delta m = 10mg = 0.01$ g

So, percentage error in measurement of mass $\frac{\Delta m}{m} \times 100\% = \frac{0.01}{10} \times 100 = 0.1\%$

The percentage error in the time period (T) is the same as the percentage error in the total time (t) measured for n oscillations.

Total time measured is, $t = 60$ s

Resolution/Error in measurement of time is, $\Delta t = 2$ s

So, percentage error in measurement of time is, $\frac{\Delta t}{t} \times 100\% = \frac{2}{60} \times 100\% = 3.33\%$

So, using propagation formula:

Percentage error in k $=$ (% error in m) $+ 2 \times$ (% error in t) $= 0.1\% + 2 \times 3.33\% = 6.76\%$

Therefore, the correct option is (D).

Question14

The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measured Young's modulus is $\alpha \times 10^9\ \mathrm{N/m^2}$. The value of α is ________.

(Ignore the contribution of the load to Young's modulus error calculation)

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Options:

A.

1.3

B.

1.65

C.

0.13

D.

0.25

Answer: B

Solution:

Young's modulus (Y) is defined as the ratio of longitudinal stress to longitudinal strain.

Let F be the force, L the original length, A the cross-sectional area, and ΔL the extension.

$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$

For a wire of diameter D, the area $A = \frac{\pi D^2}{4}$.

Substituting this into the formula:

$Y = \frac{FL}{\left(\frac{\pi D^2}{4}\right)\Delta L} = \frac{4FL}{\pi D^2 \Delta L}$

To find the absolute error in Y, we need to find the relative error.

Since F and π are constants, the relative error in Y is:

$\frac{\Delta Y}{Y} = \frac{\Delta L_{\text{scale}}}{L} + \frac{2\Delta D}{D} + \frac{\Delta(\Delta L)}{\Delta L}$

Where ΔL_{scale} , ΔD, and $\Delta(\Delta L)$ are the least counts of the respective measuring instruments.

The applied tensile force is $F = 100$ N,

The length of wire is $L = 150$ cm $= 1.5$ m,

The diameter of wire is $D = 0.08$ cm $= 8 \times 10^{-4}$ m,

The extension in the length of wire is $\Delta L = 0.5$ cm $= 5 \times 10^{-3}$ m.

$Y = \frac{4 \times 100 \times 1.5}{\pi \times (8 \times 10^{-4})^2 \times 5 \times 10^{-3}}$

$Y = \frac{600}{\pi \times 64 \times 10^{-8} \times 5 \times 10^{-3}} = \frac{600}{\pi \times 320 \times 10^{-11}} = \frac{6 \times 10^{11}}{3.2\pi}$

$Y \approx 5.96 \times 10^{10}$ N/m^2

So, the error in measurement of Young's modulus is (ΔY):

$\Delta Y = Y\left[\frac{0.1}{150} + 2\left(\frac{0.001}{0.08}\right) + \frac{0.001}{0.5}\right]$

$\Delta Y = Y\,[0.00066 + 0.025 + 0.002] = Y\,[0.02766]$

$\Delta Y = (5.96 \times 10^{10}) \times 0.02766 \approx 1.65 \times 10^9$ N/m^2

Therefore, the value of α is 1.65. Thus, the correct option is (B).

Question15

The dimensional formula of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity of vacuum and E = electric field) is $M^a L^b T^c$.

The value of $2a - b + c =$ ________.

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Options:

A.

0

B.

1

C.

-1

D.

2

Answer: B

Solution:

The expression given is $\frac{1}{2}\varepsilon_0 E^2$, which represents the energy density (U_d) of an electric field in a vacuum.

Energy density is defined as the energy stored per unit volume.

$U_d = \frac{\text{Energy} \quad (U)}{\text{Volume} \quad (V)}$

Energy is equivalent to work done, which is Force $\times$ Displacement.

The dimensions of force is $[F] = [ma] = [M\ L\ T^{-2}]$

So, the dimensions of energy is $[U] = [M\ L\ T^{-2}] \times [L] = [M\ L^2 T^{-2}]$

The dimensions of volume is $[V] = [L] \times [L] \times [L] = [L^3]$

Since energy density is $\frac{U}{V}$, its dimensions are:

$[U_d] = \frac{[M\ L^2 T^{-2}]}{[L^3]} = [M\ L^{-1} T^{-2}]$

The dimensional formula of energy density due to electric field is is $M^a L^b T^c$.

By comparing this with derived dimensional formula $[M^1 L^{-1} T^{-2}]$, we get:

$a = 1,\ b = -1,\ c = -2$

So, the value of $2a - b + c = 2(1) - (-1) + (-2) = 2 + 1 - 2 = 1$

Therefore, the value of $2a - b + c$ is 1. Thus, the correct options is (B).

Question16

In a screw gauge the zero of main scale reference line coincides with the fifth division of the circular scale when two studs are in contact. There are 100 divisions in circular scale and pitch of screw gauge is 0.1 mm. When diameter of a sphere is measured, the reading of main scale is 5 mm and 50^{th} division of circular scale coincides with the reference line of main scale. The diameter of sphere is ______ mm.

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Options:

A.

5.045

B.

5.055

C.

5.450

D.

5.550

Answer: A

Solution:

The least count is the smallest distance the screw moves when the circular scale is rotated by one division. It is defined as:

$$\text{Least Count } = \frac{\text{Pitch}}{\text{Number of divisions on circular scale}}$$

The pitch is linear distance moved in one complete rotation. The pitch of screw gauge is given as 0.1 mm and there are 100 divisions:

So, the least count is $\mathrm{LC}=\frac{0.1 \mathrm{~mm}}{100}=0.001 \mathrm{~mm}$

When the studs are in contact, the 5th division of the circular scale coincides with the reference line. Since the zero of the circular scale has passed the reference line, the error is positive.

$\text{Zero Error} = +(\text{ Coinciding division} \times \mathrm{LC})$

$\Rightarrow \text{Zero Error} = +(5 \times 0.001 \mathrm{~mm}) = +0.005 \mathrm{~mm}$

The total observed reading is the sum of the Main Scale Reading (MSR) and the Circular Scale Reading (CSR).

$\mathrm{MSR} = 5 \mathrm{~mm}$

$\mathrm{CSR} = \text{Coinciding circular division} \times \mathrm{LC} = 50 \times 0.001 \mathrm{~mm} = 0.05 \mathrm{~mm}$

So, observed reading $= 5 \mathrm{~mm} + 0.05 \mathrm{~mm} = 5.05 \mathrm{~mm}$

The true diameter is obtained by subtracting the zero error from the observed reading:

So, the true diameter = Observed Reading - Zero Error

$\text{True Diameter} = 5.05 \mathrm{~mm} - 0.005 \mathrm{~mm} = 5.045 \mathrm{~mm}$

Therefore, the correct option is (A).

Question17

Dimensions of universal gravitational constant (G) in terms of Planck's constant (h), distance (L), mass (M) and time (T) are ________.

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Options:

A.

$[hTLM^{-2}]$

B.

$[hT^{-1}LM^{-2}]$

C.

$[hTL^2M^{-2}]$

D.

$[h^{-1}T^{-1}LM^{-2}]$

Answer: B

Solution:

To find the dimensions of the universal gravitational constant G in terms of Planck's constant h , length L , mass M , and time T , we must first determine the base dimensions of both G and h in the standard MLT system.

According to Newton's law of universal gravitation, the force F between two masses m_1 and m_2 separated by a distance r is given by:

$$F = \frac{Gm_1 m_2}{r^2}$$

$$\Rightarrow G = \frac{Fr^2}{m_1 m_2}$$

Substituting the dimensions $[MLT^{-2}]$ for force, $[L]$ for distance (r), and mass(m) as $[M]$:

$$[G] = \frac{[MLT^{-2}][L^2]}{[M][M]} = [M^{-1} L^3 T^{-2}]$$

Planck's constant relates the energy E of a photon to its frequency v via the equation:

$$E = h\nu$$

Since frequency v has dimensions of inverse time [T^{-1}], and energy E (work done) has dimensions $[ML^2 T^{-2}]$, we have:

$$[h] = \frac{[E]}{[v]} = \frac{[ML^2 T^{-2}]}{[T^{-1}]} = [ML^2 T^{-1}]$$

Let assume the relationship is of the form $G = h^a L^b M^c T^d$. Comparing the dimensions:

$$[M^{-1} L^3 T^{-2}] = [ML^2 T^{-1}]^a [L]^b [M]^c [T]^d$$

$$[M^{-1} L^3 T^{-2}] = [M^{a+c} L^{2a+b} T^{-a+d}]$$

By equating the exponents of M, L, and T from both sides, we get a system of equations:

For $M : a + c = -1 \ldots$ (i)

For $L : 2a + b = 3 \ldots$ (ii)

For $T : -a + d = -2 \ldots$ (iii)

Looking at the provided options, we see that h is with a power of $1, a = 1$

Let's $a = 1$:

From eq (i): $1 + c = -1 \Rightarrow c = -2$

From eq (ii): $2(1) + b = 3 \Rightarrow b = 1$

From eq (iii): $-1+\mathrm{d}=-2 \Rightarrow \mathrm{~d}=-1$

Substituting these values, we get $G \propto h^1 L^1 M^{-2} T^{-1}$. This corresponds to the dimensional formula $\left[\mathrm{hT}^{-1} \mathrm{LM}^{-2}\right]$.

Hence, the correct option is (B).

Question18

In a screw gauge when the circular scale is given five complete rotations it moves linearly by 2.5 mm . If the circular scale has 100 divisions, the least count of screw gauge is ____ mm.

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Options:

A.

1×10^{-2}

B. 1×10^{-3}

C.

5×10^{-2}

D. 5×10^{-3}

Answer: D

Solution:

The pitch of a screw gauge is defined as the linear distance moved by the screw on the main scale when the circular head is given one complete rotation.

The screw moves linearly by 2.5 mm when given five complete rotations. Therefore, the distance moved in a single rotation is :

$$\text{Pitch } =\frac{\text{Total linear distance moved}}{\text{Number of full rotations}}$$

$$\text{Pitch } =\frac{2.5 \text{ mm}}{5}=0.5 \text{ mm}$$

The least count is the smallest distance that can be measured accurately using the instrument. It is the distance moved by the screw when the circular scale is rotated through only one of its divisions.

So, the least count of screw gauge is the ratio of the pitch to the total number of divisions on the circular scale:

It is given that the circular scale has 100 divisions,

$\text{L. C.} = \frac{0.5 \text{ mm}}{100} = 0.005 \text{ mm} = 5 \times 10^{-3} \text{ mm}$

Therefore, the least count of given screw gauge is 5×10^{-3} mm.

Thus, the correct option is (D).

Question19

Match the LIST-I with LIST-II

List - I		List - II	
A.	Planck's constant	I.	$\mathrm{ML}^2 \mathrm{~T}^{-2}$
B.	Stopping potential	II.	T^{-1}
C.	Work function	III.	$\mathrm{ML}^2 \mathrm{~T}^{-1}$
D.	Threshold frequency	IV.	$\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-1}$

Choose the correct answer from the options given below:

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Options:

A.

A-III, B-IV, C-I, D-II

B.

A-I, B-II, C-III, D-IV

C.

A-IV, B-III, C-I, D-II

D.

A-I, B-IV, C-III, D-II

Answer: A

Solution:

For the dimensions of Planck's Constant (h)

According to the photon energy equation $\mathrm{E}=\mathrm{h}\nu$, where E is energy and ν is frequency.

Planck's constant is given by $\mathrm{h}=\frac{\mathrm{E}}{\mathrm{v}}$

The dimensions of energy (E) are $[\mathrm{E}]=\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$.

The dimensions of frequency (v) are $[v]=\left[\mathrm{T}^{-1}\right]$.

Therefore, dimensions of h :

$$[\mathrm{h}]=\frac{\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]}{\left[\mathrm{T}^{-1}\right]}=\left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]$$

Thus, $\mathrm{A}\rightarrow\mathrm{III}$

For the dimensions of stopping potential (V_0)

Stopping potential is defined as the potential difference required to stop the fastest moving electrons. The work done by the potential is equal to the kinetic energy, $\mathrm{W}=\mathrm{qV}_0$, where q is charge.

So, stopping potential is $\mathrm{V}_0=\frac{\mathrm{w}}{\mathrm{q}}$.

The dimensions of work (W) are $[\mathrm{W}]=\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$.

The dimensions of charge ($\mathrm{q}=\mathrm{I}\times\mathrm{t}$) are $[\mathrm{q}]=[\mathrm{AT}]$.

Therefore, dimensions of stopping potential is:

$$[\mathrm{V}_0]=\frac{\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]}{[\mathrm{AT}]}=\left[\mathrm{ML}^2\,\mathrm{T}^{-3}\,\mathrm{A}^{-1}\right]$$

Thus, $\mathrm{B}\rightarrow\mathrm{IV}$.

For the dimensions of work function (ϕ)

The work function is the minimum energy required to eject an electron from a metal surface. Since it is a form of energy, its dimensions are the same as energy (E).

So, the dimensions of work function is $[\phi]=\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$

Thus, $\mathrm{C}\rightarrow\mathrm{I}$.

For the dimensions of threshold frequency (v_0)

Threshold frequency is the minimum frequency of incident radiation required for photoelectric emission.

Frequency is defined as the reciprocal of the time period $\left(v=\frac{1}{\mathrm{T}}\right)$.

So, the dimensions of threshold frequency is $[v_0] = [\mathrm{T}^{-1}]$

Thus, C $\rightarrow$ II.

Therefore, the correct match is A - III, B - IV, C - I, D - II.

Hence, the correct option is (A).

Question20

L, C and R represents physical quantities inductance, capacitance and resistance respectively. The dimensional formula $\mathrm{ML}^2\ \mathrm{T}^{-4}\ \mathrm{A}^{-2}$ corresponds to ____ .

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Options:

A.

$\frac{R}{\sqrt{LC}}$

B.

$\frac{R}{LC}$

C.

$\frac{C}{\sqrt{LR}}$

D.

$\frac{1}{R}\sqrt{\frac{L}{C}}$

Answer: A

Solution:

Electric power (P) is defined as work done per unit time. In a resistor of resistance R if the current is I , then electric power is given as,

$\mathrm{P} = \mathrm{I}^2\mathrm{R}$

For the dimensions of force (F) = Mass × Acceleration = $\left[\mathrm{MLT}^{-2}\right]$

For the dimensions of work (W) = Force × Distance = $\left[\mathrm{MLT}^{-2}\right] \times [\mathrm{L}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]$

Hence, the dimensions of power are,

$$[\mathrm{P}]=\left[\frac{\mathrm{W}}{\mathrm{T}}\right]=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]}{[\mathrm{T}]}=\left[\mathrm{ML}^2 \mathrm{~T}^{-3}\right]$$

From the electric power formula,

$$\mathrm{R}=\frac{\mathrm{P}}{\mathrm{I}^2}$$

$$\Rightarrow[\mathrm{R}]=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-3}\right]}{\left[\mathrm{A}^2\right]}=\left[\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-2}\right]$$

The energy (E) stored in a capacitor is $\mathrm{E}=\frac{1}{2} C V^2$.

Electric Potential (V) is defined as work per unit charge, $\mathrm{V}=\frac{\mathrm{W}}{\mathrm{Q}}$,

The dimensions of charge (Q) is $[\mathrm{Q}]=[\mathrm{AT}]$.

So, the dimensions of potential is, $[\mathrm{V}]=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]}{[\mathrm{AT}]}=\left[\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-1}\right]$

Using the definition of capacitance $\mathrm{C}=\frac{\mathrm{Q}}{\mathrm{V}}$, the dimensions of capacitance is;

$$[C]=\frac{[A T]}{\left[M L^{2-3} A^{-1}\right]}=\left[M^{-1} L^{-2} T^4 A^2\right]$$

The energy (E) stored in an inductor is $\mathrm{E}=\frac{1}{2} \mathrm{LI}^2$.

The dimensions of energy are $\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]$.

So, the dimensions of inductance are,

$$\mathrm{L}=\frac{2 \mathrm{E}}{\mathrm{I}^2}$$

$$\Rightarrow[\mathrm{L}]=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]}{\mathrm{A}^2}=\left[\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~A}^{-2}\right]$$

Hence, the dimensions of inductance, capacitance, and resistance are,

$$[\mathrm{R}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-3} \mathrm{~A}^{-2}\right],[\mathrm{C}]=\left[\mathrm{M}^{-1} \mathrm{~L}^{-2} \mathrm{~T}^4 \mathrm{~A}^2\right],[\mathrm{L}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~A}^{-2}\right]$$

Let the dimensional formula [$\mathrm{ML}^2 \mathrm{~T}^{-4} \mathrm{~A}^{-2}$] corresponds to $\mathrm{R}^x \mathrm{L}^y \mathrm{C}^z$

Substituting these into the equation :

$$\left[M^1 L^2 T^{-4} A^{-2}\right]=\left[M L^2 T^{-3} A^{-2}\right]^x\left[M L^2 T^{-2} A^{-2}\right]^y\left[M^{-1} L^{-2} T^4 A^2\right]^z$$

$$\left[M^1 L^2 T^{-4} A^{-2}\right]=\left[M^{x+y-z} L^{2 x+2 y-2 z} T^{-3 x-2 y+4 z} A^{-2 x-2 y+2 z}\right]$$

Comparing the exponents for each fundamental unit:

For mass, M : $x + y - z = 1 \ldots$ (i)

For length, L : $2x + 2y - 2z = 2 \Rightarrow x + y - z = 1 \ldots$ (ii)

For time, T : $-3x - 2y + 4z = -4 \ldots$ (iii)

For current, A : $-2x - 2y + 2z = -2 \ldots$ (iv)

From (i): $x + y = 1 + z$

Substituting this into (ii) :

$-3x - 2y + 4z = -4$

$-x - 2(x + y) + 4z = -4$

$$-x - 2(1 + z) + 4z = -4$$
$$-x - 2 - 2z + 4z = -4$$
$$-x + 2z = -2 \Rightarrow x = 2z + 2 \ldots (v)$$

From given options the possible exponents of resistance (R) are 1 (a and b), $-\frac{1}{2}$ (c) or -1 (d).

If $x = 1, z = \frac{(x-2)}{2} = \frac{(1-2)}{2} = -\frac{1}{2}$; then $x + y = 1 + z \Rightarrow 1 + y = 1 - \frac{1}{2} \Rightarrow y = -\frac{1}{2}$

Then the expression will be $R^x L^y C^z = R^1\, L^{-1/2} C^{-1/2} = \frac{R}{\sqrt{LC}} \rightarrow$ (a)

If $x = -\frac{1}{2}, z = \frac{(x-2)}{2} = \frac{(-\frac{1}{2}-2)}{2} = -\frac{5}{4}$; then $x + y = 1 + z \Rightarrow -\frac{1}{2} + y = 1 - \frac{5}{2} \Rightarrow y = -1$

Then the expression will be $R^x L^y C^z = R^{-1/2}\, L^{-1} C^{-5/4}$ (doesn't match option (c))

If $x = -1, z = \frac{(x-2)}{2} = \frac{(-1-2)}{2} = -\frac{3}{2}$; then $x + y = 1 + z \Rightarrow -1 + y = 1 - \frac{3}{2} \Rightarrow y = \frac{1}{2}$

Then the expression will be $R^x L^y C^z = R^{-1}\, L^{1/2} C^{1/2} = \frac{\sqrt{LC}}{R}$ (doesn't match option (d))

Therefore, the dimensional formula [$M^1\, L^2\, T^{-4}\, A^{-2}$] corresponds to $\frac{R}{\sqrt{LC}}$.

Question21

In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7th Vernier division coincides with a main scale reading. If the value of 1 main scale division is 1 mm and there are 10 Vernier scale divisions, then the Vernier caliper has

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Options:

A.

0.07 cm negative zero error

B.

0.7 cm negative zero error

C.

0.07 cm positive zero error

D.

0.7 cm positive zero error

Answer: C

Solution:

The Least Count is the smallest measurement that can be accurately taken using the instrument.

For a Vernier calliper, the difference between the length of one Main Scale Division (MSD) and one Vernier Scale Division (VSD) is equal to the least count.

$\mathrm{LC} = 1\mathrm{MSD} - 1\mathrm{VSD}$

It is given that $1\mathrm{MSD} = 1\ \mathrm{mm}$.

In a standard Vernier calliper arrangement, 10 divisions on the Vernier scale correspond to the length of 9 divisions on the Main scale.

$10\mathrm{VSD} = 9\mathrm{MSD}$

$\Rightarrow 1\mathrm{VSD} = \frac{9}{10}\mathrm{MSD} = 0.9\ \mathrm{mm}$

So, the least count is;

$\mathrm{LC} = 1\ \mathrm{mm} - 0.9\ \mathrm{mm} = 0.1\ \mathrm{mm}$

$\Rightarrow \mathrm{LC} = 0.1\ \mathrm{mm} = 0.01\ \mathrm{cm}$

Zero error occurs when the jaws of the calliper are completely closed but the zero mark of the main scale does not align with the zero mark of the Vernier scale.

If the zero of the Vernier scale rests to the right of the main scale zero, the instrument will naturally measure a value larger than the true length and this is positive zero error.

If the zero of the Vernier scale rests to the left of the main scale zero, it creates a negative zero error.

It is given that the zero of the Vernier scale is shifted to the right. Therefore, the given Vernier scale is with a positive zero error.

If Vernier division perfectly coincides with any main scale reading, then the zero error is,

$e = n \times LC$

Where n is the coinciding division. The problem states that the 7^{th} division coincides.

$e = 7 \times 0.01 \text{ cm} = 0.07 \text{ cm}$

Since the shift is to the right, the final error is +0.07 cm .

Thus, the correct option is (C).

Question22

In an experiment to determine the resistance of a given wire using Ohm's law, the voltmeter and ammeter readings are noted as 10 V and 5 A , respectively. The least counts of voltmeter and ammeter are 500 mV and 200 mA , respectively. The estimated error in the resistance measurement is _____ Ω

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Options:

A.

0.25

B.

2

C.

2.5

D.

0.18

Answer: D

Solution:

According to Ohm's law, the resistance R is given by the ratio of potential difference V to the current I :

$R = \frac{V}{I}$

The reading of voltmeter is $\mathrm{V}=10\ \mathrm{V}$ and that of ammeter is $\mathrm{I}=5\ \mathrm{A}$

Then resistance is $\mathrm{R}=\frac{10}{5}=2\Omega$

When a quantity is determined by the division of two other quantities $\left(\mathrm{R}=\frac{\mathrm{V}}{\mathrm{I}}\right)$, the relative error in the result is the sum of the relative errors of the individual quantities.

The relative error is,

$\frac{\Delta \mathrm{R}}{\mathrm{R}}=\frac{\Delta \mathrm{V}}{\mathrm{V}}+\frac{\Delta \mathrm{I}}{\mathrm{I}}$

Where $\Delta \mathrm{R}$ is the absolute error in resistance, $\Delta \mathrm{V}$ is the error in voltage, and $\Delta \mathrm{I}$ is the error in current.

The error in a measurement is taken as the least count of the instrument used.

So, error in voltmeter is $\Delta \mathrm{V}=500 \mathrm{mV}=0.5\ \mathrm{V}$

And the error in ammeter is $\Delta \mathrm{I}=200\ \mathrm{mA}=0.2\ \mathrm{A}$

Substituting values into the error equation

$\frac{\Delta \mathrm{R}}{2}=\frac{0.5}{10}+\frac{0.2}{5}$

$\Rightarrow \frac{\Delta \mathrm{R}}{2}=0.05+0.04=0.09$

$\Rightarrow \Delta \mathrm{R}=0.09 \times 2=0.18 \Omega$

Therefore, the estimated error in the resistance measurement is 0.18Ω.

Hence, the correct option is (D).

Question23

Match List - I with List - II.

List - I		List - II	
A.	Meter (L)	I.	$\sqrt{\frac{hc}{G}}$
B.	Second (S)	II.	$\sqrt{\frac{Gh}{c^5}}$
C.	Kilogram (M)	III.	$\sqrt{\frac{K^2L^2c^3}{Gh}}$
D.	Kelvin (K)	IV.	$\sqrt{\frac{Gh}{c^3}}$

where h (Planck's constant), G (gravitational constant) and c (speed of light in vacuum) as fundamental units.

Choose the correct answer from the options given below :

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Options:

A.

A-II, B-IV, C-I, D-III

B.

A-IV, B-II, C-I, D-III

C.

A-IV, B-I, C-II, D-III

D.

A-III, B-I, C-II, D-IV

Answer: B

Solution:

The speed of light (c) is $\frac{\text{distance}}{\text{time}}$.

So, the dimensions of speed of light is $[\mathrm{c}] = \left[\mathrm{LT}^{-1}\right]$

From $\mathrm{E} = \mathrm{h}\nu$, where E is energy $\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]$ and ν is frequency $\left[\mathrm{T}^{-1}\right]$.

So, the dimensions of Planck's constant is,

$$[\mathrm{h}] = \frac{[\mathrm{E}]}{[v]} = \frac{\left[\mathrm{ML}^2\,\mathrm{T}^{-2}\right]}{\left[\mathrm{T}^{-1}\right]} = \left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]$$

From Newton's law of gravitation, $\mathrm{F} = \frac{\mathrm{Gm}_1\,\mathrm{m}_2}{\mathrm{r}^2}$, where F is gravitational force $[\mathrm{F}] = \left[\mathrm{MLT}^{-2}\right]$, G is the universal gravitational constant, m is the mass $[\mathrm{m}] = \left[\mathrm{M}^1\,\mathrm{L}^0\,\mathrm{T}^0\right]$ and r is the distance $[\mathrm{r}] = \left[\mathrm{M}^0\mathrm{LT}^0\right]$

So, the dimensions of G is,

$$[\mathrm{G}] = \frac{[\mathrm{F}]\left[\mathrm{r}^2\right]}{\left[\mathrm{m}^2\right]} = \frac{\left[\mathrm{MLT}^{-2}\right]\left[\mathrm{L}^2\right]}{\left[\mathrm{M}^2\right]} = \left[\mathrm{M}^{-1}\,\mathrm{L}^3\,\mathrm{T}^{-2}\right]$$

Let a physical quantity Q be proportional to $\mathrm{h}^{\mathrm{x}}\mathrm{G}^{\mathrm{y}}\mathrm{c}^{\mathrm{z}}$. Using principle of dimensional homogeneity

$$[\mathrm{Q}] = [\mathrm{h}]^{\mathrm{x}}[\mathrm{G}]^{\mathrm{y}}[\mathrm{c}]^{\mathrm{z}}$$

$\Rightarrow [Q] = \left[ML^2T^{-1}\right]^x \left[M^{-1}L^3T^{-2}\right]^y \left[LT^{-1}\right]^z$

$\Rightarrow [Q] = M^{x-y}L^{2x+3y+z}T^{-x-2y-z}$

For Kilogram (mass), dimensions are $\left[\mathrm{M}^1\ \mathrm{L}^0\ \mathrm{T}^0\right]$.

On comparing the powers of dimensions,

$x - y = 1 \Rightarrow x = 1 + y \ldots$ (i)

$2x + 3y + z = 0 \ldots (ii)$

$-x - 2y - z = 0 \Rightarrow z = -x - 2y$....(iii)

Substituting z into equation (i):

$2x + 3y + (-x - 2y) = 0 \Rightarrow x + y = 0 \Rightarrow x = -y$

From equation (i),

$-y = 1 + y \Rightarrow 2y = -1 \Rightarrow y = -1/2$

Then x $= 1/2$.

And $z = -\left(\frac{1}{2}\right) - 2\left(-\frac{1}{2}\right) = -\frac{1}{2} + 1 = \frac{1}{2}$

kilogram $= h^{1/2}G^{-1/2}c^{1/2} = \sqrt{\frac{hc}{G}}$

So, $\mathrm{M} \propto \sqrt{\frac{\mathrm{hc}}{\mathrm{G}}} \Rightarrow$ (C) $\rightarrow$ (I)

For meter (length), the dimensions are $\left[\mathrm{M}^0\ \mathrm{L}^1\ \mathrm{T}^0\right] = \left[\mathrm{M}^{\mathrm{x-y}}\mathrm{L}^{2\mathrm{x}+3\mathrm{y}+\mathrm{z}}\mathrm{T}^{-\mathrm{x}-2\mathrm{y}-\mathrm{z}}\right]$

On comparing the powers of dimensions,

$x - y = 0 \Rightarrow x = y$....(i)

$2x + 3y + z = 1 \ldots$(ii)

$-x - 2y - z = 0 \Rightarrow z = -x - 2y$.....(iii)

Substituting z into equation (ii):

$2x + 3y - x - 2y = 1 \Rightarrow x + y = 1$

Since $\mathrm{x} = \mathrm{y}$, then $2\mathrm{x} = 1 \Rightarrow \mathrm{x} = \frac{1}{2}, \mathrm{y} = \frac{1}{2}$.

Then $\mathrm{z} = -\frac{1}{2} - 2\left(\frac{1}{2}\right) = -\frac{3}{2}$

Thus, Meter (L) $= [\mathrm{h}]^{1/2}[\mathrm{G}]^{1/2}[\mathrm{c}]^{-3/2}$

So, $\mathrm{L} \propto \sqrt{\frac{\mathrm{Gh}}{\mathrm{c}^3}} \Rightarrow$ (A) $\rightarrow$ (IV)

For Second (time), the dimensions are $\left[\mathrm{M}^0\ \mathrm{L}^0\ \mathrm{T}^1\right]$.

On comparing the powers of dimensions,

$x-y=0 \Rightarrow x=y$ ....(i)

$$2\mathrm{x}+3\mathrm{y}+\mathrm{z}=0 \Rightarrow 2\mathrm{x}+3\mathrm{x}+\mathrm{z}=0 \Rightarrow \mathrm{z}=-5\mathrm{x} \ldots (ii)$$
$$-\mathrm{x}-2\mathrm{y}-\mathrm{z}=1 \Rightarrow -\mathrm{x}-2\mathrm{x}-\mathrm{z}=1 \Rightarrow -3\mathrm{x}-\mathrm{z}=1 \ldots\ldots (iii)$$

Substituting z :

$-3x-(-5x)=1 \Rightarrow 2x=1 \Rightarrow x=\frac{1}{2}, y=\frac{1}{2}$

Then $\mathrm{z}=-5\left(\frac{1}{2}\right)=-\frac{5}{2}$.

Thus, Second $(S)=[h]^{1/2}[G]^{1/2}[c]^{-5/2}$

So, $\mathrm{S} \propto \sqrt{\frac{\mathrm{Gh}}{\mathrm{c}^5}} \Rightarrow (\mathrm{B}) \rightarrow (\mathrm{II})$

Let a physical quantity θ represent the Kelvin be proportional to $\mathrm{h}^{\mathrm{x}}\mathrm{G}^{\mathrm{y}}\mathrm{c}^{\mathrm{z}}\mathrm{K}^{\mathrm{w}}$. Using principle of dimensional homogeneity

$[\theta]=[\mathrm{h}]^{\mathrm{x}}[\mathrm{G}]^{\mathrm{y}}[\mathrm{c}]^{\mathrm{z}}[\mathrm{K}]^{\mathrm{w}}$

$\Rightarrow [\theta]=\left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]^{\mathrm{x}}\left[\mathrm{M}^{-1}\,\mathrm{L}^3\,\mathrm{T}^{-2}\right]^{\mathrm{y}}\left[\mathrm{LT}^{-1}\right]^{\mathrm{z}}[\mathrm{K}]^{\mathrm{w}}$

$\Rightarrow [\theta]=M^{x-y}L^{2x+3y+z}T^{-x-2y-z}K^{w}$

The dimensions of temperature is $[\theta]=\left[\mathrm{M}^0\,\mathrm{L}^0\,\mathrm{T}^0\,\mathrm{K}^1\right]=\left[\mathrm{M}^{\mathrm{x}-\mathrm{y}}\mathrm{L}^{2\mathrm{x}+3\mathrm{y}+\mathrm{z}}\mathrm{T}^{-\mathrm{x}-2\mathrm{y}-\mathrm{z}}\mathrm{K}^{\mathrm{w}}\right]$ On comparing the powers,

$x-y=0 \Rightarrow x=y \ldots (i)$

$2x+3y+z=0 \Rightarrow 2x+3x+z=0 \Rightarrow z=-5x \ldots (ii)$

$-x-2y-z=0 \ldots \text{(iii)}$

$\mathrm{w}=1 \ldots$ (iv)

On simplifying $\sqrt{\frac{\mathrm{K}^2\,\mathrm{L}^2\mathrm{c}^3}{\mathrm{Gh}}}=\sqrt{\frac{\mathrm{K}^2\,\mathrm{L}^2\left[\mathrm{LT}^{-1}\right]^2}{\left(\mathrm{M}^{-1}\,\mathrm{L}^3\,\mathrm{T}^{-2}\right)\left[\left[\mathrm{ML}^2\,\mathrm{T}^{-1}\right]\right]}}=\sqrt{\mathrm{K}^2}=\mathrm{K}$ which is kelvin. Thus, Kelvin

$(\mathrm{K})=\sqrt{\frac{\mathrm{K}^2\,\mathrm{L}^2\mathrm{c}^3}{\mathrm{Gh}}} \Rightarrow (\mathrm{D}) \rightarrow (\mathrm{III})$

A - IV, B - II, C - I, D - III

Therefore, the correct option is (B).

Question24

The potential energy of a particle changes with distance x from a fixed origin as $V=\frac{A\sqrt{x}}{x+B}$, where A and B are constant with appropriate dimensions. The dimensions of AB are ____

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Options:

A.

$\left[\mathrm{M}^1 \mathrm{~L}^{5/2} \mathrm{~T}^{-2}\right]$

B.

$\left[\mathrm{M}^{3/2} \mathrm{~L}^{5/2} \mathrm{~T}^{-2}\right]$

C.

$\left[\mathrm{M}^1 \mathrm{~L}^2 \mathrm{~T}^{-2}\right]$

D.

$\left[\mathrm{M}^1 \mathrm{~L}^{7/2} \mathrm{~T}^{-2}\right]$

Answer: D

Solution:

According to the principle of homogeneity, dimensions of each term on both sides of a equation must be the same. Also, only physical quantities of the same dimensions can be added or subtracted.

The potential energy of the particle is given as $V=\frac{A\sqrt{x}}{x+B}$, where A and B are constant.

In the denominator of the expression $\mathrm{x}+\mathrm{B}$, the constant B is added to distance x . Therefore, the dimensions of B must be the same as the dimensions of x.

$[\mathrm{B}]=[\mathrm{x}]=\left[\mathrm{L}^1\right]$

The potential energy V has the same dimensions as work or energy.

$[\mathrm{V}]=[\mathrm{F} \times \mathrm{x}]=\left[\mathrm{M}^1 \mathrm{~L}^1 \mathrm{~T}^{-2} \times \mathrm{L}^1\right]=\left[\mathrm{M}^1 \mathrm{~L}^2 \mathrm{~T}^{-2}\right]$

On rearranging the formula of potential energy for A :

$A=\frac{V(x+B)}{\sqrt{x}}$

Substituting the dimensions:

$[\mathrm{A}]=\frac{\left[\mathrm{M}^1 \mathrm{~L}^2 \mathrm{~T}^{-2}\right]\left[\mathrm{L}^1\right]}{\left[\mathrm{L}^{1/2}\right]}$

$$\Rightarrow [\mathrm{A}] = \frac{\left[\mathrm{M}^1\,\mathrm{L}^3\,\mathrm{T}^{-2}\right]}{\left[\mathrm{L}^{1/2}\right]}$$

$$\Rightarrow [\mathrm{A}] = \left[\mathrm{M}^1\,\mathrm{L}^{3-\frac{1}{2}}\,\mathrm{T}^{-2}\right] = \left[\mathrm{M}^1\,\mathrm{L}^{5/2}\,\mathrm{T}^{-2}\right]$$

To find the dimensions of the product AB :

$[\mathrm{AB}] = [\mathrm{A}] \times [\mathrm{B}]$

$$\Rightarrow [\mathrm{AB}] = \left[\mathrm{M}^1\,\mathrm{L}^{5/2}\,\mathrm{T}^{-2}\right] \times \left[\mathrm{L}^1\right]$$

$$\Rightarrow [\mathrm{AB}] = \left[\mathrm{M}^1\,\mathrm{L}^{5/2+1}\,\mathrm{T}^{-2}\right]$$

$$\Rightarrow [\mathrm{AB}] = \left[\mathrm{M}^1\,\mathrm{L}^{7/2}\,\mathrm{T}^{-2}\right]$$

Therefore, the dimensions of AB is $\left[M^1 L^{7/2} T^{-2}\right]$.

Thus, the correct option is (D).

Question25

The density ρ of a uniform cylinder is determined by measuring its mass m, length l and diameter d. The measured values of m, l and d are 97.42 ± 0.02 g, 8.35 ± 0.05 mm and 20.20 ± 0.02 mm, respectively. Calculated percentage fractional error in ρ is ____ .

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Options:

A.

0.63%

B.

0.82%

C.

0.72%

D.

0.25%

Answer: B

Solution:

The density (ρ) of a body is defined as the ratio of its mass (m) to its volume (V).

$\rho = \frac{m}{V}$

For a uniform cylinder of length l and diameter D , the volume is,

$V = \pi r^2 l = \pi\left(\frac{D}{2}\right)^2 l = \frac{\pi D^2 l}{4}$

Substituting the volume into the density formula:

$\rho = \frac{\mathrm{m}}{\mathrm{V}} = \frac{4\ \mathrm{m}}{\pi \mathrm{D}^2 \mathrm{l}}$

To find the fractional error, we take the natural logarithm $(\ln)$ on both sides:

$\ln(\rho) = \ln(4) + \ln(m) - \ln(\pi) - 2\ln(D) - \ln(l)$

Differentiating both sides to find the relative change

$\frac{\mathrm{d}\rho}{\rho} = \frac{\mathrm{dm}}{\mathrm{m}} - \frac{2\mathrm{dD}}{\mathrm{D}} - \frac{\mathrm{dl}}{\mathrm{l}}$

The maximum possible fractional error is the sum of the absolute values of the individual relative errors:

$\frac{\Delta\rho}{\rho} = \frac{\Delta \mathrm{m}}{\mathrm{m}} + \frac{2\Delta \mathrm{D}}{\mathrm{D}} + \frac{\Delta \mathrm{l}}{\mathrm{l}}$

For the percentage fractional error,

$\%\ \text{Error in}\ \rho = \left(\frac{\Delta \mathrm{m}}{\mathrm{m}} + \frac{2\Delta \mathrm{D}}{\mathrm{D}} + \frac{\Delta \mathrm{l}}{\mathrm{l}}\right) \times 100$

The measured value of mass is $97.42 \pm 0.02\ \mathrm{g} \Rightarrow\ \mathrm{m} = 97.42\ \mathrm{g}$ and the absolute error is $\Delta \mathrm{m} = 0.02\ \mathrm{g}$ So, relative error in mass is,

$\frac{\Delta \mathrm{m}}{\mathrm{m}} = \frac{0.02}{97.42} \approx 0.000205$

The measured value of length is $8.35 \pm 0.05\ \mathrm{mm} \Rightarrow \mathrm{l} = 8.35\ \mathrm{mm}$ and the absolute error is $\Delta \mathrm{l} = 0.05\ \mathrm{mm}$

So, relative error in length is,

$\frac{\Delta \mathrm{l}}{\mathrm{l}} = \frac{0.05}{8.35} \approx 0.005988$

The measured value of diameter is $20.20 \pm 0.02\ \mathrm{mm} \Rightarrow \mathrm{D} = 20.20\ \mathrm{mm}$ and the absolute error is $\Delta \mathrm{D} = 0.02\ \mathrm{mm}$

So, relative error in diameter is,

$\frac{2\Delta \mathrm{D}}{\mathrm{D}} = \frac{2\times 0.02}{20.20} = \frac{0.04}{20.20} \approx 0.001980$

Putting the values in the expression for the percentage error in density,

$\frac{\Delta\rho}{\rho} \times 100\% = (0.000205 + 0.005988 + 0.001980) \times 100\% = 0.008173 \times 100\%$

$\Rightarrow \frac{\Delta \rho}{\rho} \times 100\% = 0.8173\% \approx 0.82\%$

Therefore, the calculated percentage fraction error in density is approximately 0.82%.

Hence, the correct option is (B).

Question26

Match List - I with List - II.

List - I		List - II	
A.	Boltzmann constant	I.	$\left[M^{-1} L^3 T^{-2}\right]$
B.	Stefan's constant	II.	$\left[M L^2 T^{-1}\right]$
C.	Planck's constant	III.	$\left[ML^2 T^{-2} K^{-1}\right]$
D.	Gravitational constant	IV.	$\left[M L^0 T^{-3} K^{-4}\right]$

Choose the correct answer from the options given below :

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Options:

A.

A-I, B-II, C-III, D-IV

B.

A-IV, B-III, C-II, D-I

C.

A-III, B-IV, C-II, D-I

D.

A-II, B-I, C-IV, D-III

Answer: C

Solution:

For the dimensional equation of Boltzmann Constant (k_B)

The average kinetic energy (E) of a gas molecule is related to temperature (T) by the equation :

$\mathrm{E}=\frac{3}{2} \mathrm{k}_{\mathrm{B}} \mathrm{T}$

To find the dimensions of k_{B} :

$[\mathrm{k}_{\mathrm{B}}]=[\mathrm{E}][\mathrm{T}]$

Using the formula Work $=$ Force $\times$ Distance, dimensions of energy is $[\mathrm{E}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]$

The dimensions of temperature is $[\mathrm{T}]=[\mathrm{K}]$

So,

$[\mathrm{k}_{\mathrm{B}}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right][\mathrm{K}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~K}^{-1}\right]$

Thus, (A) Boltzmann constant $\rightarrow$ (III) [$\mathrm{ML}^2 \mathrm{~T}^{-2} \mathrm{~K}^{-1}$]

For the dimensions of Stefan's Constant (σ)

According to Stefan-Boltzmann law, the energy radiated per unit area per unit time (P) is:

$\mathrm{P}=\sigma \mathrm{T}^4$

Where $\mathrm{P}=\frac{\text{Energy}}{\text{Area} \times \text{ Time}}$

Dimensions of

$[\mathrm{P}]=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]}{[\mathrm{L}^2][\mathrm{T}]}=\left[\mathrm{MT}^{-3}\right]$

Dimensions of T^4 is $\left[\mathrm{K}^4\right]$

So, the dimensions of Stefan's constant is

$[\sigma]=\frac{[\mathrm{P}]}{[\mathrm{T}^4]}=\frac{[\mathrm{MT}^{-3}]}{[\mathrm{K}^4]}=\left[\mathrm{MT}^{-3} \mathrm{~K}^{-4}\right]=\left[\mathrm{ML}^0 \mathrm{~T}^{-3} \mathrm{~K}^{-4}\right]$

Thus, (B) Stefan's constant $\rightarrow$ (IV)[$\mathrm{ML}^0 \mathrm{~T}^{-3} \mathrm{~K}^{-4}$]

For the dimensions of Planck's Constant (h):

The energy (E) of a photon is given by $E=h \nu$, where ν is frequency (Time $^{-1}$).

So, the dimension of Planck's constant is

$[\mathrm{h}]=\frac{[\mathrm{E}]}{[v]}=\frac{\left[\mathrm{ML}^2 \mathrm{~T}^{-2}\right]}{[\mathrm{T}^{-1}]}$

$\Rightarrow[\mathrm{h}]=\left[\mathrm{ML}^2 \mathrm{~T}^{-1}\right]$

Thus, (C) Planck's constant → (II)[$\mathrm{ML^2\,T^{-1}}$]

For the dimensions of Gravitational Constant (G):

From Newton's law of gravitation $F=\frac{Gm_1m_2}{r^2}\Rightarrow G=\frac{Fr^2}{m_1m_2}$

The dimensions of Force is $[\mathrm{F}]=\left[\mathrm{MLT^{-2}}\right]$

The dimensions of Distance squared is $\left[\mathrm{r^2}\right]=\left[\mathrm{L^2}\right]$

The dimensions of mass squared is $[\mathrm{m_1\,m_2}]=\left[\mathrm{M^2}\right]$

So, the dimensions of Gravitational constant is

$[\mathrm{G}]=\frac{\left[\mathrm{MLT^{-2}}\right]\left[\mathrm{L^2}\right]}{\left[\mathrm{M^2}\right]}=\left[\mathrm{M^{-1}\,L^3\,T^{-2}}\right]$

Thus (D) Gravitational constant → (I) $\left[\mathrm{M^{-1}\,L^3\,T^{-2}}\right]$

$\mathrm{A\rightarrow III,\ B\rightarrow IV, C\rightarrow II, D\rightarrow I}$

Therefore, the correct option is (C).

Question27

The percentage error in the calculated volume of a sphere, if there is 2% error in its diameter measurement, is _____ .

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Options:

A.

1

B.

2

C.

6

D.

8

Answer: C

Solution:

The volume V of a sphere is given by the formula $V=\frac{4}{3}\pi r^3$, where r is the radius of the sphere.

Since the error is given in terms of diameter (D), we express the radius as $r=\frac{D}{2}$.

Substituting this into the volume formula:

$\mathrm{V}=\frac{4}{3}\pi\left(\frac{\mathrm{D}}{2}\right)^3$

$\Rightarrow \mathrm{V}=\frac{4}{3}\pi\left(\frac{\mathrm{D}^3}{8}\right)$

$\Rightarrow V=\frac{1}{6}\pi D^3$

To find the relative error, we take the natural logarithm $(\ln)$ on both sides of the equation :

$\ln(V)=\ln\left(\frac{\pi}{6}\right)+\ln\left(D^3\right)$

$\Rightarrow \ln(V)=\ln\left(\frac{\pi}{6}\right)+3\ln(D)$

Now, on differentiating both sides.

$\frac{\mathrm{dV}}{\mathrm{V}}=0+3\left(\frac{\mathrm{dD}}{\mathrm{D}}\right)$

$\Rightarrow \frac{\mathrm{dV}}{\mathrm{V}}=3\frac{\mathrm{dD}}{\mathrm{D}}$

To express this as a percentage error, we multiply both sides by 100 :

$\left(\frac{\mathrm{dV}}{\mathrm{V}}\times 100\right)=3\times\left(\frac{\mathrm{dD}}{\mathrm{D}}\times 100\right)$

It is given that the percentage error in the diameter is 2%.

$\frac{dD}{D}\times 100=2\%$

So, percentage error in Volume $=3\times 2\%$

Percentage error in Volume $=6\%$

Therefore, the correct option is (C).

Question28

Consider the equation $H=\frac{x^p\epsilon^q E^r}{t^s}$

Where $H=$ magnetic field; $E=$ electric field, $\epsilon=$ permittivity, $x=$ distance, $t=$ time The values of p, q, r and s respectively are :

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Options:

A.

$1, 1, 1, 1$

B.

$-1, 1, 2, 1$

C.

$1, -1, -2, 1$

D.

$-1, -2, -2, 1$

Answer: A

Solution:

To find the values of the exponents p, q, r, and s, we will use dimensional analysis. Both sides of the equation must have the same fundamental dimensions.

The dimension of distance (x) is the dimension of length

$[\mathrm{x}] = \mathrm{L}$

The dimension is time $[\mathrm{t}] = \mathrm{T}$

The dimension of electric field (E) can be derived from the formula for force on a charge, $\mathrm{F} = \mathrm{qE}$. Therefore, $\mathrm{E} = \mathrm{F}/\mathrm{q}$.

The dimensions of force is $[\mathrm{F}] = \left[\mathrm{MLT}^{-2}\right]$

Charge $(\mathrm{q}) = \text{Current} \times \text{Time}$

So, the dimensions of charge is, $[\mathrm{q}] = [\mathrm{AT}]$

So, the dimensions of electric field is, $[\mathrm{E}] = \frac{\left[\mathrm{MLT}^{-2}\right]}{[\mathrm{AT}]} = \left[\mathrm{MLT}^{-3}\ \mathrm{A}^{-1}\right]$

Using relationship between electric field (E) and and magnetic field (H) in electromagnetic waves. In a vacuum, the ratio of the magnitudes of the electric field to the magnetic field (B) is the speed of light (c),

$\mathrm{E}/\mathrm{B} = \mathrm{c}$

Using the formula of magnetic force on a current-carrying conductor:

$F = BIL$

Where, $F =$ force $B =$ magnetic field (flux density), $I =$ current and $L =$ length

$B = \frac{F}{IL}$

Substituting the dimensions,

$[B] = \frac{[MLT^{-2}]}{[LA]}$

$\Rightarrow [B] = \left[M\,T^{-2}I^{-1}\right]$

The relation between B and H is $B = \mu H \Rightarrow \frac{B}{\mu} = H$. Where $\mu =$ permeability.

Using the formula of magnetic field, $B = \frac{\mu i}{2\pi r} \Rightarrow \left[\frac{B}{\mu}\right] = \left[\frac{i}{r}\right]$, here r is distance from straight conductor carrying current i.

$\left[\frac{B}{\mu}\right] = [H] = \frac{[A]}{[L]} = \left[L^{-1}\,A\right]$

Putting the dimensions of electric field and speed of light, the dimensions of magnetic field is,

$[H] = \frac{\left[MLT^{-3}\,A^{-1}\right]}{[LT^{-1}]} = \left[MT^{-2}\,A^{-1}\right]$

We can derive the dimensions of permittivity (ϵ) from Coulomb's Law, $F = \frac{1}{4\pi\epsilon}\frac{q_1q_2}{r^2}$.

$[\epsilon] = \left[\frac{q^2}{Fr^2}\right]$

$[\epsilon] = \frac{[AT]^2}{[MLT^{-2}][L^2]} = \left[M^{-1}\,L^{-3}\,T^4\,A^2\right]$

Substituting the dimensions into the given equation,

$[H] = \frac{[x]^p[\epsilon]^q[E]^r}{[t]^s}$

$\Rightarrow \left[L^{-1}A\right] = \frac{[L]^p\left[M^{-1}L^{-3}T^4A^2\right]^q\left[MLT^{-3}A^{-1}\right]^r}{[T]^s}$

$\left[L^{-1}A\right] = \left[M^{-q+r}L^{p-3q+r}T^{4q-3r-s}A^{2q-r}\right]$

On comparing the powers of M, L, T, and A on both sides,

For M, $-q + r = 0 \ldots$ (i)

For A, $2q - r = 1$. Since $r = q$, then $2q - q = 1$, so $q = 1$ and $r = 1$.

For L, $p - 3q + r = -1 \ldots$ (ii).

$$p - 3(1) + 1 = -1$$
$$p - 2 = -1 \Rightarrow p = 1$$

For T, $4q - 3r - s = 0 \ldots$ (iii)

$$4(1) - 3(1) - s = 0$$
$$1 - s = 0 \Rightarrow s = 1$$

Therefore, the values are $\mathrm{p}=1, \mathrm{q}=1, \mathrm{r}=1,\ \mathrm{s}=1$.

Hence, the correct option is (A).

Question29

A new unit (α) of length is chosen such that it is equal to the speed of light in vacuum. What is the distance between Venus and Earth in terms of α units if light takes 6 min. 40 s to cover this distance?

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Options:

A.

200α

B.

400α

C.

300α

D.

500α

Answer: B

Solution:

To find the distance between two points when an object moves at a constant speed, we use formula,

$\text{Distance} = \text{Speed} \times \text{Time}$

α is a new unit of length such that 1 unit (α) = speed of light in vacuum (c).

The speed of light c is the distance light travels per unit of time (usually meters per second).

$1\alpha = c \times (1 \text{ second })$

The time taken for light to travel from Venus to Earth is 6 minutes and 40 seconds.

$1 \text{ minute} = 60 \text{ seconds}$

6 minutes $= 6 \times 60 = 360$ seconds

Total time $\mathrm{t} = 360\ \mathrm{s} + 40\ \mathrm{s} = 400$ seconds

Using distance formula:

Distance = Speed of light × Time

Distance $= \mathrm{c} \times 400$ seconds

Since 1α is the distance light travels in 1 second ($\mathrm{c} \times 1\ \mathrm{s}$), we can substitute c with our new unit,

Distance $= 400 \times (\mathrm{c} \times 1\ \mathrm{s})$

Distance $= 400\alpha$

Therefore, the distance between Venus and Earth is 400α. Hence, the correct option is (B).

Question30

The least count of a screw guage is 0.01 mm . If the pitch is increased by 75% and number of divisions on the circular scale is reduced by 50%, the new least count will be ________ $\times 10^{-3}$ mm

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Answer: 35

Solution:

The least count (L.C.) of a screw gauge is given by:

$$\text{L.C.} = \frac{\text{Pitch}}{\text{Number of Divisions on Circular Scale}}$$

Let the original pitch be p and the original number of divisions be n. Then, the original least count is:

$$\frac{p}{n} = 0.01\,\mathrm{mm}$$

After modifications:

The pitch is increased by 75%, so the new pitch is:

$$p_{\text{new}} = p + 0.75p = 1.75p$$

The number of divisions is reduced by 50%, so the new number of divisions is:

$n_{\text{new}} \quad = 0.5n$

The new least count is then:

$\text{L.C.}_{\text{new}} \quad = \frac{p_{\text{new}}}{n_{\text{new}}} = \frac{1.75p}{0.5n} = \frac{1.75}{0.5} \times \frac{p}{n} = 3.5 \times \frac{p}{n}$

Substitute the original least count:

$\text{L.C.}_{\text{new}} \quad = 3.5 \times 0.01\,\text{mm} = 0.035\,\text{mm}$

Expressed in the form $\times 10^{-3}$ mm, this becomes:

$0.035\,\text{mm} = 35 \times 10^{-3}\,\text{mm}$

Thus, the new least count is:

$35 \times 10^{-3}\,\text{mm}$

Question31

A tiny metallic rectangular sheet has length and breadth of 5 mm and 2.5 mm , respectively. Using a specially designed screw gauge which has pitch of 0.75 mm and 15 divisions in the circular scale, you are asked to find the area of the sheet. In this measurement, the maximum fractional error will be $\frac{x}{100}$ where x is ________ .

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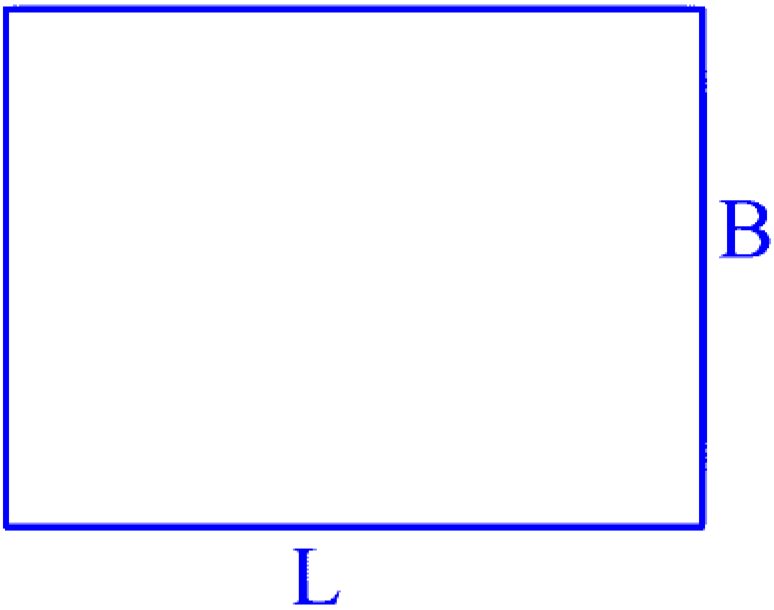
Answer: 3

Solution:



Given, L = 5 mm, B = 2.5 mm

We know, least count of a screw gauge,

$L.C. = \frac{Pitch\ length}{No.\ of\ division\ on\ circular\ scale}$

$\Rightarrow L.C. = \frac{0.75}{15} = 0.05\,mm$

We know, $A = LB$

By taking ln on both sides,

$\Rightarrow \ln A = \ln L + \ln B$

by differentiating both sides,

$\Rightarrow \frac{dA}{A} = \frac{dL}{L} + \frac{dB}{B}$

Here, $dL = dB = 0.05\,mm$ (L.C.)

So fractional error,

$\frac{dA}{A} = \frac{0.05}{5} + \frac{0.05}{2.5}$

$= \frac{1}{100} + \frac{2}{100} = \frac{3}{100} = \frac{x}{100}$ (given)

Hence, $x = 3$.

Question32

A physical quantity Q is related to four observables a,b,c,d as follows :

$$Q = \frac{ab^4}{cd}$$

where,
$\mathrm{a}=(60 \pm 3) \mathrm{Pa} ; \mathrm{b}=(20 \pm 0.1) \mathrm{m} ; \mathrm{c}=(40 \pm 0.2) \mathrm{Nsm}^{-2}$ **and** $\mathrm{d}=(50 \pm 0.1) \mathrm{m}$**, then the percentage error in Q is** $\frac{x}{1000}$**, where** $x=$ ________ **.**

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Answer: 7700

Solution:

Given, $Q=\frac{a b^4}{c d}$

$a=(60 \pm 3)\, Pa \Rightarrow a=60\, Pa,\, \Delta a=3 Pa$

$b=(20 \pm 0.1)\, m \Rightarrow b=20 m,\, \Delta b=0.1\, m$

$c=(40 \pm 0.2)\, N s m^{-2} \Rightarrow c=40\, N s m^{-2},\, \Delta c=0.2\, N s m^{-2}$

$d=(50 \pm 0.1)\, m \Rightarrow\, d=50\, m,\, \Delta d=0.1 m$

As, $Q=\frac{a b^4}{c d}$

by taking ln on both sides,

$\ln Q=\ln a+u \ln b-\ln c-\ln d$

Now, by differentiating,

$\frac{d Q}{Q}=\frac{d a}{a}+4 \frac{d b}{b}-\frac{d c}{c}-\frac{d d}{d}$

So, maximum fractional error in Q is given by,

$\frac{\Delta Q}{Q}=\frac{\Delta a}{a}+4 \frac{\Delta b}{b}+\frac{\Delta c}{c}+\frac{\Delta d}{d}$

$\Rightarrow \frac{\Delta Q}{Q}=\frac{3}{60}+4\left(\frac{0.1}{20}\right)+\frac{0.2}{40}+\frac{0.1}{40}$

$=\frac{1}{20}+\frac{1}{50}+\frac{1}{200}+\frac{1}{500}$

$\Rightarrow \frac{\Delta Q}{Q}=\frac{50+20+5+2}{1000}=\frac{77}{1000}$

Hence the % error in $Q=\frac{\Delta Q}{Q} \times 100 \%$

$=\frac{7700}{1000} \%=\frac{x}{1000}$ (given)

So, $x=7700$

Question33

A physical quantity C is related to four other quantities $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s as follows

$$C = \frac{pq^2}{r^3\sqrt{s}}$$

The percentage errors in the measurement of $\mathrm{p}, \mathrm{q}, \mathrm{r}$ and s are $1\%, 2\%, 3\%$ and 2%, respectively. The percentage error in the measurement of C will be__________%

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Answer: 15

Solution:

To determine the percentage error in the measurement of C, which is related to p, q, r, and s as:

$$C = \frac{pq^2}{r^3\sqrt{s}}$$

we first express it in terms of powers:

$$C = p^1 q^2 r^{-3} s^{-1/2}$$

The percentage error in C can be calculated using the formula for the propagation of error, which is:

$$\left(\frac{\Delta C}{C}\right)_{\max} = \frac{\Delta p}{p} + 2\left|\frac{\Delta q}{q}\right| + 3\left|\frac{\Delta r}{r}\right| + \frac{1}{2}\left|\frac{\Delta s}{s}\right|$$

Given the percentage errors for p, q, r, and s are 1%, 2%, 3%, and 2% respectively, we substitute these values into the formula:

$$\left(\frac{\Delta C}{C}\right)_{\max} = 1\% + 2 \times 2\% + 3 \times 3\% + \frac{1}{2} \times 2\%$$

Calculating each part:

Contribution from p: 1%

Contribution from q: 4% (since $2 \times 2\% = 4\%$)

Contribution from r: 9% (since $3 \times 3\% = 9\%$)

Contribution from s: 1% (since $\frac{1}{2} \times 2\% = 1\%$)

Adding these contributions together:

$1\% + 4\% + 9\% + 1\% = 15\%$

Thus, the maximum percentage error in the measurement of C is 15%.

Question34

Given below are two statements :

Statement I: In a vernier callipers, one vernier scale division is always smaller than one main scale division.

Statement II : The vernier constant is given by one main scale division multiplied by the number of vernier scale divisions.

In the light of the above statements, choose the correct answer from the options given below.

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Options:

A. Both Statement I and Statement II are false

B. Statement I is true but Statement II is false

C. Both Statement I and Statement II are true

D. Statement I is false but Statement II is true

Answer: B

Solution:

In general, one vernier scale division is smaller than one main scale division but in some modified cases it may be not correct. Also least count is given by one main scale division divided by number of vernier scale division for normal vernier calliper.

Hence, option 2 is correct.

Question35

If B is magnetic field and μ_0 is permeability of free space, then the dimensions of (B/μ_0) is

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Options:

A. $\mathrm{MT}^{-2}\ \mathrm{A}^{-1}$

B. $\mathrm{L}^{-1}\ \mathrm{A}$

C. $\mathrm{ML}^{2}\ \mathrm{T}^{-2}\ \mathrm{A}^{-1}$

D. $\mathrm{LT}^{-2}\ \mathrm{A}^{-1}$

Answer: B

Solution:

To determine the dimensions of $\left(\frac{B}{\mu_0}\right)$, we start with the formula for the magnetic field B inside a solenoid:

$B = \mu_0 n I$

where $n = \frac{N}{L}$ is the number of turns per unit length, I is the current, and μ_0 is the permeability of free space.

Rearranging the formula gives:

$\frac{B}{\mu_0} = nI = \frac{N}{L} I$

From the above, the dimensions of $\left(\frac{B}{\mu_0}\right)$ are:

$\left[\frac{B}{\mu_0}\right] = \left[L^{-1} A\right]$

Thus, the dimensional formula of $\left(\frac{B}{\mu_0}\right)$ corresponds to $L^{-1}\mathrm{A}$.

Question36

The maximum percentage error in the measurment of density of a wire is

[Given, mass of wire $= (0.60 \pm 0.003)\mathrm{g}$

radius of wire $= (0.50 \pm 0.01)\mathrm{cm}$

length of wire $= (10.00 \pm 0.05)\mathrm{cm}]$

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Options:

A. 7

B. 8

C. 5

D. 4

Answer: C

Solution:

To determine the maximum percentage error in the density of the wire, follow these steps:

The density of the wire is given by the formula for a cylinder:

$\rho = \frac{m}{\pi r^2 l}$

When calculating the maximum percentage (relative) error, you add the relative errors from each variable, taking into account the power to which each variable is raised. Thus, for density:

$\frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + 2\frac{\Delta r}{r} + \frac{\Delta l}{l}$

Here:

$\frac{\Delta m}{m}$ is the relative error in the mass.

$2\frac{\Delta r}{r}$ is due to the radius being squared.

$\frac{\Delta l}{l}$ is the relative error in the length.

Now plug in the given values:

Mass: $m = 0.60\,\text{g}$ with an error $\Delta m = 0.003\,\text{g}$

$\frac{\Delta m}{m} = \frac{0.003}{0.60} = 0.005$ or 0.5%.

Radius: $r = 0.50\,\text{cm}$ with an error $\Delta r = 0.01\,\text{cm}$

$\frac{\Delta r}{r} = \frac{0.01}{0.50} = 0.02$ or 2%. Since the radius is squared in the formula, multiply by 2:

$2\frac{\Delta r}{r} = 2 \times 0.02 = 0.04$ or 4%.

Length: $l = 10.00\,\text{cm}$ with an error $\Delta l = 0.05\,\text{cm}$

$\frac{\Delta l}{l} = \frac{0.05}{10.00} = 0.005$ or 0.5%.

Sum these contributions to find the overall maximum relative error:

$\frac{\Delta \rho}{\rho} = 0.005 + 0.04 + 0.005 = 0.05$

Converting this relative error into a percentage gives:

$0.05 \times 100\% = 5\%$

Thus, the maximum percentage error in the density of the wire is 5%, which corresponds to Option C.

Question37

The position of a particle moving on x-axis is given by $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is

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Options:

A. L

B. $\mathrm{L}^3\,\mathrm{T}^{-2}$

C. $\mathrm{L}^2\,\mathrm{T}^{-2}$

D. L^2

Answer: C

Solution:

Let's analyze the position function step by step.

The particle's position is given by:

$$x(t) = A\sin t + B\cos^2 t + Ct^2 + D,$$

where t represents time, which has the dimension T.

Since the overall dimensions of position $x(t)$ must be length (L), each term in the expression must also have dimensions of length.

Term $A\sin t$:

The sine function, $\sin t$, is dimensionless (its argument must be dimensionless, and by convention, if t is in an appropriate unit where the argument is dimensionless, the function itself is dimensionless).

Therefore, A must carry the dimension of length:

$$[A] = L.$$

Term $B\cos^2 t$:

Similarly, the cosine function is dimensionless, and so is its square.

Thus, B must have the dimension of length:

$$[B] = L.$$

Term Ct^2:

Here, t^2 has the dimension T^2.

To have the overall term with the dimension L, we require:

$$[C] \times [T^2] = L.$$

So, the dimension of C must be:

$$[C] = LT^{-2}.$$

Term D:

This is a constant term added to position, so it must also have the dimension of length:

$$[D] = L.$$

Now, we are asked to find the dimension of:

$$\frac{ABC}{D}.$$

The dimensions of A, B, and C are:

$$[A] = L, \quad [B] = L, \quad [C] = LT^{-2}.$$

Multiplying them together:

$$[ABC] = L \times L \times (LT^{-2}) = L^3T^{-2}.$$

Since $[D] = L$, dividing by D gives:

$\frac{[ABC]}{[D]} = \frac{L^3 T^{-2}}{L} = L^2 T^{-2}.$

Thus, the dimension of $\frac{ABC}{D}$ is:

$\boxed{L^2T^{-2}}$.

The correct answer is Option C.

Question38

Match List - I with List - II

	List - I		List - II
(A)	Permeability of free space	(I)	$\left[ML^2 T^{-2}\right]$
(B)	Magnetic field	(II)	$\left[MT^{-2} A^{-1}\right]$
(C)	Magnetic moment	(III)	$\left[MLT^{-2} A^{-2}\right]$
(D)	Torsional constant	(IV)	$\left[L^2 A\right]$

Choose the correct answer from the options given below :

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Options:

A. (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

B. $(A)-(I),(B)-(IV),(C)-(II),(D)-(III)$

C. $(A)-(II),(B)-(I),(C)-(III),(D)-(IV)$

D. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Answer: A

Solution:

The correct matching is:

(A) Permeability of free space has dimensions

$[MLT^{-2}A^{-2}]$

which corresponds to entry **(III)**.

(B) Magnetic field has dimensions

$[MT^{-2}A^{-1}]$

which corresponds to entry **(II)**.

(C) Magnetic moment in SI units is expressed in Ampere-square meters, i.e.,

$[L^2A]$

which corresponds to entry **(IV)**.

(D) Torsional constant (torque per unit angular displacement) has dimensions

$[ML^2T^{-2}]$

which corresponds to entry **(I)**.

Thus, the correct answer is:

Option A: (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

Question39

The energy of a system is given as $\mathrm{E(t)} = \alpha^3 \mathrm{e}^{-\beta t}$, where t is the time and $\beta = 0.3\ \mathrm{s}^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At $t = 5$ s, maximum percentage error in the energy is :

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Options:

A. 6%

B. 11.6%

C. 4%

D. 8.4%

Answer: A

Solution:

We start with the energy function:

$E(t) = \alpha^3 e^{-\beta t}$,

where $\beta = 0.3\,\mathrm{s}^{-1}$.

Step 1. Take the Logarithm

Taking the logarithm of both sides:

$\ln E = 3\ln\alpha - \beta t$.

Step 2. Differentiate to Find the Relative Error

Differentiate both sides:

$\frac{dE}{E} = 3\,\frac{d\alpha}{\alpha} - \beta\,dt$.

For maximum error estimation, we consider the absolute values and sum the contributions:

$\left|\frac{\Delta E}{E}\right| \approx 3\,\left|\frac{\Delta\alpha}{\alpha}\right| + \beta\,|\Delta t|$.

Step 3. Plug in the Given Errors

The percentage error in α is 1.2% (i.e., $\Delta\alpha/\alpha = 0.012$).

The percentage error in t is 1.6%, so for $t = 5\,\mathrm{s}$:

$\Delta t = 0.016 \times 5 = 0.08\,\mathrm{s}$.

Now substitute these values:

$\left|\frac{\Delta E}{E}\right| \approx 3(0.012) + 0.3(0.08)$.

Calculating each term:

$3(0.012) = 0.036$ (or 3.6%),

$0.3(0.08) = 0.024$ (or 2.4%).

Step 4. Compute the Total Maximum Percentage Error

Add the contributions:

$\left|\frac{\Delta E}{E}\right| \approx 0.036 + 0.024 = 0.06$,

which is 6%.

Thus, the maximum percentage error in the energy at $t = 5\,\mathrm{s}$ is:

$\boxed{6\%}$

Answer: Option A (6%).

Question40

For an experimental expression $y = \frac{32.3 \times 1125}{27.4}$, where all the digits are significant. Then to report the value of y we should write

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Options:

A. $y = 1326.186$

B. $y = 1326.2$

C. $y = 1326.19$

D. $y = 1330$

Answer: D

Solution:

$32.3 \quad (3 \text{ sig figs}), \quad 1125 \quad (4 \text{ sig figs}), \quad 27.4 \quad (3 \text{ sig figs})$

In multiplication and division, the final answer must be reported with the same number of significant figures as the factor with the fewest significant figures, which here is 3.

First, calculate the unrounded result:

$y = \frac{32.3 \times 1125}{27.4} \approx \frac{36337.5}{27.4} \approx 1326.186$

Expressing 1326.186 in scientific notation:

$1326.186 \approx 1.326186 \times 10^3$

Rounding to 3 significant figures:

The first three significant digits are 1, 3, and 2.

Considering the fourth digit (6) causes the last significant digit to round up.

Thus,

$1.326186 \times 10^3 \approx 1.33 \times 10^3$

In standard decimal form, this is written as:

1330

Hence, the final reported value is:

$y = 1330$

Question41

Match List - I with List - II.

List - I	List - II
(A) Angular Impulse	(I) $M^0 L^2 T^{-2}$
(B) Latent Heat	(II) $M L^2 T^{-3} A^{-1}$
(C) Electrical resistivity	(III) $M L^2 T^{-1}$
(D) Electromotive force	(IV) $M L^3 T^{-3} A^{-2}$

Choose the correct answer from the options given below:

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Options:

A.

(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

B.

(A)-(I), (B)-(III), (C)-(IV), (D)-(II)

C.

(A)-(III), (B)-(I), (C)-(IV), (D)-(II)

D.

(A)-(III), (B)-(I), (C)-(II), (D)-(IV)

Answer: C

Solution:

Matching the quantities from List - I with their correct dimensions from List - II is crucial to understanding their physical significance. Here is the correct matching:

(A) Angular Impulse: The dimensions of angular impulse are equivalent to those of angular momentum, represented as ML^2T^{-1}. Hence, (A) corresponds to (III).

(B) Latent Heat: Latent heat has dimensions of energy per unit mass, which simplifies to $M^0L^2T^{-2}$. Hence, (B) corresponds to (I).

(C) Electrical resistivity: Electrical resistivity has the dimensions of resistance multiplied by length, which can be represented as $ML^3T^{-3}A^{-2}$. Hence, (C) corresponds to (IV).

(D) Electromotive force: Electromotive force (emf) is similar to electric potential difference and has dimensions $ML^2T^{-3}A^{-1}$. Hence, (D) corresponds to (II).

Therefore, the correct answer is **Option C**:

(A)-(III)

(B)-(I)

(C)-(IV)

(D)-(II)

Question42

The expression given below shows the variation of velocity (*v*) with time (*t*),

$v = \mathrm{At}^2 + \frac{\mathrm{Bt}}{\mathrm{C+t}}$.

The dimension of ABC is :

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Options:

A.

$[M^0L^1T^{-2}]$

B.

$[M^0L^2T^{-3}]$

C.

$[M^0L^2T^{-2}]$

D.

$[M^0L^1T^{-3}]$

Answer: B

Solution:

$$\left[LT^{-1}\right] = [A]\left[T^2\right] = \frac{[B][T]}{[C]+[T]}$$
$$[C] = [T]$$
$$[A] = \left[LT^{-3}\right]$$
$$[B] = \left[LT^{-1}\right]$$
$$[ABC] = \left[L^2\,T^{-3}\right]$$

Question43

The pair of physical quantities not having the same dimensions is :

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Options:

A.

Angular momentum and Planck's constant

B.

Torque and energy

C.

Surface tension and impulse

D.

Pressure and Young's modulus

Answer: C

Solution:

$[\tau] = [\mathrm{E}]$
$[\sigma] \neq [\mathrm{I}]$
$[\mathrm{L}] = [\mathrm{h}]$
$[\mathrm{P}] = [\mathrm{Y}]$

Question44

Match List - I with List - II.

List - I	List - II
(A) Young's Modulus	(I) $M L^{-1} T^{-1}$
(B) Torque	(II) $M L^{-1} T^{-2}$
(C) Coefficient of Viscosity	(III) $M^{-1} L^{3} T^{-2}$
(D) Gravitational Constant	(IV) $M L^{2} T^{-2}$

Choose the correct answer from the options given below :

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Options:

A. (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

B. (A)-(IV), (B)-(II), (C)-(III), (D)-(I)

C. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

Answer: D

Solution:

We know, Young's modulus, $E = \frac{\sigma}{\varepsilon}$

$\Rightarrow E = \frac{\frac{F}{A}}{\frac{\Delta L}{L}} = \frac{F}{A}\left(\frac{L}{\Delta L}\right)$

$\Rightarrow [E] = \frac{[F]}{[A]} = \frac{[M^1 L^1 T^{-2}]}{[L^2]}$

(A) $\Rightarrow [E] = [M^1 L^{-1} T^{-2}]$

(B) we know, Torque = $r \times F$

$[\tau] = [r][F]$

$= [L][M^1 L^1 T^{-2}]$

$[\tau] = [M^1 L^2 T^{-2}]$

(C) We know, $F = \mu A \frac{u}{y}$

where, F = force, μ = coefficient of viscosity

A = area $\frac{u}{y}$ = rate of shear deformation

$\Rightarrow \mu = \frac{Fy}{Au}$

$\Rightarrow [\mu] = \frac{[M^1 L^1 T^{-2}][L]}{[L^2][LT^{-1}]}$

$\Rightarrow [\mu] = [M^1 L^{-1} T^{-1}]$

(D) We know, $F = \frac{Gm_1 m_2}{r^2}$ (Newton's law of gravitation)

$\Rightarrow G = \frac{Fr^2}{m_1 m_2} <\Rightarrow G = \frac{Fr^2}{m_1 m_2}$

$\Rightarrow [G] = \frac{[M^1 L^1 T^{-2}][L^2]}{[M^2]}$

$\Rightarrow [G] = [M^{-1} L^3 T^{-2}]$

Hence, option D is correct.

Question45

Match List I with List II.

	List - I		List - II
(A)	Coefficient of viscosity	(I)	$[ML^0 T^{-3}]$
(B)	Intensity of wave	(II)	$[ML^{-2} T^{-2}]$
(C)	Pressure gradient	(III)	$[M^{-1} LT^2]$
(D)	Compressibility	(IV)	$[ML^{-1} T^{-1}]$

Choose the correct answer from the options given below:

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Options:

A. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

B. (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

C. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

D. (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Answer: C

Solution:

Here are the dimensional formulas:

Coefficient of viscosity, μ

$$[\mu]=\frac{\text{pressure}\quad\times\text{time}}{}=\frac{[M\,L^{-1}T^{-2}]\,[T]}{}=[M\,L^{-1}T^{-1}]\quad\longrightarrow\text{(IV)}$$

Intensity of a wave, I

$$[I]=\frac{\text{power}}{\text{area}}=\frac{[M\,L^{2}T^{-3}]}{[L^{2}]}=[M\,L^{0}T^{-3}]\quad\longrightarrow\text{(I)}$$

Pressure gradient, dP/dx

$$\left[\frac{dP}{dx}\right]=\frac{[M\,L^{-1}T^{-2}]}{[L]}=[M\,L^{-2}T^{-2}]\quad\longrightarrow\text{(II)}$$

Compressibility, κ

$$[\kappa]=\frac{1}{[\text{pressure}\quad]}=[M^{-1}L\,T^{2}]\quad\longrightarrow\text{(III)}$$

Matching gives

(A)–(IV), (B)–(I), (C)–(II), (D)–(III), i.e. **Option C**.

Question46

Given a charge q , current I and permeability of vacuum μ_{o^*}. Which of the following quantity has the dimension of momentum ?

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Options:

A. qI/μ_o

B. $q^2\mu_o I$

C. $q\mu_0/I$

D. $\mathrm{q}\mu_{\mathrm{o}}\mathrm{I}$

Answer: D

Solution:

$$\begin{aligned}
&\mathrm{Q}=\mathrm{AT}\\
&\mathrm{I}=\mathrm{A}\\
&\mu_0=\mathrm{MLT}^{-2}\ \mathrm{A}^{-2}\\
&\mathrm{P}=\mathrm{Q}^{\mathrm{x}}\mu_0^{\mathrm{y}}\mathrm{I}^{\mathrm{z}}=\left[\mathrm{AT}^{\mathrm{x}}\left[\mathrm{MLT}^{-2}\ \mathrm{A}^{-2}\right]^{\mathrm{y}}[\ \mathrm{A}]^{\mathrm{z}}\right.\\
&\mathrm{MLT}^{-1}=\mathrm{M}^{\mathrm{y}}\ \mathrm{L}^{\mathrm{y}}\ \mathrm{T}^{\mathrm{x}-2\mathrm{y}}\ \mathrm{A}^{-2\mathrm{y}+\mathrm{z}+\mathrm{x}}\\
&\ \text{Now; } \mathrm{y}=1\\
&\mathrm{x}-2\mathrm{y}=-1\\
&-2\mathrm{y}+\mathrm{z}=0\\
&\therefore \mathrm{x}=\mathrm{y}=\mathrm{z}=1
\end{aligned}$$

Question47

If μ_0 and ϵ_0 are the permeability and permittivity of free space, respectively, then the dimension of $\left(\frac{1}{\mu_0\epsilon_0}\right)$ is :

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Options:

A. T^2/L

B. $\mathrm{L}^2/\mathrm{T}^2$

C. $\mathrm{T}^2/\mathrm{L}^2$

D. L/T^2

Answer: B

Solution:

The expression $\frac{1}{\mu_0 \epsilon_0}$ is related to the speed of light c, given by the equation:

$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$

Therefore, we have:

$\frac{1}{\mu_0 \epsilon_0} = c^2$

The speed of light c has the dimensions of $\mathrm{L}\,\mathrm{T}^{-1}$, where L is the dimension of length and T is the dimension of time. Thus, when squared, the dimensions become:

$c^2 = (\mathrm{L}\,\mathrm{T}^{-1})^2 = \mathrm{L}^2\mathrm{T}^{-2}$

Hence, the dimensions of $\frac{1}{\mu_0 \epsilon_0}$ are $\mathrm{L}^2\mathrm{T}^{-2}$.

Question48

A person measures mass of 3 different particles as 435.42 g, 226.3 g and 0.125 g . According to the rules for arithmetic operations with significant figures, the addition of the masses of 3 particles will be.

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Options:

A. 661.845 g

B. 661.84 g

C. 662 g

D. 661.8 g

Answer: D

Solution:

To calculate the total mass of the three particles, we perform the following addition:

$m_1 + m_2 + m_3 = 435.42\,\mathrm{g} + 226.3\,\mathrm{g} + 0.125\,\mathrm{g}$

When adding numbers, the result should be reported with the same number of decimal places as the measurement with the fewest decimal places. Here, the second measurement (226.3 g) has only one decimal place, which is the fewest among the given values.

Thus, according to the rules for significant figures in addition, the sum should be rounded to one decimal place:

Total mass $= 661.8\,\mathrm{g}$

Question49

Match the LIST-I with LIST-II Match the LIST-I with LIST-II

LIST-I		LIST-II	
A.	Boltzmann constant Boltzmann constant	I	ML2 T−1
B	Coefficient of viscosity Coefficient of viscosity	II	MLT−3 K−1
C	Planck's constant Planck's constant	III	ML2 T−2 K−1
D	Thermal conductivity Thermal conductivity	IV	ML−1 T−1

Choose the correct answer from the options given below:

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Options:

A. A - III, B - IV, C - I, D - II

B. A - III, B - IV, C - II, D - I

C. A - III, B - II, C - I, D - IV

D. A - II, B - III, C - IV, D - I

Answer: A

Solution:

Explanation of Dimensional Analysis

In order to match the quantities from LIST-I with their respective dimensions in LIST-II, we need to analyze their dimensional formulas:

(A) Boltzmann Constant [k]:

The equation relating pressure (P), volume (V), number of moles (N), and temperature (T) gives us:

$$[\mathrm{k}]=\frac{\mathrm{PV}}{\mathrm{NT}}=\frac{\mathrm{ML}^2\,\mathrm{T}^{-2}}{\mathrm{K}}=\mathrm{ML}^2\,\mathrm{T}^{-2}\,\mathrm{K}^{-1}$$

This corresponds to option III.

(B) Coefficient of Viscosity [η]:

The coefficient of viscosity can be defined using the relation:

$$[\eta]=\frac{\mathrm{F}}{6\pi \mathrm{rv}}=\frac{\mathrm{MLT}^{-2}}{\mathrm{L}^2\,\mathrm{T}^{-1}}=\mathrm{ML}^{-1}\,\mathrm{T}^{-1}$$

This corresponds to option IV.

(C) Planck's Constant [h]:

The relationship between energy (E) and frequency (f) gives the dimensional formula:

$$[\mathrm{h}]=\frac{\mathrm{E}}{\mathrm{f}}=\frac{\mathrm{ML}^2\,\mathrm{T}^{-2}}{\mathrm{T}^{-1}}=\mathrm{ML}^2\,\mathrm{T}^{-1}$$

This corresponds to option I.

(D) Thermal Conductivity [k]:

From the heat conduction equation, we get:

$$\mathrm{k}=\frac{\left(\mathrm{ML}^2\,\mathrm{T}^{-3}\right)\mathrm{L}}{\mathrm{L}^2\cdot\,\mathrm{K}}=\mathrm{MLT}^{-3}\,\mathrm{K}^{-1}$$

This corresponds to option II.

To conclude, the correct matches are:

A - III

B - IV

C - I

D - II

This analysis ensures each physical quantity is aligned with its correct dimensional representation.

Question50

In an electromagnetic system, the quantity representing the ratio of electric flux and magnetic flux has dimension of $M^P L^Q T^R A^S$, where value of ' Q ' and ' R ' are

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Options:

A. $(3, -5)$

B. $(-2, 1)$

C. $(-2, 2)$

D. $(1, -1)$

Answer: D

Solution:

To solve this problem, we need to find the dimensions of the electric flux and the magnetic flux, and then take their ratio.

Determine the dimension of electric flux (Φ_e):

Electric flux is defined as

$\Phi_E = \int_S \mathbf{E} \cdot d\mathbf{A}.$

The electric field $\mathbf{E}$ has dimensions:

$[E] = \frac{\text{Force}}{\text{Charge}}.$

Since force has dimensions MLT^{-2}, and charge (in SI) has dimensions $A\,T$, we get:

$[E] = \frac{MLT^{-2}}{A\,T} = MLT^{-3}A^{-1}.$

The area element dA has dimensions L^2.

Thus, the electric flux has dimensions:

$[\Phi_E] = [E][A] = MLT^{-3}A^{-1} \times L^2 = ML^3T^{-3}A^{-1}.$

Determine the dimension of magnetic flux (Φ_B):

Magnetic flux is defined as

$\Phi_B = \int_S \mathbf{B} \cdot d\mathbf{A}.$

The magnetic field $\mathbf{B}$ has dimensions determined via the Lorentz force law (using $F = q\,v\,B$), giving:

$[B] = \frac{F}{q\,v} = \frac{MLT^{-2}}{(A\,T)(LT^{-1})} = MT^{-2}A^{-1}.$

Again, the area element has dimensions L^2.

So the magnetic flux has dimensions:

$[\Phi_B] = [B][A] = (MT^{-2}A^{-1})(L^2) = ML^2T^{-2}A^{-1}.$

Find the ratio of electric flux to magnetic flux:

Taking the ratio:

$\frac{\Phi_E}{\Phi_B} = \frac{ML^3T^{-3}A^{-1}}{ML^2T^{-2}A^{-1}}$.

Canceling like terms:

Mass (M) cancels.

Current (A^{-1}) cancels.

For length, $L^{3-2} = L^1$.

For time, $T^{-3-(-2)} = T^{-1}$.

Therefore, the ratio has dimensions:

L^1T^{-1}.

This means in the expression

$M^PL^QT^RA^S$,

the exponents for length and time are $Q = 1$ and $R = -1$.

Looking at the options provided, this corresponds to Option D: $(1, -1)$.

Thus, the answer is Option D.

Question51

For the determination of refractive index of glass slab, a travelling microscope is used whose main scale contains 300 equal divisions equals to 15 cm . The vernier scale attached to the microscope has 25 divisions equals to 24 divisions of main scale. The least count (LC) of the travelling microscope is (in cm) :

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Options:

A. 0.002

B. 0.0025

C. 0.0005

D. 0.001

Answer: A

Solution:

Determine one main scale division (msd):

$300\,\text{msd} = 15\,\text{cm}$

Therefore,

$1\,\text{msd} = \frac{15}{300}\,\text{cm} = 0.05\,\text{cm}$

Determine one vernier scale division (vsd):

$25\,\text{vsd} = 24\,\text{msd}$

Therefore,

$1\,\text{vsd} = \frac{24}{25}\,\text{msd}$

Calculate the least count (LC):

The least count is given by the difference between one main scale division and one vernier scale division:

$\text{LC} = 1\,\text{msd} - 1\,\text{vsd}$

Substitute the expression for 1 vsd:

$\text{LC} = 1\,\text{msd} - \frac{24}{25} \times 1\,\text{msd}$

This simplifies to:

$\text{LC} = \frac{1}{25} \times 1\,\text{msd}$

Calculate the LC in cm:

Substituting the value of 1 msd:

$\text{LC} = \frac{1}{25} \times 0.05\,\text{cm} = 0.002\,\text{cm}$

Hence, the least count of the traveling microscope is 0.002 cm.

Question52

In an electromagnetic system, a quantity defined as the ratio of electric dipole moment and magnetic dipole moment has dimension of $\left[\mathrm{M}^{\mathrm{P}} \mathrm{L}^{\mathrm{Q}} \mathrm{T}^{R} A^{\mathrm{S}}\right]$. The value of P and Q are :

Options:

A. $-1, 0$

B. $0, -1$

C. $-1, 1$

D. $1, -1$

Answer: B

Solution:

Electric dipole moment $(\overrightarrow{\mathrm{P}}) = \mathrm{q} \times 2\ell$

Magnetic dipole moment $(\overrightarrow{\mathrm{M}}) = \mathrm{IA}$

$\left[\frac{\mathrm{P}}{\mathrm{M}}\right] = \left[\frac{\mathrm{LTA}}{\mathrm{L}^2\ \mathrm{A}}\right] = \mathrm{L}^{-1}\ \mathrm{T} = \mathrm{M}^0\ \mathrm{L}^{-1}\ \mathrm{T}^1\ \mathrm{A}^0$

After compering values of $\mathrm{P} \& \mathrm{Q}$ are $0, -1$

Correct Answer : Option 4

Question53

Match List - I with List - II.

List - I	List - II
(A) Mass density	(I) $[\mathrm{ML}^2\mathrm{T}^{-3}]$
(B) Impulse	(II) $[\mathrm{MLT}^{-1}]$
(C) Power	(III) $[\mathrm{ML}^2\mathrm{T}^0]$
(D) Moment of inertia	(IV) $[\mathrm{ML}^{-3}\mathrm{T}^0]$

Choose the correct answer from the options given below :

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Options:

A.

(A)-(IV), (B)-(II), (C)-(III), (D)-(I)

B.

(A)-(II), (B)-(III), (C)-(IV), (D)-(I)

C.

(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

D.

(A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Answer: D

Solution:

(A) Mass density:

Mass density is defined as mass per unit volume.

The dimension of mass is $[M]$.

The dimension of volume is $[L^3]$.

Therefore, the dimension of mass density is $\frac{[M]}{[L^3]} = [ML^{-3}T^0]$.

Matches with (IV).

(B) Impulse:

Impulse is defined as the change in momentum, or force multiplied by time.

Impulse = Force x Time

The dimension of force is $[MLT^{-2}]$.

The dimension of time is $[T]$.

Therefore, the dimension of impulse is $[MLT^{-2}] \cdot [T] = [MLT^{-1}]$.

Matches with (II).

(C) Power:

Power is defined as the rate of doing work or energy per unit time.

Power = Work / Time

The dimension of work (or energy) is $[ML^2T^{-2}]$.

The dimension of time is $[T]$.

Therefore, the dimension of power is $\frac{[ML^2T^{-2}]}{[T]} = [ML^2T^{-3}]$.

Matches with (I).

(D) Moment of inertia:

Moment of inertia is defined as $I = mr^2$, where m is mass and r is the distance from the axis of rotation.

The dimension of mass is $[M]$.

The dimension of distance squared is $[L^2]$.

Therefore, the dimension of moment of inertia is $[ML^2T^0]$.

Matches with (III).

So, the correct matches are:

(A) - (IV)

(B) - (II)

(C) - (I)

(D) - (III)

Therefore, the correct option is **D**.

Question54

The dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is equal to that of:

(μ_0 = Vacuum permeability and ϵ_0 = Vacuum permittivity)

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Options:

A.

Voltage

B.

Inductance

C.

Resistance

D.

Capacitance

Answer: C

Solution:

The dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ can be understood as follows:

Vacuum permeability (μ_0) and vacuum permittivity (ϵ_0) relate to the properties of inductance and capacitance, respectively, in a vacuum. We start by considering the formulas for inductance (L) and capacitance (C):

Inductance (L) is given by:

$$L = \frac{\mu_0 \cdot \text{Number of turns} \qquad \cdot \text{Area}}{\text{Length}}$$

Capacitance (C) is given by:

$$C = \frac{\text{Area} \quad \cdot \epsilon_0}{\text{Distance}}$$

From these, the ratio $\frac{L}{C}$ can be expressed as $\frac{\mu_0}{\epsilon_0}$:

$$\frac{L}{C} \propto \frac{\mu_0}{\epsilon_0}$$

Taking the square root of this ratio, we have:

$$\sqrt{\frac{\mu_0}{\epsilon_0}} \propto \sqrt{\frac{L}{C}}$$

This simplifies further to considering the relationship between time constant (τ) and resistance (R):

$$\frac{L}{C} = \frac{\tau R}{(\tau / R)} = R^2$$

Thus, by taking the square root:

$$\sqrt{\frac{\mu_0}{\epsilon_0}} = R$$

In conclusion, the dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is equivalent to the dimension of resistance, R.

Question55

A quantity Q is formulated as $X^{-2}Y^{+\frac{3}{2}}Z^{-\frac{2}{5}}$. X, Y, and Z are independent parameters which have fractional errors of 0.1, 0.2, and 0.5, respectively in measurement. The maximum fractional error of Q is

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Options:

A.

0.6

B.

0.8

C.

0.7

D.

0.1

Answer: C

Solution:

The quantity Q is expressed as $Q = X^{-2}Y^{+\frac{3}{2}}Z^{-\frac{2}{5}}$. The variables X, Y, and Z are independent parameters with fractional errors of 0.1, 0.2, and 0.5, respectively. To determine the maximum fractional error in Q, we can use the following method:

Fractional error in $Q = |-2|\frac{\Delta X}{X} + \left|\frac{3}{2}\right|\frac{\Delta Y}{Y} + \left|-\frac{2}{5}\right|\frac{\Delta Z}{Z}$

Substituting the fractional errors into the equation:

$= 2 \times 0.1 + \frac{3}{2} \times 0.2 + \frac{2}{5} \times 0.5$

Calculating each term gives:

$= 0.2 + 0.3 + 0.2$

Summing these values results in:

$= 0.7$

Thus, the maximum fractional error in Q is 0.7.

Question56

Identify the physical quantity that cannot be measured using spherometer :

[27-Jan-2024 Shift 1]

Options:

A.

Radius of curvature of concave surface

B.

Specific rotation of liquids

C.

Thickness of thin plates

D.

Radius of curvature of convex surface

Answer: B

Solution:

Spherometer can be used to measure curvature of surface.

Question57

Given below are two statements: one is labelled as Assertion(A) and the other is labelled as Reason (R).

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.
Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

In the light of the above statements, choose the correct answer from the options given below :

[27-Jan-2024 Shift 2]

Options:

A.

Both (A) and (R) are correct and (R) is the correct explanation of (A)

B.

Both (A) and (R) are correct but (R) is not the correct explanation of (A)

C.

(A) is true but (R) is false

D.

(A) is false but (R) is true

Answer: B

Solution:

Assertion & Reason both are correct Theory

Question58

If 50 Vernier divisions are equal to 49 main scale divisions of a travelling microscope and one smallest reading of main scale is 0.5 mm, the Vernier constant of travelling microscope is:

[30-Jan-2024 Shift 2]

Options:

A.

0.1 mm

B.

0.1 cm

C.

0.01 cm

D.

0.01 mm

Answer: D

Solution:

$50V + S = 49S + S$

$S = 50(S - V)$

$.5 = 50(S - V)$

$S - V = \frac{0.5}{50} = \frac{1}{100} = 0.01\text{ mm}$

Question59

The measured value of the length of a simple pendulum is 20cm with 2mm accuracy. The time for 50 oscillations was measured to be 40 seconds with 1 second resolution. From these measurements, the accuracy in the measurement of acceleration due to gravity is N%. The value of N is:

[31-Jan-2024 Shift 2]

Options:

A.

4

B.

8

C.

6

D.

5

Answer: C

Solution:

$$T = 2\pi\sqrt{\frac{\ell}{g}}$$

$$g = \frac{4\pi^2\ell}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta\ell}{\ell} + \frac{2\,\Delta T}{T}$$

$$= \frac{0.2}{20} + 2\left(\frac{1}{40}\right)$$

$$= \frac{0.3}{20}$$

$$\text{Percentage change} = \frac{0.3}{20} \times 100 = 6\%$$

Question60

The resistance R = V/1 where V = (200 ± 5)V and I = (20 ± 0.2)A, the percentage error in the measurement of R is :

[29-Jan-2024 Shift 1]

Options:

A.

3.5%

B.

7%

C.

3%

D.

5.5%

Answer: A

Solution:

$$R = \frac{V}{1}$$

According to error analysis

$$\frac{dR}{R} = \frac{dV}{V} + \frac{dI}{I}$$

$$\frac{dR}{R} = \frac{5}{200} + \frac{0.2}{20}$$

$$\frac{dR}{R} = \frac{7}{200}$$

$$\% \text{ error } \frac{dR}{R} \times 100 = \frac{7}{200} \times 100 = 3.5\%$$

Question61

A physical quantity Q is found to depend on quantities a, b, c by the relation $Q = a^4b^3/c^2$. The percentage error in a, b and c are 3%,4% and 5% respectively. Then, the percentage error in Q is :

[29-Jan-2024 Shift 2]

Options:

A.

66%

B.

43%

C.

34%

D.

14%

Answer: C

Solution:

Sol. $Q = \frac{a^4 b^3}{c^2}$

$\frac{\Delta Q}{Q} = 4\frac{\Delta a}{a} + 3\frac{\Delta b}{b} + 2\frac{\Delta c}{c}$

$\frac{\Delta Q}{Q} \times 100 = 4\left(\frac{\Delta a}{a} \times 100\right) + 3\left(\frac{\Delta b}{b} \times 100\right) + 2\left(\frac{\Delta c}{c} \times 100\right)$

% error in $Q = 4 \times 3\% + 3 \times 4\% + 2 \times 5\%$

$= 12\% + 12\% + 10\%$

$= 34\%$

Question62

A vector has magnitude same as that of $\vec{A} - = 3\hat{j} + 4\hat{j}$ and is parallel to $\vec{B} = 4\hat{i} + 3\hat{j}$. **The x and y components of this vector in first quadrant are x and 3 respectively where x =____**

[30-Jan-2024 Shift 2]

Answer: 4

Solution:

$$\overline{N} = |\overline{A}|\ \hat{B} = \frac{5\left(4\hat{i} + 3\hat{j}\right)}{5} = 4\hat{i} + 3\hat{j}$$

$\therefore x = 4$

Question63

If the percentage errors in measuring the length and the diameter of a wire are 0.1% each. The percentage error in measuring its resistance will be:

[31-Jan-2024 Shift 1]

Options:

A.

0.2%

B.

0.3%

C.

0.1%

D.

0.144%

Answer: B

Solution:

$$R = \frac{\rho L}{\pi \frac{d^2}{4}}$$

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + \frac{2\,\Delta\,d}{d}$$

$$\frac{\Delta L}{L} = 0.1\% \text{ and } \frac{\Delta d}{d} = 0.1\%$$

$$\frac{\Delta R}{R} = 0.3\%$$

Question64

If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at an angle θ, then

[31-Jan-2024 Shift 2]

Options:

A.

$$\left|\vec{A} - \vec{B}\right| = \sqrt{2}R\,sin\left(\frac{\theta}{2}\right)$$

B.

$$\left|\vec{A} + \vec{B}\right| = 2R\,sin\left(\frac{\theta}{2}\right)$$

C.

$$\left|\vec{A} + \vec{B}\right| = 2R\cos\left(\frac{\theta}{2}\right)$$

D.

$$\left|\vec{A} - \vec{B}\right| = 2R\cos\left(\frac{\theta}{2}\right)$$

Answer: C

Solution:

The magnitude of resultant vector

$R' = \sqrt{a^2 + b^2 + 2ab\cos\theta}$

Here $a = b = R$

Then $R' = \sqrt{R^2 + R^2 + 2R^2\cos\theta}$

$= R\sqrt{2}\sqrt{1+\cos\theta}$

$= \sqrt{2}R\sqrt{2\cos^2\frac{\theta}{2}}$

$= 2R\cos\frac{\theta}{2}$

Question65

Given below are two statements :

Statement (I) : Planck's constant and angular momentum have same dimensions.
Statement (II) : Linear momentum and moment of force have same dimensions.

In the light of the above statements, choose the correct answer from the options given below :

[27-Jan-2024 Shift 1]

Options:

A.

Statement I is true but Statement II is false

B.

Both Statement I and Statement II are false

C.

Both Statement I and Statement II are true

D.

Statement I is false but Statement II is true

Answer: A

Solution:

$[h] = ML^2T^{-1}$

$[L] = ML^2T^{-1}$

$[P] = MLT^{-1}$

$[\tau] = ML^2T^{-2}$

(Here h is Planck's constant, L is angular momentum, P is linear momentum and τ is moment of force)

Question66

The equation of state of a real gas is given by $\left(P + \frac{a}{V^2}\right)$ (V − b) = RT, where P,V and T are pressure. volume and temperature respectively and R is the universal gas constant. The dimensions of a/b^2 is similar to that of :

[27-Jan-2024 Shift 2]

Options:

A.

PV

B.

P

C.

RT

D.

R

Answer: B

Solution:

$$[P] = \left[\frac{a}{V^2}\right] \Rightarrow [a] = [PV^2]$$

And $[V] = [b]$

$$\frac{[a]}{[b^2]} = \frac{[PV^2]}{[V^2]} = [P]$$

Question67

Match List-I with List-II.

	List-I		List-II
A.	Coefficient of viscosity	I.	$\{[ML^2T^{-2}]\}$
B.	Surface Tension	II.	$\{[ML^2T^{-1}]\}$
C.	Angular momentum	III.	$\{[ML^{-1}T^{-1}]\}$
D.	Rotational kinetic energy	IV.	$\{[ML^0T^{-2}]\}$

[30-Jan-2024 Shift 1]

Options:

A.

A-II, B-I, C-IV, D-III

B.

A-I, B-II, C-III, D-IV

C.

A-III, B-IV, C-II, D-I

D.

A-IV B-III C-II. D-I

Answer: C

Solution:

$$F = \eta A \frac{dv}{dy}$$

$$[MLT^{-2}] = \eta[L^2][T^{-1}]$$

$$\eta = [ML^{-1}T^{-1}]$$

$$S.T = \frac{F}{\ell} = \frac{[MLT^{-2}]}{[L]} = [ML^0T^{-2}]$$

$$L = mvr = [ML^2T^{-1}]$$

$$K.E = \frac{1}{2}I\omega^2 = [ML^2T^{-2}]$$

Question68

If mass is written as $m = kc^P G^{-1/2} h^{1/2}$ then the value of P will be : (Constants have their usual meaning with k a dimensionless constant)

[30-Jan-2024 Shift 2]

Options:

A.

1/2

B.

1/3

C.

2

D.

1/4

Answer: A

Solution:

$m = kc^{P}G^{-1/2}h^{1/2}$

$M^{1}L^{0}T^{0} = [LT^{-1}]^{P}[M^{-1}L^{3}T^{-2}]^{-1/2}\ [ML^{2}T^{-1}]^{1/2}$

By comparing $P = 1/2$

Question69

A force is represented by $F = ax^{2} + bt^{1/2}$

Where x = distance and t = time. The dimensions of b^{2}/a are :

[31-Jan-2024 Shift 1]

Options:

A.

$[ML^{3}T^{-3}]$

B.

$[MLT^{-2}]$

C.

$[ML^{-1}T^{-1}]$

D.

$[ML^{2}T^{-3}]$

Answer: A

Solution:

$$F = ax^2 + bt^{1/2}$$

$$[a] = \frac{[F]}{[x^2]} = [M^1L^{-1}T^{-2}]$$

$$[b] = \frac{[F]}{[t^{1/2}]} = [M^1L^1T^{-5/2}]$$

$$\left[\frac{b^2}{a}\right] = \frac{[M^2L^2T^{-5}]}{[M^1L^{-1}T^{-2}]} = [M^1L^3T^{-3}]$$

Question70

Consider two physical quantities A and B related to each other as E $= \frac{B-x^2}{At}$ where E, x and t have dimensions of energy,

[31-Jan-2024 Shift 2]

Options:

A.

$L^{-2}M^1T^0$

B.

$L^2M^{-1}T^1$

C.

$L^{-2}M^{-1}T^1$

D.

$L^0M^{-1}T^1$

Answer: B

Solution:

$[B] = L^2$

$$A = \frac{x^2}{tE} = \frac{L^2}{TML^2T^{-2}} = \frac{1}{MT^{-1}}$$

$[A] = M^{-1}T$

$[AB] = [L^2M^{-1}T^1]$

Question71

The dimensional formula of angular impulse is :

[1-Feb-2024 Shift 1]

Options:

A.

$[ML^{-2}T^{-1}]$

B.

$[ML^2T^{-2}]$

C.

$[MLT^{-1}]$

D.

$[ML^2T^{-1}]$

Answer: D

Solution:

Angular impulse = change in angular momentum.

[Angular impulse] = [Angular momentum] = $[mvr]$

$= [ML^2T^{-1}]$

Question72

10 divisions on the main scale of a Vernier calliper coincide with 11 divisions on the Vernier scale. If each division on the main scale is of 5 units, the least count of the instrument is :

[1-Feb-2024 Shift 1]

Options:

A.

1/2

B.

10/11

C.

50/11

D.

5/11

Answer: D

Solution:

$10\,\text{MSD} = 11\,\text{VSD}$

$1\,\text{VSD} = \frac{10}{11}\text{MSD}$

$\text{LC} = 1\,\text{MSD} - 1\,\text{VSD}$

$= 1\,\text{MSD} - \frac{10}{11}\text{MSD}$

$= \frac{1\,\text{MSD}}{11}$

$= \frac{5}{11}$ units

Question73

The radius (r), length (l) and resistance (R) of a metal wire was measured in the laboratory as

$r = (0.35 \pm 0.05)$ cm

$R = (100 \pm 10)$ ohm

$l = (15 \pm 0.2)$ cm

The percentage error in resistivity of the material of the wire is :

[1-Feb-2024 Shift 1]

Options:

A.

25.6%

B.

39.9%

C.

37.3%

D.

35.6%

Answer: B

Solution:

$$\rho = R\frac{\rho}{\ell}$$

$$\frac{\Delta\rho}{\rho} = \frac{\Delta R}{R} + 2\frac{\Delta r}{r} + \frac{\Delta\ell}{\ell}$$

$$= \frac{10}{100} + 2\times\frac{0.05}{0.35} + \frac{0.2}{15}$$

$$= \frac{1}{10} + \frac{2}{7} + \frac{1}{75}$$

$$\frac{\Delta\rho}{\rho} = 39.9\%$$

Question74

Choose the correct answer from the options given below :

Match List - I with List - II.

List - I (Number)	List - II (Significant figure)
(A) 1001	(I) 3
(B) 010.1	(II) 4
(C) 100.100	(III) 5
(D) 0.0010010	(IV) 6

[1-Feb-2024 Shift 2]

Options:

A.

(A)-(III), (B)-(IV), (C)-(II), (D)-(I)

B.

(A)-(IV), (B)-(III), (C)-(I), (D)-(II)

C.

(A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D.

(A)-(I), (B)-(II), (C)-(III), (D)-(IV)

Answer: C

Solution:

Theoretical

Question75

Match List I with List II

	LIST I		LIST II
A.	Planck's constant (h)	I.	$[M^1L^2T^{-2}]$
B.	Stopping potential (Vs)	II.	$[M^1L^1T^{-1}]$
C.	Work function (Ø)	III.	$[M^1L^2T^{-1}]$
D.	Momentum (p)	IV.	$[M^1L^2T^{-3}A^{-1}]$

[24-Jan-2023 Shift 1]

Options:

A. A-III, B-I, C-II, D-IV

B. A-III, B-IV, C-I, D-II

C. A-II, B-IV, C-III, D-I

D. A-I, B-III, C-IV, D-II

Answer: B

Solution:

(A) Planck's constant

$h\nu = E$

$$h = \frac{E}{\nu} = \frac{M^1L^2T^{-2}}{T^{-1}} = M^1L^2T^{-1}$$

(B) $E = qV$

$$V = \frac{E}{q} = \frac{M^1L^2T^{-2}}{A^1T^1} = M^1L^2T^{-3}A^{-1}(IV)$$

(C) φ(work function) = energy

$= M^1L^2T^{-2}$

(D) Momentum (p) = F.t

$= M^1L^1T^{-2}T^1$

$= M^1L^1T^{-1}$

Question76

The frequency (v) of an oscillating liquid drop may depend upon radius (r) of the drop, density (ρ) of liquid and the surface tension (s) of the liquid as : $v = r^a\rho^b s^c$. The values of a, b and c respectively are
[24-Jan-2023 Shift 2]

Options:

A. $\left(-\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

B. $\left(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}\right)$

C. $\left(\frac{3}{2}, \frac{1}{2}, -\frac{1}{2}\right)$

D. $\left(-\frac{3}{2}, \frac{1}{2}, \frac{1}{2}\right)$

Answer: A

Solution:

$$[T^{-1}] = [L^1]^a[M^1L^{-3}]^b\left[\frac{MLT^{-2}}{L}\right]^c$$

$$\Rightarrow T^{-1} = M^{b+c}\cdot L^{a-3b}\cdot T^{-2c}$$

$c = \frac{1}{2}, b = -\frac{1}{2}, \quad a - 3b = 0$

$a + \frac{3}{2} = 0 \Rightarrow a = -\frac{3}{2}$

Question77

Match List I with List II

List - I	List - II
A. Surface tension	I. $\mathrm{Kg\,m^{-1}}\ s^{-1}$
B. Pressure	II. $\mathrm{Kg\,ms^{-1}}$
C. Viscosity	III. $\mathrm{Kg\,m^{-1}}\ s^{-2}$
D. Impulse	IV. $\mathrm{Kg}\ s^{-2}$

Choose the correct answer from the options given below : [25-Jan-2023 Shift 1]

Options:

A. A-IV, B-III, C- II, D- I

B. A-IV, B-III, C-I, D-II

C. A-III, B-IV, C-I, D-II

D. A-II, B-I, C-III, D-IV

Answer: B

Solution:

$$\text{(A) Surface Tension} = \frac{F}{\ell} = \frac{MLT^{-2}}{L} = ML^{-1}T^{-2}$$

$= \mathrm{Kgs^{-2}}$(IV)

$$\text{(B) Pressure} = \frac{F}{A} = \frac{MLT^{-2}}{L^2}$$

$= \mathrm{kg\,m^{-1}\ s^{-2}}$ (III)

(C) Viscosity $= = \dfrac{F}{A\left(\dfrac{dV}{dz}\right)} = \dfrac{MLT^{-2}}{L^2\left(\dfrac{LT^{-1}}{L}\right)}$

$= ML^{-1}T^{-1} = kg\,m^{-1}\ s^{-1}$(I)

(D) Impulse $= \int Fdt = MLT^{-2} \times T$

$= MLT^{-1} = Kgms^{-1}$ (II)

So A-(IV), B-(III), C-(I), D- (II)

Question78

Match List I with List II

List	List II
A. Young's Modulus (Y)	I. $[ML^{-1}T^{-1}]$
B. Co-efficient of Viscosity (η)	II. $[ML^2T^{-1}]$
C. Planck's Constant (h)	III. $[ML^{-1}T^{-2}]$
D. Work Function (φ)	IV. $[ML^2T^{-2}]$

Choose the correct answer from the options given below: [25-Jan-2023 Shift 2]

Options:

A. A-II, B-III, C-IV, D-I

B. A-III, B-I, C-II, D-IV

C. A-I, B-III, C-IV, D-II

D. A-I, B-II, C-III, D-IV

Answer: B

Solution:

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{F/A}{\Delta\ell/\ell} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$$

$$F = 6\pi\eta rv \Rightarrow \eta = \frac{F}{6\pi rv}$$

$$[\eta] = \frac{[MLT^{-2}]}{[L][LT^{-1}]} = [ML^{-1}T^{-1}]$$

$$E = h\nu \Rightarrow h = \frac{E}{\nu} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

Work function has same dimension as that of energy, so $[\varphi] = [ML^2T^{-2}]$

Question79

Match List I with List II :

List-I (PhysicalQuantity)	List-II (Dimensional Formula)
A Pressure gradient	I $[M^0L^2T^{-2}]$
B Energy density	II $[M^1L^{-1}T^{-2}]$
C Electric Field	III $[M^1L^{-2}T^{-2}]$
D Latent heat	IV $[M^1L^1T^{-3}A^{-1}]$

Choose the correct answer from the options given below: [29-Jan-2023 Shift 1]

Options:

A. A-III, B-II, C-I, D-IV

B. A-II, B-III, C-IV, D-I

C. A-III, B-II, C-IV, D-I

D. A-II, B-III, C-I, D-IV

Answer: C

Solution:

Pressure gradient $= \frac{dp}{dx} = \frac{[ML^{-1}T^{-2}]}{[L]}$

$= [M^1L^{-2}T^{-2}]$

Energy density $= \frac{\text{energy}}{\text{volume}} = \frac{[ML^2T^{-2}]}{[L^3]}$

$= [M^1L^{-1}T^{-2}]$

Electric field $= \frac{\text{Force}}{\text{charg e}} = \frac{[MLT^{-2}]}{[A \cdot T]}$

$= [M^1L^1T^{-3}A^{-1}]$

Latent heat $= \frac{\text{heat}}{\text{mass}} = \frac{[ML^2T^{-2}]}{[M]}$

$= [M^0L^2T^{-2}]$

Question80

The equation of a circle is given by $x^2 + y^2 = a^2$, where a is the radius. If the equation is modified to change the origin other than (0, 0), then find out the correct dimensions of A and B in a new equation: $(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$.

The dimensions of t is given as $[T^{-1}]$.
[29-Jan-2023 Shift 2]

Options:

A. $A = [L^{-1}T], B = [LT^{-1}]$

B. $A = (LT], B = [L^{-1}T^{-1}]$

C. $A = [L^{-1}T^{-1}], B = [LT^{-1}]$

D. $A = [L^{-1}T^{-1}], B = [LT]$

Answer: B

Solution:

$$(x - At)^2 + \left(y - \frac{t}{B}\right)^2 = a^2$$

$$[At] = A \times \frac{1}{T} = L$$

$\therefore \quad [A] = T^1 L^1$

$\frac{t}{B}$ is in meters

$$\therefore \quad \frac{1}{T[B]} = L$$

$\therefore \quad [B] = T^{-1} L^{-1}$

$\therefore$ Correct Ans. (2)

Question81

Electric field in a certain region is given by $\vec{E} = \left(\frac{A}{x^2}\hat{i} + \frac{B}{y^3}\hat{j}\right)$. The SI unit of A and B are :
[30-Jan-2023 Shift 1]

Options:

A. Nm^3C^{-1}; Nm^2C^{-1}

B. Nm^2C^{-1}; Nm^3C^{-1}

C. Nm^3C; Nm^2C

D. Nm^2C; Nm^3C

Answer: B

Solution:

$$\vec{E} = \frac{A}{x^2}\hat{i} + \frac{B}{y^3}\hat{j}$$

$$\left[\frac{A}{x^2}\right] = NC^{-1} \Rightarrow [A] = Nm^2C^{-1}$$

$$\left[\frac{B}{y^3}\right] = NC^{-1} \Rightarrow [B] = Nm^3C^{-1}$$

Question82

Match List I with List II.

List I	List II
A. Torque	I. $kg\,m^{-1}\,s^{-2}$
B. Energy density	II. $kg\,ms^{-1}$
C. Pressure gradient	III. $kg\,m^{-2}\,s^{-2}$
D. Impulse	IV. $kg\,m^{2}\,s^{-2}$

Choose the correct answer from the options given below :
[30-Jan-2023 Shift 2]

Options:

A. A-IV, B-III, C-I, D-II

B. A-I, B-IV, C-III, D-II

C. A-IV, B-I, C-II, D-III

D. A-IV, B-I, C-III, D-II

Answer: D

Solution:

Question83

If R, X_L and X_C represent resistance, inductive reactance and capacitive reactance. Then which of the following is dimensionless:
[31-Jan-2023 Shift 1]

Options:

A. RX_LX_C

B. $\frac{R}{\sqrt{X_LX_C}}$

C. $\frac{R}{X_LX_C}$

D. $R\frac{X_L}{X_C}$

Answer: B

Solution:

All three have same dimension therefore $\frac{R}{\sqrt{X_LX_C}}$ is dimensionless.

Option 2

Question84

Match List-I with List-II.

List-I	List-II
A. Angular momentum	I. $[ML^2T^{-2}]$.
B. Torque	II. $[ML^{-2}T^{-2}]$
C. Stress	III. $[ML^2T^{-1}]$
D. Pressure gradient	IV. $[ML^{-1}T^{-2}]$.

Choose the correct answer from the options given below: [31-Jan-2023 Shift 2]

Options:

A. A-I, B-IV, C-III, D-II

B. A-III, B-I, C-IV, D-II

C. A-II, B-III, C-IV, D-I

D. A-IV, B-II, C-I, D-III

Answer: B

Solution:

$L = \vec{r} \times \vec{p} \Rightarrow [L] = [M^0L^1T^0][M^1L^1T^{-1}]$
$= [M^1L^2T^{-1}]$
$\vec{\tau} = \vec{r} \times \vec{F} \Rightarrow [\tau] = [L^1][MLT^{-2}]$
$= [ML^2T^{-2}]$
$\text{Stress} \equiv \text{Pressure} = \frac{F}{A} \Rightarrow [\text{Stress}] = [ML^{-1}T^{-2}]$
Pressure Gradient $= \frac{dP}{dx} \Rightarrow$ [Pressure Gradient] $= [ML^{-2}T^{-2}]$

Question85

Vectors $a\hat{i} + b\hat{j} + \hat{k}$ and $2\hat{i} - 3\hat{j} + 4\hat{k}$ are perpendicular to each other when $3a + 2b = 7$, the ratio of a to b is $\frac{x}{2}$. The value of x is____ [24-Jan-2023 Shift 1]

Answer: 1

Solution:

For two perpendicular vectors
$\left(a\hat{i} + b\hat{j} + \hat{k}\right) \cdot \left(2\hat{i} - 3\hat{j} + 4\hat{k}\right) = 0$
$2a - 3b + 4 = 0$
On solving, $2a - 3b = -4$
Also given
$3a + 2b = 7$
We get $a = 1, b = 2$
$\frac{a}{b} = \frac{x}{2} \Rightarrow x = \frac{2a}{b} = \frac{2 \times 1}{2}$
$\Rightarrow x = 1$

Question86

If two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other.
Then, the value of m will be :
[24-Jan-2023 Shift 2]

Options:

A. 1

B. -1

C. -3

D. 2

Answer: D

Solution:

$$\vec{P} \cdot \vec{Q} = 0$$
$$\left(\hat{i} + 2m\hat{j} + m\hat{k}\right) \cdot \left(4\hat{i} - 2\hat{j} + m\hat{k}\right) = 0$$
$$\Rightarrow 4 - 4m + m^2 = 0$$
$$\Rightarrow (m-2)^2 = 0 \Rightarrow m = 2$$

Question87

If $\vec{P} = 3\hat{i} + \sqrt{3}\hat{j} + 2\hat{k}$ and $\vec{Q} = 4\hat{i} + \sqrt{3}\hat{j} + 2.5\hat{k}$ then, The unit vector in the direction of $\vec{P} \times \vec{Q}$ is $\frac{1}{x}\left(\sqrt{3}\hat{i} + \hat{j} - 2\sqrt{3}\hat{k}\right)$. The value of x is
[25-Jan-2023 Shift 1]

Answer: 4

Solution:

$$\vec{P}\times\vec{Q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & \sqrt{3} & 2 \\ 4 & \sqrt{3} & 2.5 \end{vmatrix} = \sqrt{3}\,\frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3}\hat{k}$$

$$\Rightarrow \frac{\vec{P}\times\vec{Q}}{\left|\vec{P}\times\vec{Q}\right|} = \frac{1}{2}\left(\sqrt{3}\,\frac{\hat{i}}{2} + \frac{\hat{j}}{2} - \sqrt{3}\hat{k}\right)$$

$$= \frac{1}{4}\left(\sqrt{3}\,\hat{i} + \hat{j} - 2\sqrt{3}\hat{k}\right) \quad x = 4$$

Question88

In an experiment of measuring the refractive index of a glass slab using travelling microscope in physics lab, a student measures real thickness of the glass slab as 5.25 mm and apparent thickness of the glass slab at 5.00 mm. Travelling microscope has 20 divisions in one cm on main scale and 50 divisions on Vernier scale is equal to 49 divisions on main scale. The estimated uncertainty in the measurement of refractive index of the slab is $\frac{x}{10}\times 10^{-3}$, where x is

____

[29-Jan-2023 Shift 2]

Answer: 41

Solution:

$$u = \frac{h}{h'} = \frac{5.25}{5.00}$$

$$\text{Least count} = \frac{1}{20}\,\text{cm} - \frac{49}{50}\cdot\frac{1}{20}\,\text{cm}$$

$$= \frac{1}{50}\times\frac{1}{20}\,\text{cm} = 0.01\text{ mm}$$

$$\ln u = \ln h - \ln h'$$

$$\frac{du}{u} = \frac{dh}{h} - \frac{dh'}{h'}$$

$$du = \left[\frac{0.01}{5.25} + \frac{0.01}{5.00}\right]\frac{5.25}{5.00}$$
$$= \frac{41}{10} \times 10^{-3}$$
Ans. $= 41$

Question89

In a metre bridge experiment the balance point in obtained if the gaps are closed by 2Ω and 3Ω. A shunt of X Ω is added to 3Ω resistor to shift the balancing point by 22.5 cm. The value of X is _______
[29-Jan-2023 Shift 1]

Answer: 2

Solution:

$$\frac{2}{\left(\frac{3x}{3+x}\right)} = \frac{40+22.5}{60-22.5} = \frac{62.5}{37.5} = \frac{5}{3}$$
$$\frac{6}{5} = \frac{3x}{3+x}$$
$$6 + 2x = 5x \Rightarrow x = 2$$

Question90

A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω. When a resistance of 15Ω is used for shunting null point moves to 300 cm. The internal resistance of the cell is ______ Ω.
[29-Jan-2023 Shift 2]

Answer: 5

Solution:

$$\frac{\varepsilon}{r+5} \times 5 = 200x \; . \; . \; . \text{ (1)}$$
$$\frac{\varepsilon \times 15}{r+15} = 300x \; . \; . \; . \text{ (2)}$$
$$\Rightarrow r = 5\Omega$$
Ans. 5

Question91

In a screw gauge, there are 100 divisions on the circular scale and the main scale moves by 0.5 mm on a complete rotation of the circular scale. The zero of circular scale lies 6 divisions below the line of graduation when two studs are brought in contact with each other. When a wire is placed between the studs, 4 linear scale divisions are clearly visible while 46 $^{\text{th}}$ division the circular scale coincide with the reference line. The diameter of the wire is _______ $\times 10^{-2}$ mm.
[30-Jan-2023 Shift 1]

Answer: 220

Solution:

$$\text{Least count} = \frac{\text{Pitch}}{\text{No. of circular divisions}}$$
$$= \frac{0.5 \text{ mm}}{100}$$
$$\text{Least count} = 5 \times 10^{-3} \text{ mm}$$
$$\text{Positive Error} = \text{MSR} + \text{CSR(LC)}$$
$$= 0 \text{ mm} + 6(5 \times 10^{-3} \text{ mm})$$
Reading of Diameter = MSR + CSR (LC)− Positive zero error
$$= 4 \times 0.5 \text{ mm} + (46(5 \times 10^{-3})) - 6(5 \times 10^{-3}) \text{ mm}$$
$$= 2 \text{ mm} + 40 \times 5 \times 10^{-3} \text{ mm} = 2.2 \text{ mm} \;\; \text{(Ans.)}$$

Question92

$\left(P+\frac{a}{V^2}\right)(V-b)=RT$ represents the equation of state of some gases. Where P is the pressure, V is the volume, T is the temperature and a, b, R are the constants. The physical quantity, which has dimensional formula as that of $\frac{b^2}{a}$, will be :
[1-Feb-2023 Shift 1]

Options:

A. Bulk modulus

B. Modulus of rigidity

C. Compressibility

D. Energy density

Answer: C

Solution:

$$[b]=[V]$$
$$\left[\frac{a}{b^2}\right]=[P] \quad \therefore \quad \left[\frac{b^2}{a}\right]=\frac{1}{[P]}=\frac{1}{[B]}=[K]$$

Question93

If the velocity of light c, universal gravitational constant G and planck's constant h are chosen as fundamental quantities. The dimensions of mass in the new system is:
[1-Feb-2023 Shift 2]

Options:

A. $\left[h^{\frac{1}{2}}c^{\frac{1}{2}}G^{1}\right]$

B. $[h^1 c^1 G^{-1}]$

C. $\left[h^{\frac{1}{2}} c^{\frac{1}{2}} G^{\frac{1}{2}}\right]$

D. $\left[h^{\frac{1}{2}} c^{\frac{1}{2}} G^{-\frac{1}{2}}\right]$

Answer: D

Solution:

Say dimensional formale of mass is $H^x C^y G^z$

$M^1 = (ML^2T^{-1})^x(LT^{-1})(M^{-1}L^3T^{-2})^z$

$M^1L^0T^0 = M^{x-z}L^{2x+y+3z}T^{-x-y-2z}$

on comparing both side

$x - z = 1$

$2x + y + 3z = 0$

$-x - y - 2z = 0$

On solving above equations we get

$x = \frac{1}{2} \quad y = \frac{1}{2} \quad z = \frac{-1}{2}$

Question94

In an experiment to find emf of a cell using potentiometer, the length of null point for a cell of emf 1.5V is found to be 60 cm. If this cell is replaced by another cell of emf E . the length-of null point increases by 40 cm. The value of E is $\frac{x}{10}$V. The value of x is ________.
[1-Feb-2023 Shift 1]

Answer: 25

Solution:

$$\frac{E_1}{E_2} = \frac{l_1}{l_2}$$

$$\frac{1.5}{E_2} = \frac{60}{60+40} = \frac{6}{10} = \frac{3}{5}$$

$$E_2 = \frac{5}{2} = \frac{x}{10}$$

$x = 25$

Question95

Dimension of $\frac{1}{\mu_0 \in_0}$ should be equal to [8-Apr-2023 shift 1]

Options:

A. T/L

B. T^2/L^2

C. L/T

D. L^2/T^2

Answer: D

Solution:

Dimension of $\frac{1}{\mu_0 \varepsilon_0}$

$$C = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \Rightarrow \frac{1}{\mu_0 \varepsilon_0} = c^2$$

$$\left[\frac{1}{\mu_0 \varepsilon_0}\right] = [c^2]$$

$$= \left[\frac{L^2}{T^2}\right]$$

Question96

Match List I with List II

LIST - I	LIST - II
A. Torque	I. $ML^{-2}T^{-2}$
B. Stress	II. $ML^{2}T^{-2}$
C. Pressure gradient	III. $MLL^{-1}t^{-1}$
D. Coefficient of viscosity	IV. $ML^{-1}T^{-2}$

Choose the correct answer from the options given below: [8-Apr-2023 shift 2]

Options:

A. A-III, B-IV, C-I, D-II

B. A-II, B-I, C-IV, D-III

C. A-IV, B-II, C-III, D-I

D. A-II, B-IV, C-I, D-III

Answer: D

Solution:

$[\text{ Torque }] = F.L$

$MLT^{-2} \cdot L = ML^{2}T^{-2}$

$[\text{ Stress }] = \frac{F}{A}$

$\frac{M\,LT^{-2}}{L^{2}} = ML^{-1}T^{-2}$

$[\text{ Pressure gradient }] = \frac{\Delta P}{\Delta L} = \frac{F}{A^{2}.L}$

$= \frac{MLT^{-2}}{L^{3}}$

$= ML^{-2}T^{-2}$

$F = nA\frac{dv}{dy}$

$\eta = ML^{-1}T^{-1}$

Question97

Given below are two statements :
Statements I : Astronomical unit (Au), Parsec (Pc) and Light year (ly) are units for measuring astronomical distances.
Statements II : Au< Parsec (Pc)< ly
In the light of the above statements, choose the most appropriate answer from the options given below :
[11-Apr-2023 shift 1]

Options:

A. Both Statements I and Statements II are incorrect.

B. Both Statements I and Statements II are correct.

C. Statements I is incorrect but Statements II are correct.

D. Statements I is correct but Statements II are incorrect.

Answer: D

Solution:

A.V., Par sec and light year are the unit of distance
Light year → distance travelled by light in one year
$1\text{ ly} = 9.5 \times 10^{15}\text{m}$
parcec $= 3.262$ light year
A.V. $= 1.58 \times 10^{-5}$ light year
A. V. $< 1\text{y} <$ Parsec.
Statement I correct and statement II incorrect.

Question98

If force (F), velocity (V) and time (T) are considered as fundamental physical quantity, then dimensional formula of density will be:
[11-Apr-2023 shift 2]

Options:

A. $FV^{-2}T^{2}$

B. FV^4T^{-6}

C. $FV^{-4}T^{-2}$

D. $F^2V^{-2}T^6$

Answer: C

Solution:

$$\rho = F^xV^yT^z$$
$$[ML^{-3}] = [MLT^{-2}]^x[LT^{-1y}]^y[T]^z$$
$$1 = x \qquad x = 1$$
$$-3 = x + y \qquad y = -4$$
$$0 = -2x - y + z \qquad z = 2x + 4 = 2 - 4 = -2$$
$$\rho = FV^{-4}T^{-2} \quad \text{Ans. (3)}$$

Question99

LIST I	LIST II
A. Spring constant	I. $[T^{-1}]$
B. Angular speed	II. $[MT^{-2}]$
C. Angular momentum	III. $[ML^2]$
D. Moment of Inertia	IV. $[ML^2T^{-1}]$

Choose the correct answer from the options given below : [12-Apr-2023 shift 1]

Options:

A. A-II, B-III, C-I, D-IV

B. A-IV, B-I, C-III, D-II

C. A-I, B-III, C-II, D-IV

D. A-II, B-I, C-IV, D-III

Answer: D

Solution:

(i) Spring constant

$F = Kx$

$K = \frac{F}{x} = \frac{\text{kgm sec}}{\text{m}} = \text{kg} \cdot \text{sec}^{-2} = [ML^0T^{-2}]$

(ii) Angular speed : (ω)

$\omega = \frac{2\pi}{T}$

$\omega = [M^0L^0T^{-1}]$

(iii) Angular momentum

$L = MVR \Rightarrow \text{Kg} \cdot \text{m sec}^{-1}\text{m} = \text{kgm}^2\ \text{sec}^{-1}$

$= [ML^2T^{-1}]$

(iv) Moment of Inertia

$I = MR^2 = \text{Kg} \cdot \text{m}^2$

$= [ML^2T^0]$

Question100

In the equation $\left[x + \frac{a}{y^2}\right][Y - b] = RT$, X is pressure, Y is volume, R is universal gas constant ant T is temperature. The physical quantity equivalent to the ratio $\frac{a}{b}$ is :
[13-Apr-2023 shift 2]

Options:

A. Coefficient of viscosity

B. Energy

C. Impulse

D. Pressure gradient

Answer: B

Solution:

x and $\frac{a}{y^2}$ have same dimensions
y and b have same dimensions
$[a] = [ML^5T^{-2}]$
$[b] = [L^3]$
$\frac{[a]}{[b]} = ML^2T^{-2}$ has dimension of energy

Question101

The speed of a wave produced in water is given by $v = \lambda^a g^b \rho^c$. Where λ, g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a, b and c respectively, are : [15-Apr-2023 shift 1]

Options:

A. $\frac{1}{2}, 0, \frac{1}{2}$

B. $1, -1, 0$

C. $\frac{1}{2}, \frac{1}{2}, 0$

D. $1, 1, 0$

Answer: C

Solution:

By dimensional analysis,
$V = \lambda^a g^b \rho^c$
$[L^1T^{-1}] = [L^1]^a[L^1T^{-2}]^b[M^1L^{-3}]^c$
$[L^1T^{-1}] = [M^cL^{a+b-3c}T^{-2b}]$
On comparing respective powers,
$c = 0, -2b = -1, a + b - 3c = 1,$
$b = \frac{1}{2}, a = \frac{1}{2}$

Question102

Two resistances are given as $R_1 = (10 \pm 0.5)\Omega$ and $R_2 = (15 \pm 0.5)\Omega$. The percentage error in the measurement of equivalent resistance when they are connected in parallel is -
[6-Apr-2023 shift 1]

Options:

A. 2.33

B. 4.33

C. 5.33

D. 6.33

Answer: B

Solution:

In parallel combination, $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{10} + \frac{1}{15} = \frac{5}{30} = \frac{1}{6}$

Now, for error calculation,

$\frac{dR_{eq}}{R_{eq}^2} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2}$

$\Rightarrow \frac{dR_{eq}}{36} = \frac{0.5}{100} + \frac{0.5}{225}$

$dR_{eq} = 36 \times 0.5 \times \left(\frac{13}{900}\right) = 18 \times \frac{13}{900} = \frac{26}{100} = 0.26$

Now, $\frac{dR_{eq}}{R_{eq}} \times 100 = \frac{0.26}{6} \times 100 = \frac{26}{6} = 4.33$

Question103

The length of a metallic wire is increased by 20% and its area of cross section is reduced by 4%. The percentage change in resistance of the metallic wire is ________
[6-Apr-2023 shift 1]

Answer: 25

Solution:

$$R = \frac{\rho l}{A}$$
$$R' = \frac{\rho \times 1.2l}{0.96A} = \frac{10}{8} \times R = 1.25R$$

It means 25% increase in Resistance.

Question104

Two forces having magnitude A and $\frac{A}{2}$ are perpendicular to each other. The magnitude of their resultant is:
[8-Apr-2023 shift 1]

Options:

A. $\frac{5A}{2}$

B. $\frac{\sqrt{5}A^2}{2}$

C. $\frac{\sqrt{5}A}{4}$

D. $\frac{\sqrt{5}A}{2}$

Answer: D

Solution:

$$\left|\vec{F}_1\right| = A, \left|\vec{F}_2\right| = \frac{A}{2} \quad \theta = \frac{\pi}{2}$$
$$\left|\vec{F}_{net}\right| = \sqrt{F_1^2 + F_2^2}$$

$$= \sqrt{A^2 + \left(\frac{A}{2}\right)^2}$$

$$\left|\vec{F}_{net}\right| = \frac{\sqrt{5}A}{2}$$

Question105

A physical quantity P is given as

P = $\frac{a^2b^3}{c\sqrt{d}}$

The percentage error in the measurement of a, b, c and d are 1%, 2%, 3% and 4% respectively. The percentage error in the measurement of quantity P will be
[10-Apr-2023 shift 1]

Options:

A. 14%

B. 13%

C. 16%

D. 12%

Answer: B

Solution:

$$\frac{dP}{P} \times 100 = \left(2\frac{da}{a} + 3\frac{db}{b} + \frac{dc}{c} + \frac{1}{2}\frac{d(d)}{d}\right) \times 100$$

$$= 2 \times 1 + 3 \times 2 + 3 + \frac{1}{2} \times 4$$

$$= 2 + 6 + 3 + 2$$

$$= 13\%$$

Question106

Three forces $F_1 = 10N$, $F_2 = 8N$, $F_3 = 6N$ are acting on a particle of mass 5 kg. The forces F_2 and F_3 are applied perpendicularly so that

particle remains at rest. If the force F_1 is removed, then the acceleration of the particle is :
[12-Apr-2023 shift 1]

Options:

A. $4.8 ms^{-2}$

B. $0.5 ms^{-2}$

C. $7 ms^{-2}$

D. $2 ms^{-2}$

Answer: D

Solution:

$F_1 = 10N$
$F_2 = 8N$
$F_3 = 6N$

$$\vec{a} = \frac{F_2\hat{i} + F_3\hat{j}}{5}$$

$$\vec{a} = \frac{8\hat{i} + 6\hat{j}}{5}$$

$$|\vec{a}| = \frac{\sqrt{64 + 36}}{5}$$

$$= \frac{10}{5} = 2m/\sec^2$$

OR

$F_1^{\,2} = F_2^{\,2} + F_3^{\,2}$

Now F1 is removed then net force

$$F_{net} = \sqrt{F_2^{\,2} + F_3^{\,2}}$$
$$= \sqrt{64 + 36}$$

Question107

A vector in $x - y$ plane makes an angle of 30° with y-axis. The magnitude of y-component of vector is $2\sqrt{3}$. The magnitude of x-

component of the vector will be :
[15-Apr-2023 shift 1]

Options:

A. 2

B. $\sqrt{3}$

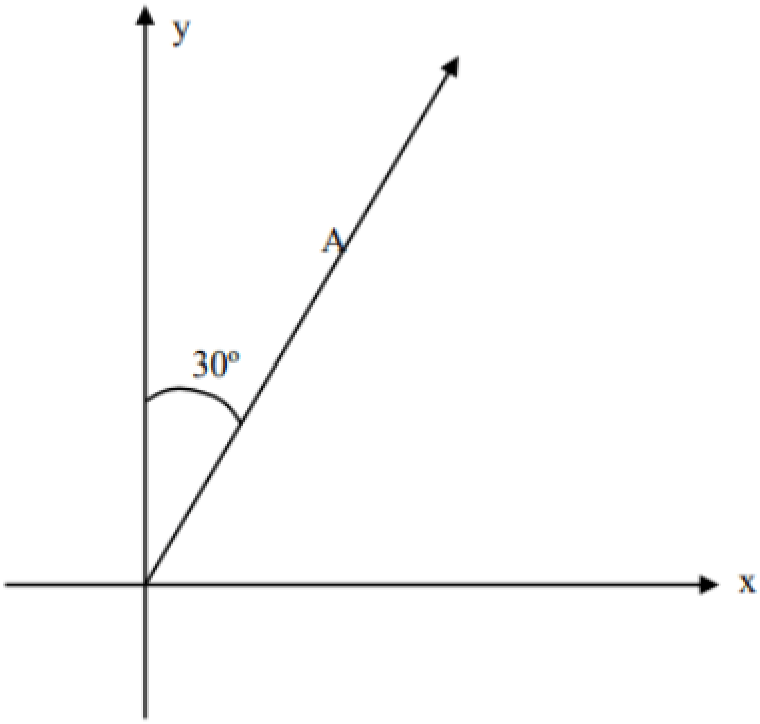
C. $\frac{1}{\sqrt{3}}$

D. 6

Answer: A

Solution:

$A_y = A\cos 30^\circ = 2\sqrt{3}$
$A = 4$
$A_x = A\sin 30^\circ = 2$



Question108

In an experiment with vernier calipers of least count 0.1 mm, when two jaws are joined together the zero of vernier scale lies right to the

zero of the main scale and 6 th division of vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between 3.2 cm and 3.3 cm marks, and 4 th division of vernier scale coincides with the main scale division. The diameter of bob is measured as [10-Apr-2023 shift 2]

Options:

A. 3.25 cm

B. 3.22 cm

C. 3.18 cm

D. 3.26 cm

Answer: C

Solution:

The zero error in verniel scale is $= 6 \times 0.1\text{ mm} = 0.6\text{ mm}$ (+ve zero error)
Note: +ve zero error will have to be subtracted
From the reading of the object.
Now, the diameter measured with the help of Vernier scale is
Given by → M.S.D + V.S.D × L.S
$\Rightarrow 3.2\text{ cm} + 0.1\text{ mm} \times 4$
$= 3.24\text{ cm}$
The actual diameter is ⇒3.24 mm - (zero error) $= 3.24 - 0.6 = 3.18\text{ cm}$

Question109

Identify the pair of physical quantities which have different dimensions: [24-Jun-2022-Shift-1]

Options:

A. Wave number and Rydberg's constant

B. Stress and Coefficient of elasticity

C. Coercivity and Magnetisation

D. Specific heat capacity and Latent heat

Answer: D

Solution:

$[S] = \frac{[C]}{[m] \times [\Delta T]}$

and, $[L] = \frac{[Q]}{[m]}$

⇒ They have different dimensions.

Question110

Identify the pair of physical quantities that have same dimensions: [24-Jun-2022-Shift-2]

Options:

A. velocity gradient and decay constant

B. Wien's constant and Stefan constant

C. angular frequency and angular momentum

D. wave number and Avogadro number

Answer: A

Solution:

Velocity gradient $= \frac{dv}{dx}$

⇒ Dimensions are $\frac{[LT^{-1}]}{[L]} = [T^{-1}]$

Decay constant λ has dimensions of $[T^{-1}]$ because of the relation $\frac{dN}{dt} = -\lambda N$

⇒ Velocity gradient and decay constant have same dimensions.

Question111

An expression for a dimensionless quantity P is given by

P = $\frac{\alpha}{\beta}\log_e\left(\frac{kt}{\beta x}\right)$; where α and β are constants, x is distance; k is Boltzmann constant and t is the temperature. Then the dimensions of α will be :
[26-Jun-2022-Shift-1]

Options:

A. $[M^0L^{-1}T^0]$

B. $[ML^0T^{-2}]$

C. $[MLT^{-2}]$

D. $[ML^2T^{-2}]$

Answer: C

Solution:

$[P] = \frac{[\alpha]}{[\beta]} \Rightarrow 1 \frac{[\alpha]}{[\beta]}$
$\Rightarrow [\alpha] = [\beta] \ldots\ldots (1)$
Now, $\left[\frac{kt}{\beta t}\right] = 1$
We know, E $= \frac{3}{2}kT$ (K T G)
$\therefore$[Energy] $= [KT]$
$\therefore [\beta x] = [kT]$
$[\beta] = \frac{ML^2T-2}{[x]} = \frac{ML^2T^{-2}}{L}$
$\therefore [\alpha] = [\beta] = MLT^{-2}$

Question112

The dimension of mutual inductance is :
[26-Jun-2022-Shift-2]

Options:

A. $[ML^2T^{-2}A^{-1}]$

B. $[ML^2T^{-3}A^{-1}]$

C. $[ML^2T^{-2}A^{-2}]$

D. $[ML^2T^{-3}A^{-2}]$

Answer: C

Solution:

$$\because U = \frac{1}{2}M i^2$$
$$\Rightarrow [M] = \frac{[U]}{[i^2]} = \frac{ML^2T^{-2}}{A^2}$$
$$= [ML^2T^{-2}A^{-2}]$$

Question113

The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be : [27-Jun-2022-Shift-2]

Options:

A. $[ML^{-1}T^{-1}]$

B. $[ML^{-1}T^{-2}]$

C. $[ML^2T^{-1}]$

D. $[M^{-1}L^3T^0]$

Answer: A

Solution:

$$[\text{pascal-second}] = \frac{MLT^{-2}}{L^2} \times T$$
$$= ML^{-1}T^{-1}$$

Question114

If L, C and R are the self inductance, capacitance and resistance respectively, which of the following does not have the dimension of time?
[27-Jun-2022-Shift-2]

Options:

A. RC

B. $\frac{L}{R}$

C. $\sqrt{LC}$

D. $\frac{L}{C}$

Answer: D

Solution:

$\left(\frac{L}{C}\right)$ does not have dimension of time.

RC, $\frac{L}{R}$ are time constant while $\sqrt{LC}$ is reciprocal of angular frequency or having dimension of time.

$U = \frac{1}{2}Li^2 = \frac{1}{2}CV^2$

So, $\left[\frac{L}{C}\right] = \frac{V^2}{i^2} = R^2$ is not the dimension of time.

Question115

In van der Waal equation $\left[P + \frac{a}{V^2}\right][V - b] = RT$; P is pressure, V is volume, R is universal gas constant and T is temperature. The ratio of constants $\frac{a}{b}$ is dimensionally equal to:
[29-Jun-2022-Shift-1]

Options:

A. $\frac{P}{V}$

B. $\frac{V}{P}$

C. PV

D. PV^3

Answer: C

Solution:

From the equation
$[a] \equiv [PV^2]$
$[b] \equiv [V]$
$\Rightarrow \left[\frac{a}{b}\right] \equiv [PV]$

Question116

If $Z = \frac{A^2B^3}{C^4}$, then the relative error in Z will be : [25-Jun-2022-Shift-1]

Options:

A. $\frac{\Delta A}{A} + \frac{\Delta B}{B} + \frac{\Delta C}{C}$

B. $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} - \frac{4\Delta C}{C}$

C. $\frac{2\Delta A}{A} + \frac{3\Delta B}{B} + \frac{4\Delta C}{C}$

D. $\frac{\Delta A}{A} + \frac{\Delta B}{B} - \frac{\Delta C}{C}$

Answer: C

Solution:

$$Z = \frac{A^2B^3}{C^4}$$
$$\therefore \frac{\Delta Z}{Z} = \frac{2\,\Delta\,A}{A} + 3 \times \frac{\Delta B}{B} + \frac{4\,\Delta\,C}{C}$$

Question117

$\vec{A}$ is a vector quantity such that $|\vec{A}| =$ non-zero constant. Which of the following expression is true for $\vec{A}$?
[25-Jun-2022-Shift-1]

Options:

A. $\vec{A} \cdot \vec{A} = 0$

B. $\vec{A} \times \vec{A} < 0$

C. $\vec{A} \times \vec{A} = 0$

D. $\vec{A} \times \vec{A} > 0$

Answer: C

Solution:

$$\vec{A} \times \vec{A} = A \times A \times \sin 0^\circ$$
$$= 0$$

Question118

Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle θ to each other?
[25-Jun-2022-Shift-1]

Options:

A. $|\hat{A} + \hat{B}| = |\hat{A} - \hat{B}| \tan \frac{\theta}{2}$

B. $\left|\hat{A}-\hat{B}\right| = \left|\hat{A}+\hat{B}\right| \tan \frac{\theta}{2}$

C. $\left|\hat{A}+\hat{B}\right| = \left|\hat{A}-\hat{B}\right| \cos \frac{\theta}{2}$

D. $\left|\hat{A}-\hat{B}\right| = \left|\hat{A}+\hat{B}\right| \cos \frac{\theta}{2}$

Answer: B

Solution:

$\because \left|\hat{A}-\hat{B}\right| = 2\sin\left(\frac{\theta}{2}\right)$

and, $\left|\hat{A}+\hat{B}\right| = 2\cos\left(\frac{\theta}{2}\right)$

$\Rightarrow \frac{\left|\hat{A}-\hat{B}\right|}{\left|\hat{A}+\hat{B}\right|} = \tan\left(\frac{\theta}{2}\right)$

Question119

For $z = a^2x^3y^{\frac{1}{2}}$, where 'a' is a constant. If percentage error in measurement of ' x ' and ' y ' are 4% and 12% respectively, then the percentage error for 'z' will be ______ %
[25-Jun-2022-Shift-2]

Answer: 18

Solution:

% error in $z = 3 \times 4 + \frac{1}{2} \times 12$

$= 12 + 6 = 18\%$

Question120

A silver wire has a mass (0.6 ± 0.006)g, radius (0.5 ± 0.005) mm and length (4 ± 0.04) cm. The maximum percentage error in the measurement of its density will be :
[27-Jun-2022-Shift-1]

Options:

A. 4%

B. 3%

C. 6%

D. 7%

Answer: A

Solution:

$\rho = \frac{m}{V} = \frac{m}{\pi r^2 \times l}$

$\therefore$% error in $\rho = \left(\frac{0.006}{0.6} + 2 \times \frac{0.005}{0.5} + \frac{0.04}{4} \right) \times 100$

$= 4\%$

Question121

The distance of the Sun from earth is 1.5×10^{11}m and its angular diameter is (2000) s when observed from the earth. The diameter of the Sun will be :
[27-Jun-2022-Shift-2]

Options:

A. 2.45×10^{10}m

B. 1.45×10^{10}m

C. 1.45×10^{9}m

D. 0.14×10^{9}m

Answer: C

Solution:

Diameter $= r \times \delta$

$= 1.5 \times 10^{11} \times (2000) \times \left(\frac{1}{3600}\right) \times \left(\frac{\pi}{180}\right)$

$= 1.45 \times 10^{9}$m

Question122

Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A} + \vec{B}$ is equal to two times the magnitude of $\vec{A} - \vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be :
[29-Jun-2022-Shift-1]

Options:

A. $\sin^{-1}\left(\frac{3}{5}\right)$

B. $\sin^{-1}\left(\frac{1}{3}\right)$

C. $\cos^{-1}\left(\frac{3}{5}\right)$

D. $\cos^{-1}\left(\frac{1}{3}\right)$

Answer: C

Solution:

$\sqrt{A^2 + A^2 + 2A^2 \cos\theta} = 2\sqrt{A^2 + A^2 + 2A^2(-\cos\theta)}$

$\Rightarrow 2A^2 + 2A^2 \cos\theta = 8A^2 + 8A^2 - (-\cos\theta)$

$\Rightarrow 5\cos\theta = 3$

$\Rightarrow \theta = \cos^{-1}\left(\frac{3}{5}\right)$

Question123

In a vernier callipers, each cm on the main scale is divided into 20 equal parts. If tenth vernier scale division coincides with nineth main scale division. Then the value of vernier constant will be _____ $\times 10^{-2}$ mm
[26-Jun-2022-Shift-1]

Answer: 5

Solution:

$1\text{ MSD} = \frac{1}{20}\text{ cm}$

$\because 10\text{ VSD} = 9\text{ MSD}$

$1\text{ VSD} = \frac{9}{10} \times \frac{1}{20}\text{ cm} = \frac{9}{200} \times 10\text{ mm} = 0.45\text{ mm}$

Now, $1\text{ MSD} = \frac{1}{20} \times 10\text{ mm} = 0.50\text{ mm}$

$LC = (0.50 - 0.45)\text{ mm} = 0.05\text{ mm}$

$= 5 \times 10^{-2}\text{ mm}$

Question124

A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then least count of the travelling microscope is ____ $\times 10^{-6}$m.
[26-Jun-2022-Shift-2]

Answer: 5

Solution:

$40M = 1\text{ cm}$
$\Rightarrow M = 0.025\text{ cm}\ldots\ldots\ldots(1)$
Also, $50V = 49M$
$\Rightarrow$ Least count $= M - V = M - \frac{49}{50}M = \frac{M}{50}$
$\Rightarrow LC = \frac{0.025}{50}\text{ cm}$
$= \frac{250}{50} \times 10^{-6}\text{m}$
$\Rightarrow LC = 5 \times 10^{-6}\text{m}$

Question125

A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 mm, 1.23 mm, 1.19 mm and 1.20 mm. The percentage error is $\frac{x}{121}$%. The value of x is
[28-Jun-2022-Shift-2]

Answer: 150

Solution:

$I_{mean} = \frac{1.22 + 1.23 + 1.19 + 1.20}{4} = 1.21$
$\Delta I_{mean} = \frac{0.01 + 0.02 + 0.02 + 0.01}{4} = 0.015$
So $\%I = \frac{\Delta I_{mean}}{I_{mean}} \times 100 = \frac{0.015}{1.21} \times 100 = \frac{150}{121}\%$
$x = 150$

Question126

The Vernier constant of Vernier callipers is 0.1 mm and it has zero error of (−0.05) cm. While measuring diameter of a sphere, the main scale reading is 1.7 cm and coinciding vernier division is 5 . The corrected diameter will be ____ $\times 10^{-2}$ cm
[29-Jun-2022-Shift-2]

Answer: 180

Solution:

Since zero error is negative, we will add 0.05 cm.
$\Rightarrow$ Corrected reading $= 1.7\text{ cm} + 5 \times 0.1\text{ mm} + 0.05\text{ cm}$
$= 180 \times 10^{-2}\text{ cm}$

Question127

If momentum [P], area [A] and time [T] are taken as fundamental quantities, then the dimensional formula for coefficient of viscosity is:
[25-Jul-2022-Shift-1]

Options:

A. $[PA^{-1}T^{0}]$

B. $[PAT^{-1}]$

C. $[PA^{-1}T]$

D. $[PA^{-1}T^{-1}]$

Answer: A

Solution:

$[\eta] = [M L^{-1} T^{-1}]$
Now if $[\eta] = [P]^a [A]^b [T]^c$
$\Rightarrow [M L^{-1} T^{-1}] = [M L^{1} T^{-1}]^a [L^2]^b [T]^c$
$\Rightarrow a = 1, a + 2b = -1, -a + c = -1$
$\Rightarrow a = 1, b = -1, c = 0$
$\Rightarrow [\eta] = [P][A]^{-1}[T]^0$
$= [PA^{-1}T^0]$

Question128

A torque meter is calibrated to reference standards of mass, length and time each with 5% accuracy. After calibration, the measured torque with this torque meter will have net accuracy of : [27-Jul-2022-Shift-1]

Options:

A. 15%

B. 25%

C. 75%

D. 5%

Answer: B

Solution:

Dimensional formula for Torque
$[\tau] = [ML^2T^{-2}]$
Now
Percentage error in torque $= \%\tau = \%M + 2\%L$
$2\%T$
$\%\tau = 25\%$

Question129

An expression of energy density is given by $u = \frac{\alpha}{\beta} \sin\left(\frac{\alpha x}{kt}\right)$, where α, β are constants, x is displacement, k is Boltzmann constant and t is the temperature. The dimensions of β will be :
[27-Jul-2022-Shift-2]

Options:

A. $[ML^2T^{-2}\theta^{-1}]$

B. $[M^0L^2T^{-2}]$

C. $[M^0L^0T^0]$

D. $[M^0L^2T^0]$

Answer: D

Solution:

$$u = \frac{\alpha}{\beta} \sin\left(\frac{\alpha x}{kt}\right)$$
$$[\alpha] = \left[\frac{kt}{x}\right] = \frac{[\text{ Energy }]}{[\text{ Distance}]}$$
$$[\beta] = \frac{[\alpha]}{[u]}$$
$$= \frac{[\text{ Energy }]/[\text{ Distance}]}{[\text{ Energy }]/\text{ Volume }]}$$
$$= [L^2]$$

Question130

The dimensions of $\left(\frac{B^2}{\mu_0}\right)$ will be :

(if μ_0 : permeability of free space and B : magnetic field)

[28-Jul-2022-Shift-1]

Options:

A. $[ML^2T^{-2}]$

B. $[MLT^{-2}]$

C. $[ML^{-1}T^{-2}]$

D. $[ML^2T^{-2}A^{-1}]$

Answer: C

Solution:

$$\left[\frac{B^2}{\mu_0}\right] = \text{[Energy density]} = \frac{ML^2T^{-2}}{L^3} = ML^{-1}T^{-2}$$

Question131

Consider the efficiency of carnot's engine is given by $\eta = \frac{\alpha\beta}{\sin\theta} \log e \frac{\beta x}{kT}$, where α and β are constants. If T is temperature, k is Boltzmann constant, θ is angular displacement and x has the dimensions of length. Then, choose the incorrect option:
[28-Jul-2022-Shift-2]

Options:

A. Dimensions of β is same as that of force.

B. Dimensions of $\alpha^{-1}x$ is same as that of energy.

C. Dimensions of $\eta^{-1}\sin\theta$ is same as that of $\alpha\beta$

D. Dimensions of α is same as that of β.

Answer: D

Solution:

(A) $[\beta] = \left[\frac{kT}{x}\right] = \left[\frac{E}{x}\right] = [MLT^{-2}] = [F]$

(B) $[\alpha\beta] = [M^0L^0T^0]$

$[\alpha]^{-1} = [\beta] = \left[\frac{kT}{x}\right]$

So $[\alpha]^{-1}[x] = [kT] = [ML^2T^{-2}]$
(C) $\eta \sin\theta = \alpha\beta$
So $[\eta \sin\theta] = [\alpha\beta]$
$[\eta] = [M^0L^0T^0]$ it is dimensionless quantity
(D) $[\alpha] \neq [\beta]$

Question132

Match List I with List II.

List I	List II
A.Torque	I. Nms^{-1}
B.Stress	II. Jkg^{-1}
C.Latent Heat	III.Nm
D.Power	IV. Nm^{-2}

Choose the correct answer from the options given below : [29-Jul-2022-Shift-2]

Options:

A. A-III, B-II, C-I, D-IV

B. A-III, B-IV, C-II, D-I

C. A-IV, B-I, C-III, D-II

D. A-II, B-III, C-I, D-IV

Answer: B

Solution:

Torque $\rightarrow Nm$
Stress $\rightarrow N/m^2$
Latent heat $\rightarrow J/kg$
Power $\rightarrow Nm/s$
A-III, B-IV, C-II, D-I

Question133

The maximum error in the measurement of resistance, current and time for which current flows in an electrical circuit are 1%, 2% and 3% respectively. The maximum percentage error in the detection of the dissipated heat will be :
[25-Jul-2022-Shift-2]

Options:

A. 2

B. 4

C. 6

D. 8

Answer: D

Solution:

$\because H = i^2Rt$
$\because$% error in $H = 2 \times 2\% + 1\% + 3\% = 8\%$

Question134

If $\vec{A} = \left(2\hat{i} + 3\hat{j} - \hat{k}\right)$ m and $\vec{B} = \left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$ m. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be m.
[26-Jul-2022-Shift-2]

Answer: 2

Solution:

$\vec{A} = (2\hat{i} + 3\hat{j} - \hat{k})$ m and $\vec{B} = (\hat{i} + 2\hat{j} + 2\hat{k})$ m

Component of $\vec{A}$ along $\vec{B} = \vec{A} \cdot \hat{B}$

$$= \vec{A} \cdot \frac{\vec{B}}{\left|\vec{B}\right| = \frac{2+6-2}{\sqrt{1^2+2^2+2^2}}}$$

$$= \frac{6}{3} = 2$$

Question135

The one division of main scale of Vernier callipers reads 1 mm and 10 divisions of Vernier scale is equal to the 9 divisions on main scale. When the two jaws of the instrument touch each other, the zero of the Vernier lies to the right of zero of the main scale and its fourth division coincides with a main scale division. When a spherical bob is tightly placed between the two jaws, the zero of the Vernier scale lies in between 4.1 cm and 4.2 cm and 6 th Vernier division coincides scale division. The diameter of the bob will be ________ $\times 10^{-2}$ cm. [27-Jul-2022-Shift-1]

Answer: 412

Solution:

$10 \text{ VSD} = 9 \text{ MSD}$

$1 \text{ VST} = .9 \text{ MSD}$

L.C. $= .1 \text{ mm} = .01 \text{ cm}$

+ ve zero error $= .4 \text{ mm}$

$= 0.04 \text{ cm}$

Negative zero error $= 4.1 \text{ cm} + 6 \times .01 \ = 4.12 \text{ cm}$

$= 412 \times 10^{-2} \text{ cm}$

Question136

If the projection of $2\hat{i} + 4\hat{j} - 2\hat{k}$ on $\hat{i} + 2\hat{j} + \alpha\hat{k}$ is zero. Then, the value of α will be ________.
[28-Jul-2022-Shift-1]

Answer: 5

Solution:

$$\vec{a} \cdot b = 0$$
$$\therefore \vec{a} \cdot \vec{b} = 0$$
$$\therefore 2 \times 1 + 4 \times 2 - 2 \times \alpha = 0$$
$$\therefore \alpha = 5$$

Question137

In an experiment to find acceleration due to gravity (g) using simple pendulum, time period of 0.5 s is measured from time of 100 oscillation with a watch of 1 s resolution. If measured value of length is 10 cm known to 1 mm accuracy, The accuracy in the determination of g is found to be x%. The value of x is________.
[28-Jul-2022-Shift-2]

Answer: 5

Solution:

$$T = 2\pi\sqrt{\frac{l}{g}}$$
$$\frac{dg}{g} \times 100 = \frac{2dT}{T} \times 100 + \frac{dl}{l} \times 100$$
$$= 2 \times \frac{1}{50} \times 100 + \frac{1}{100} \times 100 = 5\%$$

Question138

A screw gauge of pitch 0.5 mm is used to measure the diameter of uniform wire of length 6.8 cm, the main scale reading is 1.5 mm and circular scale reading is 7 . The calculated curved surface area of wire to appropriate significant figures is :
[Screw gauge has 50 divisions on its circular scale]
[26-Jul-2022-Shift-1]

Options:

A. $6.8 \mathrm{~cm}^{2}$

B. $3.4 \mathrm{~cm}^{2}$

C. $3.9 \mathrm{~cm}^{2}$

D. $2.4 \mathrm{~cm}^{2}$

Answer: B

Solution:

Least count $= \frac{0.5}{50} \mathrm{~mm} = 0.01 \mathrm{~mm}$
$\therefore$ Diameter, $d = 1.5 \mathrm{~mm} + 7 \times 0.01$
$= 1.57 \mathrm{~mm}$
$\therefore$ Surface area $= (2\pi r) \times l$
$= \pi d l$
$= 3.142 \times \frac{1.57}{10} \times 6.8 \mathrm{~cm}^{2}$
$= 3.354 \mathrm{~cm}^{2} = 3.4 \mathrm{~cm}^{2}$

Question139

In a Vernier Calipers, 10 divisions of Vernier scale is equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4^{th} Vernier scale division exactly coincides with

the main scale reading. One main scale division is equal to 1 mm. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6 $^{\text{th}}$ Vernier scale division exactly coincides with the main scale reading. The diameter of the spherical body will be :
[26-Jul-2022-Shift-2]

Options:

A. 3.02 cm

B. 3.06 cm

C. 3.10 cm

D. 3.20 cm

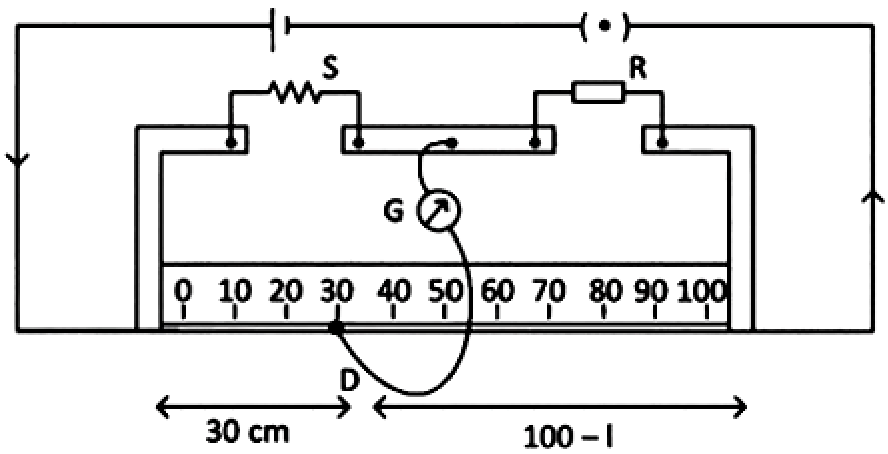
Answer: C

Solution:

$1 \text{ M.S.D } = 1 \text{ mm}$
$9 \text{ M.S.D } = 10 \text{V.S.D}$
$1 \text{ V.S.D } = 0.9 \text{ M.S.D } = 0.9 \text{ mm}$
$\text{L.C of vernier caliper } = 1 - 0.9 = 0.1 \text{ mm} = 0.01 \text{ cm}$
$\text{zero error } = (10 - 4) \times 0.1 \text{ mm} = -0.6 \text{ mm}$
$\text{Reading } = \text{ M.S.R } + \text{ V.S.R } - \text{ Zero error}$
$= 3 \text{ cm} + 6 \times 0.01 - [-0.06]$
$= 3 + 0.06 + 0.06$
$= 3.12 \text{ cm}$
Nearest given answer in the options is 3.10

Question140

In a meter bridge experiment, for measuring unknown resistance ' S ', the null point is obtained at a distance 30 cm from the left side as shown at point D. If R is 5.6kΩ, then the value of unknown resistance 'S' will be _______ Ω.



[27-Jul-2022-Shift-1]

Answer: 2400

Solution:

$$\frac{S}{30} = \frac{5.6 \times 10^3}{70}$$

$$S = \frac{3}{7} \times 5.6 \times 10^3 = 2400$$

Question141

In an experiment with a convex lens, The plot of the image distance (v') against the object distance (μ') measured from the focus gives a curve $v'\mu' = 225$. If all the distances are measured in cm. The magnitude of the focal length of the lens is ____cm
[28-Jul-2022-Shift-2]

Answer: 15

Solution:

$ru = f^2$ (by Newton's formula)
$f^2 = 225$
$f = 15$ cm

Question142

A travelling microscope has 20 divisions per cm on the main scale while its vernier scale has total 50 divisions and 25 vernier scale divisions are equal to 24 main scale divisions, what is the least count of the travelling microscope?
[29-Jul-2022-Shift-1]

Options:

A. 0.001 cm

B. 0.002 mm

C. 0.002 cm

D. 0.005 cm

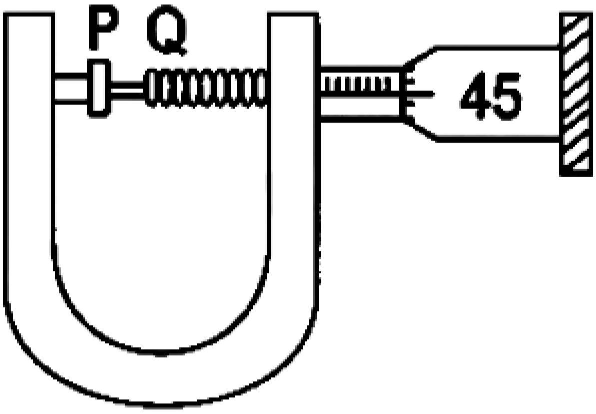
Answer: C

Solution:

$1 \text{ MSD} = \frac{1}{20} \text{ cm}$
$1 \text{ VSD} = \frac{24}{25} \text{ MSD} = \frac{24}{25} \times \frac{1}{20} \text{ cm}$
$\therefore$ Least count $= \frac{1}{20}\left(1 - \frac{24}{25}\right) \text{ cm}$
$= \frac{1}{20} \times \frac{1}{25} = \frac{1}{500} \text{ cm}$
$= 0.002$ cm

Question143

In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(A) Screw moves 0.5 mm on main scale in one complete rotation
(B) Total divisions on circular scale = 50
(C) Main scale reading is 2.5 mm
(D) 45 th division of circular scale is in the pitch line
(E) Instrument has 0.03 mm negative error
Then the diameter of wire is :
[29-Jul-2022-Shift-1]

Options:

A. 2.92 mm

B. 2.54 mm

C. 2.98 mm

D. 3.45 mm

Answer: C

Solution:

MSR = 2.5 mm

$$\text{CSR} = 45 \times \frac{0.5}{50} \text{ mm}$$

$= 0.45$ mm

Diameter reading = MSR + CSR − zero error

$= 2.5 + 0.45 - (-0.03)$

$= 2.98$ mm

Question144

The pitch of the screw gauge is 1mm and there are 100 divisions on the circular scale. When nothing is put in between the jaws, the zero of the circular scale lies 8 divisions below the reference line. When a wire is placed between the jaws, the first linear scale division is clearly visible while 72 nd division on circular scale coincides with the reference line. The radius of the wire is
[25 Feb 2021 Shift 1]

Options:

A. 1.64mm

B. 0.82mm

C. 1.80mm

D. 0.90mm

Answer: B

Solution:

Given, pitch of screw gauge, $\mathrm{P} = 1\mathrm{mm}$
Number of division, $\mathrm{n} = 100$
∴ Least count $(\mathrm{LC}) = \frac{\mathrm{P}}{\mathrm{n}} = 1/100 = 0.01\mathrm{mm}$
As, zero of circular division lies 8 divisions below.
∴ Zero error $= 8 \times \mathrm{LC} = 8 \times 0.01 = 0.08\mathrm{mm}$
Since, 1st linear scale division coinside with 72nd circular scale division.

$$\therefore \text{Rad ius(r)} = \frac{[\mathrm{P} + (72 \times \mathrm{LQ} - \text{ Zero error }]}{2}$$
$$= [1 + (72 \times 0.01) - 0.08]/2$$
$$= (1.72 - 0.08)/2 = 1.64/2$$
$$= 0.82\mathrm{mm}$$

Question145

A large number of water drops, each of radius r, combine to have a drop of radius R. If the surface tension is T and mechanical equivalent of heat is J, the rise in heat energy per unit volume will be
[26 Feb 2021 Shift 1]

Options:

A. $\frac{2T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

B. $\frac{2T}{r□}$

C. $\frac{3T}{rJ}$

D. $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

Answer: D

Solution:

Given, radius of small drop $= r$
Radius of big drop $= R$
Surface tension $= T$
and mechanical equivalent of heat $= J$
As, small drops combine to form big drop.
$\therefore$ Volume of big drop $(V_B) = n\times$ Volume of small drop (V_S)

$\Rightarrow \frac{4}{3}\pi R^3 = n\cdot\frac{4}{3}\pi r^3$

$\Rightarrow nr^3 = R^3$

$\Rightarrow r = \frac{R}{n^{1/3}}$. . . (i)

Surface energy of small drop $(E_S) =$ Surface tension $(T)\times$ Area (A)

$\Rightarrow E_S = n\times 4\pi r^2 T$ and $E_B = 4\pi R^2 T$

Now, change in energy will be

$\Delta E = E_B - E_S = 4\pi T(nr^2 - R^2)$

$\therefore$ Heat energy per unit volume $= \frac{\Delta E}{V} = \frac{4\pi T(nr^2-R^2)}{J\times\frac{4}{3}\pi R^3}$

$= \frac{3T}{J}\left(\frac{nr^2}{R^3}-\frac{1}{R}\right) = \frac{3T}{J}\left(n\frac{R^2}{n^{2/3}R^3}-\frac{1}{R}\right)$

$= \frac{3T}{J}\left[\frac{n^{1/3}}{R}-\frac{1}{R}\right]$ [from Eq. (i)]

$= \frac{3T}{J}\left[\frac{1}{r}-\frac{1}{R}\right]$

Question146

The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$.

Measured value of L is 1.0m from metre scale having a minimum division of 1mm and time of one complete oscillation is 1.95s measured from stopwatch of 0.01s resolution. The percentage error in the determination of g will be
[24 Feb 2021 Shift 2]

Options:

A. 1.13%

B. 1.03%

C. 1.33%

D. 1.30%

Answer: A

Solution:

Given, $T = 2\pi\sqrt{\frac{L}{g}}$ ... (i)
where, time period, $T = 1.95\text{s}$
Length of string, $I = 1\text{m}$
Acceleration due to gravity $= g$
Error in time period, $\Delta T = 0.01\text{s} = 10^{-2}\text{s}$
Error in length, $\Delta L = 1\text{mm} = 1 \times 10^{-3}\text{m}$
Squaring Eq. (i) on both sides, we get

$$T^2 = 4\pi^2 \frac{L}{g}$$

$$g = 4\pi^2 \frac{L}{T^2}$$

$$\Rightarrow \frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2\Delta T}{T} = \frac{10^{-3}}{1} + \frac{2 \times 10^{-2}}{1.95}$$

$$= 10^{-3} + 1.025 \times 10^{-2}$$

$$= 10^{-3} + 10.25 \times 10^{-3}$$

$$= 11.25 \times 10^{-3}$$

$$\because \Delta g/g \times 100 = 11.25 \times 10^{-3} \times 10^2$$

$$= 1.125\% = 1.13\%$$

Question147

If C and V represent capacity and voltage respectively, then what are the dimensions of λ, where $\frac{C}{V} = \lambda$?
[26 Feb 2021 Shift 2]

Options:

A. $[M^{-2}L^{-3}I^{2}T^{6}]$

B. $[M^{-3}L^{-4}I^{3}T^{7}]$

C. $[M^{-1}L^{-3}I^{-2}T^{-1}]$

D. $[M^{-2}L^{-4}I^{3}T^{7}]$

Answer: D

Solution:

Given, C and V represent capacity and voltage, respectively.
Dimensional formula of $[C] = [M^{-1}L^{-2}T^{4}I^{2}]$
Dimensional formula of $[V] = [ML^{2}T^{-3}I^{-1}]$
Therefore, dimensional formula of $[C/V] = \frac{[M^{-1}L^{-2}T^{4}I^{2}]}{[ML^{2}T^{-3}I^{-1}]}$
$= [M^{-2}L^{-4}T^{7}I^{3}]$

Question148

In a typical combustion engine, the work done by a gas molecule is given $W = \alpha^{2}\beta e^{\frac{-\beta x^{2}}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If α and β are constants, dimensions of α will be
[26 Feb 2021 Shift 1]

Options:

A. $[MLT^{-2}]$

B. $[M^{0}LT^{0}]$

C. $[M^2LT^{-2}]$

D. $[MLT^{-1}]$

Answer: B

Solution:

Given, work done by gas molecule,
$W = \alpha^2\beta e^{-\beta x^2/kT}$
Here, x is displacement, k is Boltzmann constant, α and β are constants and T is temperature.
Dimensional formula of $[W] = [ML^2T^{-2}]$
∴ Dimensions of $[\alpha^2\beta] = [ML^2T^{-2}]$
$\Rightarrow \alpha = \left[\frac{ML^2T^{-2}}{\beta}\right]^{1/2}$ . . . (i)
The term $\left[e^{-\beta x^2/kT}\right]$ should be dimensionless, i.e. $[M^0L^0T^0]$.
$\Rightarrow \left[\frac{\beta x^2}{kT}\right] = [M^0L^0T^0]$
$\Rightarrow [\beta] = \frac{[k][T]}{[x^2]}$ . . . (ii)
Energy of gaseous molecule $(E) = \frac{7}{2}kT$
$[k] = [E]/[T] = [ML^2T^{-2}K^{-1}]$
Substituting the value of k in Eq. (ii), we get
$[\beta] = \frac{[ML^2T^{-2}K^{-1}][K]}{[L^2]} = [MT^{-2}]$
Substituting the value of β in Eq. (i), we get
$[\alpha] = \left\{\frac{[ML^2T^{-2}]}{[MT^{-2}]}\right\}^{1/2} = [ML^1T^0]$

Question149

Match List-I with List-II

List-I	List-II
A. h (Planck's constant)	1. $[MLT^{-1}]$
B. E (kinetic energy)	2$[ML^2T^{-1}]$
C. V (electric potential)	3. $[ML^2T^{-2}]$
D. P (linear momentum)	4. $[ML^2I^{-1}T^{-3}]$

Choose the correct answer from the options given below.
[25 Feb 2021 Shift 1]

Options:

A. (A-3), (B-4), (C-2), (D-1)

B. (A-2), (B-3), (C-4), (D-1)

C. (A-1), (B-2), (C-4), (D-3)

D. (A-3), (B-2), (C-4), (D-1)

Answer: B

Solution:

The dimensional formulae of given terms are
Planck's constant (h) = $[M L^2 T^{-1}]$
Kinetic energy (E) = $[M L^2 T^{-2}]$
Electric potential (V) = $[M L^{2-11} T^{-3}]$
Linear momentum (p) = $[M L T^{-1}]$
So, the correct match is
$A \to 2$, $B \to 3$, $C \to 4$, $D \to 1$.

Question150

The work done by a gas molecule in an isolated system is given by, $W = \alpha\beta^2 e^{-\frac{x^2}{\alpha kT}}$, where x is the displacement k is the Boltzmann constant and T is the temperature, α and β are constants. Then the dimension of β will be :
24Feb2021

Options:

A. $[M L^2 T^{-2}]$

B. $[M LT^{-2}]$

C. $[M^2 LT^2]$

D. $[M^0 LT^0]$

Answer: B

Solution:

$\frac{x^2}{\alpha kT} \rightarrow$ dimensionless

$\Rightarrow [\alpha] = \frac{[x^2]}{[kT]} = \frac{L^2}{M L^2 T^{-2}} = M^{-1} T^2$

Now, $[W] = [\alpha][\beta]^2$

$[\beta]^2 = \frac{[W]}{[\alpha]}$

$[\beta] = \left(\frac{M L^2 T^{-2}}{M^{-1} T^2}\right)^{\frac{1}{2}} = M LT^{-2}$

Question151

Suppose you have taken a dilute solution of oleic acid in such a way that its concentration becomes 0.01cm^3 of oleic acid per cm^3 of the solution. Then, you make a thin film of this solution (monomolecular thickness) of area 4cm^2 by considering 100 spherical drops of radius $\left(\frac{3}{40\pi}\right)^{\frac{1}{3}} \times 10^{-3}$ cm^2. Then, the thickness of oleic acid layer will be

$x \times 10^{-14}$m, where x is
[17 Mar 2021 Shift 2]

Answer: 25

Solution:

Given, the radius of the spherical drop
$r = \left(\frac{3}{40\pi}\right)^{1/3} \times 10^{-3}$cm
The volume of 100 spherical drops, $V = 100 \times \frac{4\pi r^3}{3}$
Substituting the value of the radius in the above equation, we get
$V = 100 \times \frac{4\pi}{3} \times \frac{3}{40\pi} \times 10^{-9}$
$V = 10^{-8}cm^3$
Now, Volume of the film = Area of the film × Thickness of the film
⇒ Volume of the film $= 4 \times t$
The film is made up of 100 spherical drops.
∴ The volume of the film = Volume of 100 spherical drops
$4t = 10^{-8} \Rightarrow t = 25 \times 10^{-12}$m
The thickness of the oleic layer,
$t_0 = 0.01 \times t = 25 \times 10^{-14}$m
Comparing with $x \times 10^{-14}$, we get $x = 25$

Question152

In order to determine the Young's modulus of a wire of radius 0.2cm (measured using a scale of least count = 0.001cm) and length 1m (measured using a scale of least count = 1mm), a weight of mass 1kg (measured using a scale of least count = 1g) was hanged to get the elongation of 0.5cm (measured using a scale of least count 0.001cm). What will be the fractional error in the value of Young's modulus determined by this experiment? [16 Mar 2021 Shift 2]

Options:

A. 0.14%

B. 0.9%

C. 9%

D. 1.4%

Answer: D

Solution:

Young's modulus, $Y = \frac{\text{Stress}}{\text{Strain}} = \frac{FL}{Al}$

$\Rightarrow Y = \frac{mgL}{\pi R^2 I}$.....(i)

[$\because F = mg$ and $A = \pi R^2$]

To determine the fractional errors, we can write Eq. (i) as follows

$$\frac{\Delta Y}{Y} = \frac{\Delta m}{m} + \frac{\Delta L}{I} + \frac{2\cdot\Delta R}{R} + \frac{\Delta l}{I}$$

$$\Rightarrow \frac{\Delta Y}{Y} \times 100 = \frac{\Delta m}{m} \times 100 + \frac{\Delta L}{L} \times 100 + \frac{2\Delta R}{R} \times 100 + \frac{\Delta l}{l} \times 100$$

$$\Rightarrow \frac{\Delta Y}{Y} \times 100 = 100\left[\frac{\Delta m}{m} + \frac{\Delta L}{L} + \frac{2\Delta R}{R} + \frac{\Delta l}{I}\right]$$

$$= 100\left[\frac{1}{1000} + \frac{1}{1000} + 2\left(\frac{0.001}{0.2}\right) + \frac{0.001}{0.5}\right]$$

$$= \frac{1}{10} + \frac{1}{10} + 1 + \frac{1}{5} = \frac{14}{10} = 1.4\%$$

Question153

One main scale division of a vernier callipers is acm and nth division of the vernier scale coincide with (n − 1) th division of the main scal [16 Mar 2021 Shift 1]

Options:

A. The least count of the callipers (in mm) is

B. $\frac{10na}{(n-1)}$

C. $\frac{10a}{(n-1)}$

D. $\left(\frac{n-1}{10n}\right)a$

E. $\frac{10a}{n}$

Answer: E

Solution:

According to the question,
One division of main scale reading = a cm
nth vernier scale division $= (n-1)$ th main scale division
$\therefore$ One division of vernier scale reading $= \frac{(n-1) \times a}{n}$...(i)
We know that,
Least count $(LC) =$
[1 main scale division -1 vernier scale division] cm
$= a - \frac{(n-1)a}{n}$ [using Eq. (i)]
$= \frac{a(n-n+1)}{n} = \frac{a}{n} cm = \frac{a}{n} \times 10 mm$
$\Rightarrow LC = \frac{10a}{n} mm$

Question154

The resistance $R = \frac{V}{I}$, where $V = (50 \pm 2) V$ and $I = (20 \pm 0.2)$ A. The percentage error in R is x%. The value of x to the nearest integer is.........
[16 Mar 2021 Shift 1]

Answer: 5

Solution:

Given, $R = \frac{V}{I}$....(i)
where, $\& V = (50 \pm 2) V$
$I = (20 \pm 0.2) A$
From Eq. (i)

$\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 100 + \frac{\Delta l}{l} \times 100$

% error in $R = \left[\frac{2}{50} \times 100 + \frac{0.2}{20} \times 100 \right] \%$

$= [2 \times 2 + 0.2 \times 5]\%$

$= 5\%$

Question155

A person is swimming with a speed of 10m/s at an angle of 120° with the flow and reaches to a point directly opposite on the other side of the river. The speed of the flow is xm/s. The value of x to the nearest integer is ....
[18 Mar 2021 Shift 1]

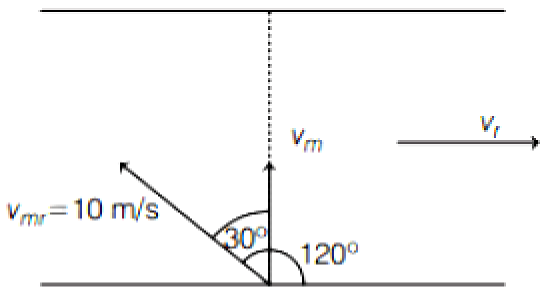
Answer: 5

Solution:

Given,

A person's swimming speed with respect to the river, $v_{mr} = 10 m/s$

As shown in the figure,



Let's determine the speed of the river flow, v_r

In X-direction,

$v_r = v_{mr} \sin 30^{\circ}$

$v_r = 10 \times \sin 30^{\circ}$

$v_r = 5 m/s$

Hence, the speed of the river flow is $5 m/s$. So, the value of the x to the nearest integer is 5 .

Question156

The vernier scale used for measurement has a positive zero error of 0.2mm. If while taking a measurement, it was noted that ' 0 ' on the vernier scale lies between 8.5cm and 8.6cm, vernier coincidence is 6 , then the correct value of measurement is .... cm.
[17 Mar 2021 Shift 1]

Options:

A. 8.36

B. 8.54

C. 8.58

D. 8.56

Answer: B

Solution:

Given, positive zero error $= 0.2\text{mm} = 0.02\text{cm}$
[∴ Least count $\text{LC} = 0.01\text{cm}$]
Main scale reading $= 8.5\text{cm}$
Vernier scale reading
= Vernier scale coincidence × Least count
$= 6 \times 0.01 = 0.06\text{cm}$
Final reading = Main scale reading + Vernier scale reading - Zero error
$= 8.5 + 0.06 - 0.02$
$= 8.54\text{cm}$

Question157

Assertion A : If in five complete rotations of the circular scale, the distance travelled on main scale of the screw gauge is 5 mm and there are 50 total divisions on circular scale, then least count is 0.001 cm.
Reason R :
Least Count $= \dfrac{\text{Pitch}}{\text{Total divisionson circular scale}}$
In the light of the above statements, choose the most appropriate answer from the options given below :
[27 Jul 2021 Shift 1]

Options:

A. A is not correct but R is correct.

B. Both A and R are correct and R is the correct explanation of A.

C. A is correct but R is not correct.

D. Both A and R are correct and R is NOT the correct explanation of A.

Answer: A

Solution:

Least Count $= \dfrac{\text{Pitch}}{\text{Total divisionson circular scale}}$

In 5 revolution, distance travel, 5 mm

In 1 revolution, it will travel 1 mm.

So least count $= \dfrac{1}{50} = 0.02$

Question158

The force is given in terms of time t and displacement x by the equation

$F = A\cos Bx + C\sin Dt$

The dimensional formula of $\frac{AD}{B}$ is :

[25 Jul 2021 Shift 2]

Options:

A. $[M^0LT^{-1}]$

B. $[ML^2T^{-3}]$

C. $[M^1L^1T^{-2}]$

D. $[M^2L^2T^{-3}]$

Answer: B

Solution:

$[A] = [MLT^{-2}]$
$[B] = [L^{-1}]$
$[D] = [T^{-1}]$
$\left[\frac{AD}{B}\right] = \frac{[MLT^{-2}][T^{-1}]}{[L^{-1}]}$
$\left[\frac{D}{B}\right] = [ML^{2}T^{-3}]$

Question159

If time (t), velocity (v), and angular momentum (l) are taken as the fundamental units. Then thedimension of mass (m) in terms of t, v and l is :
[20 Jul 2021 Shift 2]

Options:

A. $[t^{-1}v^{1}l^{-2}]$

B. $[t^{1}v^{2}l^{-1}]$

C. $[t^{-2}v^{-1}l^{1}]$

D. $[t^{-1}v^{-2}l^{1}]$

Answer: D

Solution:

$m \propto t^{a}v^{b}l^{c}$
$m \propto [T]^{a}[LT^{-1}]^{b}[ML^{2}T^{-1}]^{c}$
$M^{1}L^{0}T^{0} = M^{c}L^{b+2c}T^{a-b-c}$
comparing powers
$c = 1, b = -2, a = -1$
$m \propto t^{-1}v^{-2}l^{1}$

Question160

The entropy of any system is given by

$$S = \alpha^2 \beta \ln\left[\frac{\mu k R}{J \beta^2} + 3\right]$$

where α and beta are the constants. mu, J , k and R are no. of moles, mechanical equivalent of heat, Boltzmann constant and gas constant respectively.

$$\left[\text{Take } S = \frac{dQ}{T}\right]$$

Choose the incorrect option from the following :
[20 Jul 2021 Shift 1]

Options:

A. α and J have the same dimensions.

B. S, β, k and μR have the same dimensions.

C. S and α have different dimensions.

D. α and k have the same dimensions.

Answer: D

Solution:

$S = \alpha^2 \beta \ln\left(\frac{\mu K R}{J \beta^2} + 3\right)$

$S = \frac{Q}{T} =$ joulek /k

$[\alpha^2 \beta] =$ Joule /k

$PV = nRT \quad \left[\frac{\mu K R}{J \beta^2}\right] = 1$

$R = \frac{\text{Joule}}{K}$

$\Rightarrow R = \frac{\text{Joule}}{K}, K = \frac{\text{Joule}}{R}$

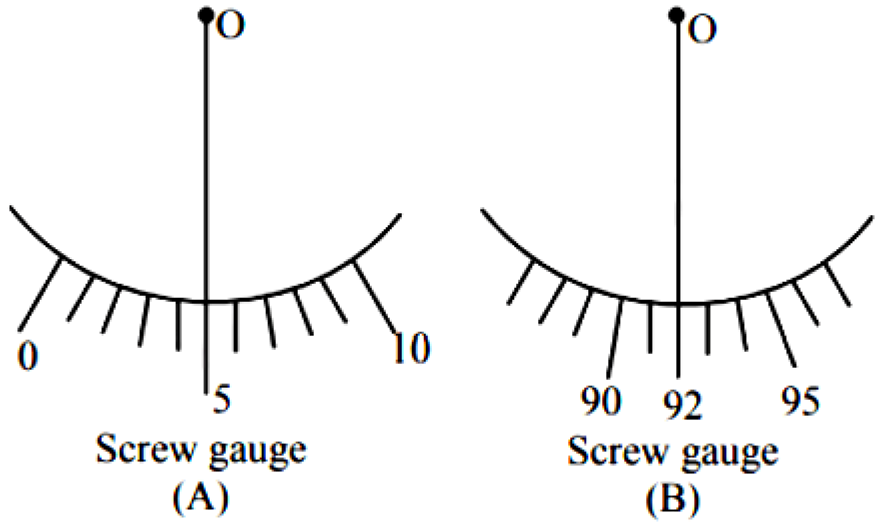
$\Rightarrow \beta = \left(\frac{\text{J oul e}}{K}\right)$

$\alpha^2 \beta = \left(\frac{J_0 \text{cl e}}{K}\right)$

$\Rightarrow \alpha =$ dimensionless

Question161

Student A and Student B used two screw gauges of equal pitch and 100 equal circular divisions to measure the radius of a given wire. The actual value of the radius of the wire is 0.322 cm. The absolute value of the difference between the final circular scale readings observed by the students A and B is ________.
[Figure shows position of reference 'O' when jaws of screw gauge are closed]
Given pitch = 0.1 cm.



[25 Jul 2021 Shift 1]

Answer: 13

Solution:

For (A)
Reading = MSR + CSR + Error
0.322 = 0.300 + CSR + 5 × LC
0.322 = 0.300 + CSR + 0.005
CSR = 0.017
For B
Reading = MSR + CSR + Error
0.322 = 0.200 + CSR + 0.092
CSR = 0.030
Difference = 0.030 – 0.017 = 0.013 cm
Division on circular scale $= \frac{0.013}{0.001} = 13$

Question162

Three students S_1, S_2 and S_3 perform an experiment for determining the acceleration due to gravity (g) using a simple pendulum. They use different lengths of pendulum and record time for different number of oscillations. The observation same as shown in the table.

Student No.	Length of pendulum (cm	No. of oscillations (n)	Total time for n oscillations	Time period (s)
1.	64.0	8	128.0	16.0
2.	64.0	4	64.0	16.0
3.	20.0	4	36.0	9.0

(Least count of length = 0.1 m
least count for time = 0.1 s)
If E_1, E_2 and E_3 are the percentage errors in 'g' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by student no._____.
[22 Jul 2021 Shift 2]

Answer: 1

Solution:

$$T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow g = \frac{4\pi^2 l}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$$

$$\Delta T = \frac{\text{least count of time }(\Delta T_0)}{\text{number of oscillations(n)}}$$

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T_0}{nT}$$

As Δl and ΔT_0 are same for all observations so $\frac{\Delta g}{g}$ is minimum for highest value of l, n and T

⇒Minimum percentage error in mathrm g is for student number-1

Question163

The diameter of a spherical bob is measured using a Vernier callipers. 9 divisions of the main scale, in the vernier calipers, are equal to 10 divisions of vernier scale. One main scale division is 1 mm. The main scale reading is 10 mm and 8thdivision of vernier scale was found to coincide exactly with one of the main scale division. If the given vernier callipers has positive zero error of 0.04 cm, then the radius of the bob is .... $\times 10^{-2}$ cm.
[31 Aug 2021 Shift 2]

Answer: 52

Solution:

Given, 9 divisions of main scale are equal to 10 divisions of Vernier scale.
i.e. 9 MSD = 10 VSD
$\Rightarrow \mathrm{VSD} = \frac{9}{10}\mathrm{MSD}$ ...(i)
Size of 1 main scale division,
1 MSD = 1 mm
Now, least count, LC = 1MSD − 1VSD ...(ii)
Using Eqs. (i) and (ii), we get
$\mathrm{LC} = 1\ \mathrm{MSD} - \frac{9}{10}\mathrm{MSD} = \frac{1}{10}\mathrm{MSD} = \frac{1}{10}\mathrm{mm}$
While measuring the diameter of bob.
Main Scale Reading, MSR = 10 mm
Vernier Scale Reading, VSR = 8
Zero error, e = 0.04 cm
Now, diameter, $\mathrm{d} = [\mathrm{MSR} + \mathrm{LC} \times \mathrm{VSR}] - \mathrm{e}$
$= \left(10\ \mathrm{mm} + \frac{1}{10} \times 8\ \mathrm{mm}\right) - 0.04\ \mathrm{cm}$
$= (10.8)\ \mathrm{mm} - 0.04\ \mathrm{cm}$
$= 1.08\ \mathrm{cm} - 0.04\ \mathrm{cm} = 1.04\ \mathrm{cm}$
Radius, $\mathrm{r} = \frac{\mathrm{d}}{2} = \frac{1.04}{2}\mathrm{cm} = 0.52\ \mathrm{cm} = 52 \times 10^{-2}\ \mathrm{cm}$
∴ Correct answer is 52.

Question164

If the length of the pendulum in pendulum clock increases by 0.1%, then the error in time per day is
[26 Aug 2021 Shift 2]

Options:

A. 86.4s

B. 4.32s

C. 43.2s

D. 8.64s

Answer: C

Solution:

Increase in length of pendulum is 0.1%.
i.e. $\frac{\Delta L}{L} \times 100 = 0.1$
Time period of pendulum is given by
$T = 2\pi\sqrt{\frac{L}{g}}$
Here, 2 π and g are constant.
$\frac{\Delta T}{T} \times 100 = \frac{1}{2}\frac{\Delta L}{L} \times 100$
$\Rightarrow \frac{\Delta T}{T} \times 100 = \frac{1}{2} \times 0.1$
$\Delta T = 0.05 \times \frac{T}{100}$
In one single day, the time in seconds is
$T = 24 \times 60 \times 60 = 86400s$
$\therefore \Delta T = 0.05 \times \frac{86400}{100} = 43.2s$
Thus, the error in time per day is 43.2 s.

Question165

In a screw gauge, 5th division of the circular scale coincides with the reference line when the ratchet is closed. There are 50 divisions on the circular scale, and the main scale moves by 0.5 mm on a complete rotation. For a particular observation the reading on the main scale is 5 mm and the 20th division of the circular scale

coincides with reference line. Calculate the true reading.
[26 Aug 2021 Shift 1]

Options:

A. 5.00 mm

B. 5.25 mm

C. 5.15 mm

D. 5.20 mm

Answer: C

Solution:

Least count of a screw gauge is given by

Least count (LC) = $\frac{\text{Pitch}}{\text{Number of circular scale divisions}}$

$\Rightarrow \text{LC} = \frac{0.5}{50}$ ...(i)

The actual reading (true reading)
= Main scale reading + Least count × Circular division − Zero error

$= 5 + \left(\frac{0.5}{50}\right) \times 20 - \frac{0.5}{50} \times 5$

[∴ Zero error = Least count × Coinciding division]

$= 5 + \frac{0.5}{50}(20 - 5)$

$= 5 + \frac{0.5}{50}(15)$

$= 5 + \frac{15}{100}$

$= \frac{515}{100} \text{ mm} = 5.15 \text{ mm}$

Question166

Which of the following equations is dimensionally incorrect? Where, t = time, h = height, s = surface tension, θ = angle, ρ = density, a, r = radius, g = acceleration due to gravity, V = volume, p = pressure, W = work done, τ = torque,ε = permittivity, E = electric field, J = current density, L = length.
[31 Aug 2021 Shift 1]

Options:

A. $V = \frac{\pi p a^4}{8\eta L}$

B. $h = \frac{2s\cos\theta}{\rho r g}$

C. $J = \varepsilon\frac{\partial E}{\partial t}$

D. $W = \tau\theta$

Answer: A

Solution:

As we know that,
Dimensional formula of volume $= [M^0L^3T^0]$ ...(i)
Since, $F = 6\pi\eta rv$
$\therefore \eta \propto \frac{F}{rv}$
where, η is viscosity and F is force.
$\therefore [\eta] = \frac{[MLT^{-2}]}{[L.LT^{-1}]} = [ML^{-1}T^{-1}]$
So, $\left[\frac{\pi p a^4}{8\eta L}\right] = \frac{[ML^{-1}T^{-2}][L^4]}{[ML^{-1}T^{-1}][L^1]}$
$= [M^0L^3T^{-1}]$ ...(ii)
Since, Eq. (i) is not equal to Eq (ii), so option (a) is wrong.
Now, since formula of capillary rise in tube, $h = \frac{2s\cos\theta}{\rho g r}$
Dimensional formula of LHS part,
∴ [h] = [L]
Dimensional formula of RHS part
$= \frac{[s]}{[\rho][g][r]} = \frac{[MT^{-2}]}{[ML^{-3}][LT^{-2}][L]} = [L]$
Hence, $h = \frac{2s\cos\theta}{\rho r g}$ is dimensionally correct.
So, option (b) will also be dimensionally correct.
In option (c),
$J = \varepsilon\frac{dE}{\partial t}$ ...(iii)
$\Rightarrow J = \varepsilon\frac{E}{t}$
Dimension of current density J is calculated as
Since, $J = \frac{I}{A}$
$\therefore [J] = \frac{[I]}{[A]} = \frac{[A]}{[L^2]}$
$\Rightarrow [J] = [AL^{-2}]$ ...(iv)
Again, we know that

$E = \frac{1}{4\pi\varepsilon} \cdot \frac{q}{r^2}$
$\Rightarrow \varepsilon E = \frac{1}{4\pi} \cdot \frac{q}{r^2}$
$\frac{\Rightarrow \varepsilon E}{t} = \frac{1}{4\pi} \cdot \frac{q}{tr^2}$
$\Rightarrow \left[\frac{\varepsilon E}{t}\right] = \frac{[q]}{[t][r^2]} = \frac{[AT]}{[T][L^2]}$
$\left[\frac{\varepsilon E}{t}\right] = [AL^{-2}]$ ...(v)
Form Eqs. (iv) and (v), we see that Eq. (iii) is dimensionally correct.
In option (d) W = τθ
and $\tau = r \times F$
So, dimensional formula of
$[\tau][\theta] = [r][F] = [L][MLT^{-2}] = [ML^2T^{-2}] = [W]$

Question167

If velocity [v], time [T] and force [F] are choosen as the base quantities, the dimensions of the mass will be [31 Aug 2021 Shift 2]

Options:

A. $[FT^{-1}v^{-1}]$

B. $[FTv^{-1}]$

C. $[FT^2v]$

D. $[FvT^{-1}]$

Answer: B

Solution:

When the velocity (v), time (T) and force (F) are chosen as base quantities. Then, mass is given by
$m \propto v^x T^y F^z$ ...(i)
Using dimensional formula of all quantities,
$[ML^0T^0] = [LT^{-1}]^x[T]^y[MLT^{-2}]^z$
$[M^1L^0T^0] = [M^Z L^{x+z} T^{-x+y-2z}]$
Comparing the powers of dimensions on both sides, we get
z = 1
$x + z = 0$ and $-x + y - 2z = 0$
$\Rightarrow x + 1 = 0$

$\Rightarrow x = -1$
and $-(-1) + y - 2(1) = 0$
$\Rightarrow 1 + y - 2 = 0$
$\Rightarrow y = 1$
Substituting these values in Eq. (i), we get
$m \propto v^{-1}T^{1}F^{1}$
$\Rightarrow m = [FTv^{-1}]$

Question168

If force (F), length (L) and time (T) are taken as the fundamental quantities. Then what will be the dimension of density: [27 Aug 2021 Shift 2]

Options:

A. $[FL^{-4}T^{2}]$

B. $[FL^{-3}T^{2}]$

C. $[FL^{-5}T^{2}]$

D. $[FL^{-3}T^{3}]$

Answer: A

Solution:

As we know that, the dimensional formula of
density $[D] = [ML^{-3}T^{0}]$
Since, dimensional formula of force
$[F] = [MLT^{-2}]$
Dimensional formula of length
$[L] = [M^{0}L^{1}T^{0}]$
Dimensional formula of time
$[T] = [M^{0}L^{0}T^{1}]$
$\therefore \ [D] = [F]^{a}[L^{b}[T]^{c}$
$\Rightarrow \ [ML^{-3}T^{0}] = [ML^{-2}]^{a}$
$[M^{0}L^{1}T^{0}]^{b}[M^{0}L^{0}T^{1}]^{c}$
$= [M^{a}L^{a+b}T^{-2a+c}]$
Comparing powers of dimensions on both sides, we get
$ca = 1$
$a + b = -3$
$\Rightarrow \ b = -3 - 1 = -4$
and $\ -2a + c = 0$

$c = 2a$
$\Rightarrow \; c = 2 \times 1 = 2$
∴ Dimensional formula of density will be
$[F^{1}L^{-4}T^{2}]$.

Question169

Which of the following is not a dimensionless quantity? [27 Aug 2021 Shift 1]

Options:

A. Relative magnetic permeability (μ_r)

B. Power factor

C. Permeability of free space (μ_0)

D. Quality factor

Answer: C

Solution:

Relative magnetic permeability (μ_r) It is the ratio of permeability of medium to the permeability of free space i.e., $\mu_r = \frac{\mu_m}{\mu_0}$
As, it is a ratio of permeabilities. So, it is unitless and dimensionless quantity.
Power factor $(\cos\phi)$ It is cosine of phase difference between alternating current and alternating voltage. Hence, it has only a numerical value, so it is also a unitless and dimensionless quantity.
Permeability of free space (μ_0) It is the ratio of magnetising field induction (B) to magnetising field intensity (H)
i.e. $\mu_0 = \frac{B}{H}$
$\Rightarrow [\mu_0] = \frac{[B]}{[H]}$ ...(i)
Since, $H = nI$
where, n = number of turns per unit length
and I = current
$\therefore [H] = [n][I]$
$= [L^{-1}][A] = [AL^{-1}]$
We know that, force on a current carrying conductor in magnetic field B is given as
F = IBl
where, l = length of the conductor,
B = magnetic field
and I = current.

$\therefore B = \frac{F}{Il}$

$\Rightarrow [B] = \frac{[F]}{[I][l]} = \frac{[MLT^{-2}]}{[A][L]} = [MT^{-2}A^{-1}]$

$\therefore$From Eq. (i), we have

$[\mu_0] = \frac{[MT^{-2}A^{-1}]}{[AL^{-1}]} = [MLT^{-2}A^{-2}]$

Hence, permeability of free space is not a dimensionless quantity.

Quality factor It is the ratio of energy stored to the energy dissipated per cycle.

Quality factor $= \frac{\text{Energy stored}}{\text{Energy dissipated per c ycle}}$

As, it is ratio of energies so, it will be unitless and dimensionless quantity.

Question170

Match List-I with List-II.

List - I	List - II
A. Magnetic induction	1.$[ML^2T^{-2}A^{-1}]$
B. Magnetic flux	2.$[ML^{-1}A]$
C. Magnetic permeability	3.$[MT^{-2}A^{-1}]$
D. Magnetisation	4.$[MLT^{-2}A^{-2}]$

Choose the most appropriate answer from the options given below. [26 Aug 2021 Shift 2]

Options:

A. A-2 B-4 C-1 D-3

B. A-2 B-1 C-4 D-3

C. A-3 B-2 C-4 D-1

D. None of the above

Answer: D

Solution:

Magnetic induction
Magnetic force of induction can be given as
$F = qvB$
$\Rightarrow B = \frac{F}{qv}$
$[B] = \frac{[F]}{[q][v]} = \frac{[MLT^{-2}]}{[AT][LT^{-1}]} = [M^1T^{-2}A^{-1}]$
Magnetic Flux Magnetic flux is the product of magnetic inductionand area.
$\phi = B \cdot A$
$[\phi] = [B][A] = [M^1T^{-2}A^{-1}][L^2]$
$= [ML^2T^{-2}A^{-1}]$
Magnetic permeability
We know that, magnetic field inside the solenoid.
$B = \mu_0 nI$
$\Rightarrow \mu_0 = \frac{B}{nI}$ (where, n is number of turns per unit length)
$\Rightarrow [\mu_0] = \frac{[B]}{[n][I]} = \frac{[MT^{-2}A^{-1}]}{[L^{-1}][A]} = [MLT^{-2}A^{-2}]$
Magnetisation The magnetic dipole moment acquired per unitvolume is known as magnetisation.
$I = \frac{M}{V}$
$[I] = \frac{[L^2A]}{[L^3]} = [L^{-1}A]$
Hence, no option is correct.

Question171

If E, L, M and G denote the quantities as energy, angular momentum, mass and constant of gravitation respectively, then the dimension of P in the formula $P = EL^2M^{-5}G^{-2}$ is [26 Aug 2021 Shift 1]

Options:

A. $[M^0L^1T^0]$

B. $[M^{-1}L^{-1}T^2]$

C. $[M^1L^1T^{-2}]$

D. $[M^0L^0T^0]$

Answer: D

Solution:

Dimension of energy, $E = [ML^2T^{-2}]$
Angular momentum, $L = [ML^2T^{-1}]$
Mass, M = [M]
and gravitational constant, $G = [M^{-1}L^3T^{-2}]$
The dimension of P in formula,
$P = EL^2M^{-5}G^{-2}$ is given as follows

$$[P] = \frac{[E][L^2]}{[M^5][C^2]}$$
$$= \frac{[ML^2T^{-2}] \times [M^2L^4T^{-2}]}{[M^5][M^{-2}L^6T^{-4}]}$$
$$= [M^0L^0T^0]$$

Question172

If E and H represent the intensity of electric field and magnetising field respectively, then the unit of E / H will be
[27 Aug 2021 Shift 1]

Options:

A. ohm

B. mho

C. joule

D. newton

Answer: A

Solution:

Unit of intensity of electric field E is Vm^{-1}.
Unit of intensity of magnetising field H is Am^{-1}.
Unit of E / H can be calculated as
$$\frac{\text{Unit of E}}{\text{Unit of H}} = \frac{Vm^{-1}}{Am^{-1}} = VA = \text{ohm.}$$
Thus, the unit of E / H will be ohm.

Question173

Match List-I with List-II.

List-I	List-II
A. R_H (Rydberg constant)	1. $kg\,m^{-1}\,s^{-1}$
B. h (Planck's constant)	2. $kg\,m^{2}\,s^{-1}$
C. μ_B (Magnetic field energy density)	3. m^{-1}
D. η (Coefficient of viscosity)	4. $kg\,m^{-1}\,s^{-2}$

Choose the most appropriate answer from the options given below. [27 Aug 2021 Shift 2]

Options:

A. A-2 B-3 C-4 D-1

B. A-3 B-2 C-4 D-1

C. A-4 B-2 C-1 D-3

D. A-3 B-2 C-1 D-4

Answer: B

Solution:

As we know that, SI unit of following terms are
(a) R_H (Rydberg constant) $= m^{-1}$
(b)h(Planck's constant)
$= J - s = kg - m^2 \cdot s^{-2} \cdot s = kg - m^2 s^{-1}$
(c) Magnetic field energy density
$(\mu_B) = \frac{\text{Energy}}{\text{Volume}}$
$= \frac{kg - m^2 s^{-2}}{m^3}$

$= kgm^{-1}s^{-2}$

(d) η (coefficient of viscosity)

$\because\ F = \eta A \frac{dv}{dx}$

$\therefore\ \eta = \frac{F\,dx}{A\,dv} = \frac{kg - ms^{-2} \cdot m}{m^2 \cdot ms^{-1}} = kgm^{-1}s^{-1}$

So, the correct match is $A - 3, B - 2, C - 4$ and D-1.

Question174

Two resistors $R_1 = (4 \pm 0.8)\Omega$ and $R_2 = (4 \pm 0.4)\Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be
[1 Sep 2021 Shift 2]

Options:

A. $(4 \pm 0.4)\Omega$

B. $(2 \pm 0.4)\Omega$

C. $(2 \pm 0.3)\Omega$

D. $(4 \pm 0.3)\Omega$

Answer: C

Solution:

Given,

$R_1 = (4 \pm 0.8)\Omega$

$R_2 = (4 \pm 0.4)\Omega$

Equivalent resistance when the resistors are connected in parallel isgiven by

$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$

$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{4} + \frac{1}{4}$

$R_{eq} = 2\Omega$

Now,

$\frac{\Delta R_{eq}}{R_{eq}^2} = \frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}$

Substituting the values in the above equation, we get

$\frac{\Delta R_{eq}}{4} = \frac{0.8}{16} + \frac{0.4}{16}$

$\Delta R_{eq} = 0.3\Omega$
∴ The equivalent resistance in parallel combination is
$R_{eq} = (2 \pm 0.3)\Omega$.

Question175

A student determined Young's modulus of elasticity using the formula $Y = \frac{MgL^3}{4bd^3\delta}$.

The value of g is taken to be $9.8 m/s^2$, without any significant error, his observations are as following.

Physical quantity	Least count of the equipment used for measurement	Observed value
Mass (M)	1 g	2 kg
Length of bar (L)	1mm	1m
Breadth of bar (b)	0.1 mm	4 cm
Thickness of bar (d)	0.01 mm	0.4 cm
Depression (δ)	0.01 mm	5 mm

Then, the fractional error in the measurement of Y is [1 Sep 2021 Shift 2]

Options:

A. 0.0083

B. 0.0155

C. 0.155

D. 0.083

Answer: B

Solution:

The given formula of Young's modulus of elasticity,
$$Y = \frac{mgL^3}{4bd^3\delta}$$
where, Y = Young's modulus of elasticity,
m = mass of the bar,
L = length of the bar

b = breadth of the bar
d = thickness of the bar
and δ = depression of the bar.
There is no error in the value of the g.
The fractional error in the measurement of Y,

$$\frac{\Delta Y}{Y}=\frac{\Delta M}{M}+3\frac{\Delta L}{L}+\frac{\Delta b}{b}+3\frac{\Delta d}{d}+\frac{\Delta\delta}{\delta}$$

Substituting the values in the above expression, we get

$$\frac{\Delta Y}{Y}=\frac{10^{-3}}{2}+3\frac{(1\times10^{-3})}{1}+\frac{0.1\times10^{-3}}{4\times10^{-2}}+3\frac{(0.01\times10^{-3})}{0.4\times10^{-2}}+\frac{(0.01\times10^{-3})}{5\times10^{-3}}$$

$$\frac{\Delta Y}{Y}=0.0155$$

The fractional error in the measurement of the Young's modulus is 0.0155.

Question176

If the screw on a screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is:
[9 Jan. 2020 I]

Options:

A. 0.001 cm

B. 0.02 mm

C. 0.01 cm

D. 0.001 mm

Answer: A

Solution:

When screw on a screw-gauge is given six rotations, it moves by 3mm on the main scale

$\therefore$ Pitch $=\frac{3}{6}=0.5\text{ mm}$

$\therefore$ Least count L.C. $=\frac{\text{Pitch}}{\text{CSD}}=\frac{0.5\text{ mm}}{50}$

$=\frac{1}{100}\text{mm}=0.01\text{ mm}=0.001\text{ cm}$

Question177

For the four sets of three measured physical quantities as given below. Which of the following options is correct?
(A) $A_1 = 24.36$, $B_1 = 0.0724$, $C_1 = 256.2$
(B) $A_2 = 24.44$, $B_2 = 16.082$, $C_2 = 240.2$
(C) $A_3 = 25.2$, $B_3 = 19.2812$, $C_3 = 236.183$
(D) $A_A = 25$, $B_A = 236.191$, $C_A = 19.5$
[9 Jan. 2020 II]

Options:

A.

$A_4 + B_4 + C_4 < A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2$

B.

$A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3 = A_4 + B_4 + C_4$

C.

$A_4 + B_4 + C_4 < A_1 + B_1 + C_1 = A_2 + B_2 + C_2 = A_3 + B_3 + C_3$

D. $A_1 + B_1 + C_1 < A_3 + B_3 + C_3 < A_2 + B_2 + C_2 < A_4 + B_4 + C_4$

E. None of the above

Answer: E

Solution:

$$
\begin{aligned}
D_1 &= A_1 + B_1 + C_1 = 24.36 + 0.0724 + 256.2 = 280.6 \\
D_2 &= A_2 + B_2 + C_2 = 24.44 + 16.082 + 240.2 = 280.7 \\
D_3 &= A_3 + B_3 + C_3 = 25.2 + 19.2812 + 236.183 = 280.7 \\
D_4 &= A_4 + B_4 + C_4 = 25 + 236.191 + 19.5 = 281
\end{aligned}
$$

Question178

A simple pendulum is being used to determine the value of gravitational acceleration g at a certain place. The length of the

pendulum is 25.0 cm and a stop watch with 1 s resolution measures the time taken for 40 oscillations to be 50 s. The accuracy in g is: [8 Jan. 2020 II]

Options:

A. 5.40%

B. 3.40%

C. 4.40%

D. 2.40%

Answer: C

Solution:

Given, Length of simple pendulum, l = 25.0 cm
Time of 40 oscillation, T = 50s

Time period of pendulum $T = 2\pi\sqrt{\frac{l}{g}}$

$\Rightarrow T^2 = \frac{4\pi^2 l}{g} \Rightarrow g = \frac{4\pi^2 l}{T^2}$

$\Rightarrow$ Fractional error in $g = \frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$

$\Rightarrow \frac{\Delta g}{g} = \left(\frac{0.1}{25.0}\right) + 2\left(\frac{1}{50}\right) = 0.044$

$\therefore$ Percentage error in $g = \frac{\Delta g}{g} \times 100 = 4.4\%$

Question179

Quantities The quantities $x = 1\sqrt{\mu_0\varepsilon_0}$, $y = \frac{E}{B}$ and $z = \frac{1}{CR}$ are defined where C -capacitance, R -Resistance, l -length, E -Electric field, B -magnetic field and ε_0, μ_0, − free space permittivity and permeability respectively. Then : [Sep. 05,2020 (II)]

Options:

A. x, y and z have the same dimension.

B. Only x and z have the same dimension.

C. Only x and y have the same dimension.

D. Only y and z have the same dimension.

Answer: A

Solution:

(a) We know that

Speed of light, $c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} = x$

Also, $c = \frac{E}{B} = y$

Time constant, $\tau = Rc = t$

$\therefore z = \frac{1}{Rc} = \frac{1}{t} =$ Speed

Thus, x, y, z will have the same dimension of speed.

Question180

Dimensional formula for thermal conductivity is (here K denotes the temperature :
[Sep. 04, 2020 (I)]

Options:

A. $MLT^{-2}K$

B. $MLT^{-2}K^{-2}$

C. $MLT^{-3}K$

D. $MLT^{-3}K^{-1}$

Answer: D

Solution:

From formula, $\frac{dQ}{dt} = kA\frac{dT}{dx}$

$$\Rightarrow k = \frac{\left(\frac{dQ}{dt}\right)}{A\left(\frac{dT}{dx}\right)}$$

$$[k] = \frac{[ML^2 \sim T^{-3}]}{[L^2][KL^{-1}]} = [MLT^{-3}K^{-1}]$$

Question181

A quantity x is given by $\left(\frac{IFv^2}{WL^4}\right)$ in terms of moment of inertia I , force F , velocity v, work W and Length L. The dimensional formula for x is same as that of : [Sep. 04, 2020 (II)]

Options:

A. planck's constant

B. force constant

C. energy density

D. coefficient of viscosity

Answer: C

Solution:

Dimension of Force $F = M^1L^1T^{-2}$

Dimension of velocity $V = L^1T^{-1}$

Dimension of work $= M^1L^2T^{-2}$

Dimension of length $= L$

Moment of inertia $= ML^2$

$$\therefore x = \frac{IFv^2}{WL^4}$$

$$= \frac{(M^1L^2)(M^1L^1T^{-2})(L^1T^{-2})^2}{(M^1L^2T^{-2})(L^4)}$$

$$= \frac{M^1L^{-2}T^{-2}}{L^3} = M^1L^{-1}T^{-2} = \text{ Energy density}$$

Question182

Amount of solar energy received on the earth's surface per unit area per unit time is defined a solar constant. Dimension of solar constant is :
[Sep. 03, 2020 (II)]

Options:

A. $M L^2 T^{-2}$

B. $M L^0 T^{-3}$

C. $M^2 L^0 T^{-1}$

D. $M L T^{-2}$

Answer: B

Solution:

Solar constant $= \frac{\text{Energy}}{\text{Time Area}}$
Dimension of Energy, $E = M L^2 T^{-2}$
Dimension of Time $= T$
Dimension of Area $= L^2$
$\therefore$ Dimension of Solar constant $= \frac{M^1 L^2 T^{-2}}{T L^2} = M^1 L^0 T^{-3}$

Question183

If speed V, area A and force F are chosen as fundamental units, then the dimension of Young's modulus will be :
[Sep. 02, 2020 (I)]

Options:

A. $F A^2 V^{-1}$

B. $F A^{2} V^{-3}$

C. $F A^{2} V^{-2}$

D. $F A^{-1} V^{0}$

Answer: D

Solution:

Young's modulus, $Y = \frac{\text{stress}}{\text{strain}}$
$\Rightarrow Y = \frac{F/A}{\Delta l_0 / l_0} = F A^{-1} V^{0}$

Question184

If momentum (P), area (A) and time (T) are taken to be the fundamental quantities then the dimensional formula for energy is : [Sep. 02, 2020 (II)]

Options:

A. $[P^{2} A T^{-2}]$

B. $[P A^{-1} T^{-2}]$

C. $\left[P A^{\frac{1}{2}} T^{-1}\right]$

D. $\left[P^{\frac{1}{2}} A T^{-1}\right]$

Answer: C

Solution:

Energy, $E \propto A^{a} T^{b} P^{c}$
Or, $E = k A^{a} T^{b} P^{c}$ ....(i)
where k is a dimensionless constant and a, b and c are the exponents.
Dimension of momentum, $P = M^{1} L^{1} T^{-1}$
Dimension of area, $A = L^{2}$

Dimension of time, $T = T^1$
Putting these value in equation (i), we get
$M^1L^2T^{-2} = M^cL^{2a+c}T^{b-c}$
by comparison
$c = 1$
$2a + c = 2$
$b - c = -2$
$c = 1, a = \frac{1}{2}, b = -1$
$\therefore E = A^{\frac{1}{2}}T^{-1}P^1$

Question185

A screw gauge has 50 divisions on its circular scale. The circular scale is 4 units ahead of the pitch scale marking, prior to use. Upon one complete rotation of the circular scale, a displacement of 0.5 mm is noticed on the pitch scale. The nature of zero error involved, and the least count of the screw gauge, are respectively : [Sep. 06, 2020 (I)]

Options:

A. Negative, 2 mm

B. Positive, 10 mm

C. Positive, 0.1 mm

D. Positive, 0.1 mm

Answer: B

Solution:

Given : No. of division on circular scale of screw gauge = 50
Pitch = 0.5 mm
Least count of screw gauge $= \frac{\text{Pitch}}{\text{No. of division on circular scale}}$
$= \frac{0.5}{50} \text{mm} = 1 \times 10^5 \text{m} = 10 \mu\text{m}$
And nature of zero error is positive.

Question186

The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)\%$. If the relative errors in measuring the mass and the diameter are 6.0% and 1.5% respectively, the value of x is ______.
[NA Sep. 06, 2020 (I)]

Answer: 1050

Solution:

Density, $\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi\left(\frac{D}{2}\right)^3} \Rightarrow \rho = \frac{6}{\pi} M D^{-3}$

$\therefore \%\left(\frac{\Delta\rho}{\rho}\right) = \frac{\Delta m}{m} + 3\left(\frac{\Delta D}{D}\right) = 6 + 3 \times 1.5 = 10.5\%$

$\%\left(\frac{\Delta\rho}{\rho}\right) = \frac{1050}{100}\% = \left(\frac{x}{100}\right)\%$

$\therefore x = 1050.00$

Question187

A student measuring the diameter of a pencil of circular cross-section with the help of a vernier scale records the following four readings 5.50 mm, 5.55 mm, 5.45 mm, 5.65 mm, The average of these four reading is 5.5375 mm and the standard deviation of the data is 0.07395 mm. The average diameter of the pencil should therefore be recorded as :
[Sep. 06, 2020 (II)]

Options:

A. (5.5375 ± 0.0739) mm

B. (5.5375 ± 0.0740) mm

C. (5.538 ± 0.074) mm

D. (5.54 ± 0.07) mm

Answer: D

Solution:

Average diameter, d_{av} = 5.5375 mm
Deviation of data, Δd = 0.07395 mm
As the measured data are upto two digits after decimal, therefore answer should be in two digits after decimal.
∴ d = (5.54 ± 0.07) mm

Question188

A physical quantity z depends on four observables a, b, c and d , as $z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c}\,d^3}$. The percentages of error in the measurement of a, b, c and d are 2%, 1.5%, 4% and 2.5% respectively. The percentage of error in z is :
[Sep. 05, 2020 (I)]

Options:

A. 12.25%

B. 16.5%

C. 13.5%

D. 14.5%

Answer: D

Solution:

Given : $z = \frac{a^2 b^{\frac{2}{3}}}{\sqrt{c}\, d^3}$

Percentage error in Z, $= \frac{\Delta Z}{Z} = \frac{2\Delta a}{a} + \frac{2}{3}\frac{\Delta b}{b} + \frac{1}{2}\frac{\Delta c}{c} + \frac{3\Delta d}{d}$

$= 2 \times 2 + \frac{2}{3} \times 1.5 + \frac{1}{2} \times 4 + 3 \times 2.5 = 14.5\%$

Question189

Using screw gauge of pitch 0.1 cm and 50 divisions on its circular scale, the thickness of an object is measured. It should correctly be recorded as :
[Sep. 03, 2020 (I)]

Options:

A. 2.121 cm

B. 2.124 cm

C. 2.125 cm

D. 2.123 cm

Answer: A

Solution:

Thickness = M.S. Reading + Circular Scale Reading (L.C.)

Here LC $= \frac{\text{Pitch}}{\text{Circular scale division}} = \frac{0.1}{50}$ = 0.002 cm per division

So, correct measurement is measurement of integral multiple of L.C.

Question190

The least count of the main scale of a vernier callipers is 1 mm. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main

scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 cm and 3.2 cm and 4th VSD coincides with a main scale division. The length of the cylinder is : (VSD is vernier scale division)
[Sep. 02, 2020 (I)]

Options:

A. 3.2 cm

B. 3.21 cm

C. 3.07 cm

D. 2.99 cm

Answer: C

Solution:

L.C. of vernier callipers = 1 MSD – 1 VSD $= \left(1 - \frac{9}{10}\right) \times 1 = 0.1 \text{ mm} = 0.01 \text{ cm}$

Here 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale.
Zero error = 7 × 0.1 = 0.7 mm = 0.07 cm.
Length of the cylinder = measured value – zero error = (3.1 + 4 × 0.01) – 0.07 = 3.07 cm.

Question191

The density of a material in SI unit is 128 kg m^{-3}. In certain units in which the unit of length is 25 cm and the unit of mass is 50g, the numerical value of density of the material is:
[10 Jan. 2019 I]

Options:

A. 40

B. 16

C. 640

D. 410

Answer: A

Solution:

Density of material in SI unit, $= 128\,\text{kg/m}^3$

Density of material in new system $= \dfrac{128(50\text{g})(20)}{(25\,\text{cm})^3(4)^3} = \dfrac{128}{64}(20) = 40$ units

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Question192

Let l , r, c and v represent inductance, resistance, capacitance and voltage, respectively. The dimension of $\frac{1}{\text{rcv}}$ in SI units will be:
[12 Jan. 2019 II]

Options:

A. $[\text{LA}^{-2}]$

B. $[\text{A}^{-1}]$

C. $[\text{LT A}]$

D. $[\text{LT}^{2}]$

Answer: B

Solution:

As we know, $\left[\frac{1}{\text{r}}\right] = [\text{T}]$ and $[\text{cv}] = [\text{AT}]$

$\therefore \left[\dfrac{1}{\text{rcv}}\right] = \left[\dfrac{\text{T}}{\text{AT}}\right] = [\text{A}^{-1}]$

Question193

The force of interaction between two atoms is given by

$\text{F} = \alpha\beta \exp\left(-\frac{x^2}{\alpha \text{kT}}\right)$; where x is the distance, k is the Boltzmann

constant and T is temperature and alpha and β are two constants. The dimensions of beta is:
[11 Jan. 2019 I]

Options:

A. $M^{0}L^{2}T^{-4}$

B. $M^{2}LT^{-4}$

C. MLT^{-2}

D. $M^{2}L^{2}T^{-2}$

Answer: B

Solution:

Force of interaction between two atoms, $F = \alpha\beta e^{\left(\frac{-x^2}{\alpha kT}\right)}$
Since exponential terms are dimensionless

$\therefore \left[\frac{x^2}{\alpha kT}\right] = M^0L^0T^0$

$\Rightarrow \frac{L^2}{[\alpha]MLT^{-2}} = M^0L^0T^0$

$\Rightarrow [\alpha] = M^{-1}T^2$

$[F] = [\alpha][\beta]$

$MLT^{-2} = M^{-1}T^2[\beta]$

$\Rightarrow [\beta] = M^2LT^{-4}$

Question194

If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of Young's modulus will be:
[11 Jan. 2019 II]

Options:

A. $V^{-2}A^{2}F^{-2}$

B. $V^{-2}A^{2}F^{2}$

C. $V^{-4}A^{-2}F$

D. $V^{-4}A^{2}F$

Answer: D

Solution:

Let $[Y] = [V]^a[F]^b[A]^c$
$[ML^{-1}T^{-2}] = [LT^{-1}]^a[MLT^{-2}]^b[LT^{-2}]^c$
$[ML^{-1}T^{-2}] = [M^bL^{a+b+c}T^{-a-2b-2c}]$
Comparing power both side of similar terms we get,
$b = 1, a + b + c = -1, -a - 2b - 2c = -2$
solving above equations we get:
$a = -4, b = 1, c = 2$
so $[Y] = [V^{-4}FA^2] = [V^{-4}A^2F]$

Question195

A quantity f is given by $f = \sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant and h is the Planck's constant. Dimension of f is that of :
[9 Jan. 2019 I]

Options:

A. area

B. energy

C. momentum

D. volume

Answer: B

Solution:

Dimension of $[h] = [ML^2T^{-1}]$
$[C] = [LT^{-1}]$

$[G] = [M^{-1}L^3T^{-2}]$

Hence dimension of $\left[\sqrt{\frac{hC^5}{G}}\right] = \frac{[ML^2T^{-1}].[L^5T^{-5}]}{[M^{-1}L^3T^{-2}]}$

$= [ML^2T^{-2}] =$ energy

Question196

Expression for time in terms of G (universal gravitational constant), h (Planck's constant) and c (speed of light) is proportional to: [9 Jan. 2019 II]

Options:

A. $\sqrt{\frac{hc^5}{G}}$

B. $\sqrt{\frac{c^3}{Gh}}$

C. $\sqrt{\frac{Gh}{c^5}}$

D. $\sqrt{\frac{Gh}{c^3}}$

Answer: C

Solution:

Let $t \propto G^x h^y C^z$

Dimensions of $G = [M^{-1}\sim L^3T^{-2}]$

$h = [ML^2T^{-1}]$ and $C = [LT^{-1}]$

$[T] = [M^{-1}L^3T^{-2}]^x[ML^2T^{-1}]^y[LT^{-1}]^z$

$[M^0L^0T^1] = [M^{-x+y}L^{3x+2y+z}T^{-2x-y-z}]$

By comparing the powers of M, L, T both the sides

$-x+y=0 \Rightarrow x=y$

$3x+2y+z=0 \Rightarrow 5x+z=0$ ......(i)

$-2x-y-z=1 \Rightarrow 3x+z=-1$ .....(ii)

Solving eqns. (i) and (ii),

$x = y = \frac{1}{2}, z = -\frac{5}{2}$

$\therefore t \propto \sqrt{\frac{Gh}{C^5}}$

Question197

The dimensions of stopping potential V_0 in photoelectric effect in units of Planck's constant ' h ', speed of light ' c ' and Gravitational constant ' G ' and ampere A is:
[8 Jan. 2019 I]

Options:

A. $h^{\frac{1}{3}}G^{\frac{2}{3}}c^{\frac{1}{3}}A^{-1}$

B. $h^{\frac{2}{3}}c^{\frac{5}{3}}G^{\frac{1}{3}}A^{-1}$

C. $h^{-\frac{2}{3}}e^{-\frac{1}{3}}G^{\frac{4}{3}}A^{-1}$

D. $h^0A^{-1}G^{-1}C^5$

Answer: D

Solution:

Stopping potential $(V_0) \propto h^xA^yG^ZC^r$

Here, $h =$ Planck's constant $= [ML^2T^{-1}]$

$I =$ current $= [A]$

$G =$ Gravitational constant $= [M^{-1}L^3T^{-2}]$

and $c =$ speed of light $= [LT^{-1}]$

$V_0 =$ potential $= [ML^2T^{-3}A^{-1}]$

$\therefore [ML^2T^{-3}A^{-1}] = [ML^2T^{-1}]^x[A]^y[M^{-1}L^3T^{-2}]^z[LT^{-1}]r$

$M^{x-z}; L^{2x+3z+r}; T^{-x-2z-r}; A^y$

Comparing dimension of M, L, T, A, we get

$y = -1, x = 0, z = -1, r = 5$

$\therefore V_0 \propto h^0A^{-1}G^{-1}C^5$

Question198

The dimensions of $\frac{B^2}{2\mu_0}$, where B is magnetic field and μ_0 is the magnetic permeability of vacuum, is:
[8 Jan. 2019 II]

Options:

A. MLT^{-2}

B. ML^2T^{-1}

C. ML^2T^{-2}

D. $ML^{-1}T^{-2}$

Answer: D

Solution:

The quantity $\frac{B^2}{2\mu_0}$ is the energy density of magnetic field

$$\Rightarrow \left[\frac{B^2}{2\mu_0}\right] = \frac{\text{Energy}}{\text{Volume}} = \frac{\text{Force} \times \text{displacement}}{(\text{displacement})^3} = \left[\frac{ML^2T^{-2}}{L^3}\right] = ML^{-1}T^{-2}$$

Question199

The least count of the main scale of a screw gauge is 1 mm. The minimum number of divisions on its circular scale required to measure 5 μm diameter of a wire is:
[12 Jan. 2019 I]

Options:

A. 50

B. 200

C. 100

D. 500

Answer: B

Solution:

Least count of main scale of screw gauge = 1 mm

Least count of screw gauge $= \dfrac{\text{Pitch}}{\text{Number of division on circular scale}}$

$$5 \times 10^{-6} = \frac{10^{-3}}{N}$$

$$\Rightarrow N = 200$$

Question200

The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm, respectively. What will be the value of its volume in appropriate significant figures? [10 Jan. 2019 II]

Options:

A. $4264 \pm 81\ \text{cm}^3$

B. $4264.4 \pm 81.0\ \text{cm}^3$

C. $4260 \pm 80\ \text{cm}^3$

D. $4300 \pm 80\ \text{cm}^3$

Answer: C

Solution:

$$v = \frac{\pi d^2}{4} h = 4260 \text{cm}^3$$

$$\frac{\Delta v}{v} = \frac{2\,\Delta\, d}{d} + \frac{\Delta h}{h}$$

$$\Delta v = 2 \times \frac{0.1 v}{12.6} + \frac{0.1 v}{34.2} = \frac{0.2}{126} \times 4260 + \frac{0.1 \times 4260}{34.2} = 80$$

$\therefore$ Volume $= 4260 \pm 80 \text{cm}^3.$

Question201

The pitch and the number of divisions, on the circular scale for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 division below the mean line.
The readings of the main scale and the circular scale, for a thin sheet, are 5.5 mm and 48 respectively, the thickness of the sheet is:
[9 Jan. 2019 II]

Options:

A. 5.755 mm

B. 5.950 mm

C. 5.725 mm

D. 5.740 mm

Answer: C

Solution:

Least count of screw gauge,

LC $= \dfrac{\text{Pitch}}{\text{No. of division}}$

$= 0.5 \times 10^{-3} \;\; = 0.5 \times 10^{-2}\,\text{mm} + \text{ve error}$

$= 3 \times 0.5 \times 10^{-2}\,\text{mm} \;\; = 1.5 \times 10^{-2}\,\text{mm} = 0.015\,\text{mm}$

Reading = MSR + CSR – (+ve error) $= 5.5\,\text{mm} + (48 \times 0.5 \times 10^{-2}) - 0.015$

= 5.5 + 0.24 – 0.015 = 5.725 mm

Question202

Which of the following combinations has the dimension of electrical resistance (ϵ_0 is the permittivity of vacuum and μ_0 is the permeability of vacuum)?
[12 April 2019 I]

Options:

A. $\sqrt{\frac{\mu_0}{\varepsilon_0}}$

B. $\frac{\mu_0}{\varepsilon_0}$

C. $\sqrt{\frac{\varepsilon_0}{\mu_0}}$

D. $\frac{\varepsilon_0}{\mu_0}$

Answer: A

Solution:

$$\sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{\mu_0^2}{\varepsilon_0\mu_0}} = \mu_0 c \left(\frac{1}{\sqrt{\mu_0\varepsilon_0}} = c\right)$$

$\mu_0 c \rightarrow M\,LT^{-2}A^{-2} \times LT^{-1}$

$M\,L^2T^{-3}A^{-2}$

Dimensions of resistance

Question203

In the formula $X = 5YZ^2$, X and Z have dimensions of capacitance and magnetic field, respectively. What are the dimensions of Y in SI units ?
[10 April 2019 II]

Options:

A. $[M^{-3}L^{-2}T^{8}A^{4}]$

B. $[M^{-1}L^{-2}T^{4}A^{2}]$

C. $[M^{-2}L^{0}T^{-4}A^{-2}]$

D. $[M^{-2}L^{-2}T^{6}A^{3}]$

Answer: A

Solution:

$X = 5YZ^2$
$\Rightarrow Y \propto \frac{X}{Z^2}$ ....(i)
$X = \text{Capacitance} = \frac{Q}{V} = \frac{Q^2}{W} = \frac{[A^2T^2]}{[ML^2T^{-2}]}$
$X = [M^{-1}L^{-2}T^4A^2]$
$Z = B = \frac{F}{IL}[\because F = ILB]$
$Z = [MT^{-2}A^{-1}]$
$Y = \frac{[M^{-1}L^{-2}T^4A^2]}{[MT^{-2}A^{-1}]^2}$
$Y = [M^{-3}L^{-2}T^8A^4]$ (Using (i))

Question204

In SI units, the dimensions of $\sqrt{\frac{\epsilon_0}{\mu_0}}$ is:

[8 April 2019 I]

Options:

A. $A^{-1}TML^3$

B. $AT^2M^{-1}L^{-1}$

C. $AT^{-3}ML^{3/2}$

D. $A^2T^3M^{-1}L^{-2}$

Answer: D

Solution:

$$\left[\sqrt{\frac{\varepsilon_0}{\mu_0}}\right] = \sqrt{\frac{\varepsilon_0{}^2}{\mu_0\varepsilon_0}} = \left[\frac{\varepsilon_0}{\sqrt{\mu_0\varepsilon_0}}\right] = \varepsilon_0 C[LT^{-1}] \times [\varepsilon_0]\left[\because \frac{1}{\sqrt{\mu_0\varepsilon_0}} = C\right]$$

$\because F = \frac{q^2}{4\pi\varepsilon_0 r^2}$

$$\Rightarrow[\varepsilon_0] = \frac{[AT]^2}{[MLT^{-2}]\times[L^2]} = [A^2M^{-1}L^{-3}T^4]$$

$$\therefore \left[\sqrt{\frac{\varepsilon_0}{\mu_0}}\right] = [LT^{-1}]\times[A^2M^{-1}L^{-3}T^4]$$

$$= [M^{-1}L^{-2}T^3A^2]$$

Question205

In the density measurement of a cube, the mass and edge length are measured as (10.00 ± 0.10) kg and (0.10 ± 0.01) m, respectively. The error in the measurement of density is:
[9 April 2019 I]

Options:

A. 0.01 kg/m^3

B. 0.10 kg/m^3

C. 0.013 kg/m^3

D. 0.31 kg/m^3

Answer: D

Solution:

$$\delta = \frac{M}{V} = \frac{M}{l^3} = Ml^{-3}$$

$$\frac{\Delta\delta}{\delta} = \frac{\Delta M}{M} + 3\frac{\Delta l}{l} = \frac{0.10}{10.00} + 3\left(\frac{0.01}{0.10}\right) = 0.31\ kg/m^3$$

Question206

The area of a square is 5.29 cm^2. The area of 7 such squares taking into account the significant figures is:
[9 April 2019 II]

Options:

A. 37 cm^2

B. 37.030 cm^2

C. 37.03 cm^2

D. 37.0 cm^2

Answer: D

Solution:

$A = 7 \times 5.29 = 37.03\ cm^2$
The result should have three significant figures, so $A = 37.0\ cm^2$

Question207

In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30 s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to : [8 April 2019 II]

Options:

A. 0.7%

B. 0.2%

C. 3.5%

D. 6.8%

Answer: D

Solution:

We have

$$T = 2\pi\sqrt{\frac{l}{g}} \text{ or } g = 4\pi^2\frac{l}{T^2}$$

$$\frac{\Delta g}{g}\times 100 = \frac{\Delta R}{Q}\times 100 + 2\frac{\Delta T}{T}\times 100$$

$$= \frac{0.1}{55}\times 100 + 2\left(\frac{1}{30}\right)\times 100$$

$$= 0.18 + 6.67 = 6.8\%$$

Question208

The characteristic distance at which quantum gravitational effects are significant, the Planck length, can be determined from a suitable combination of the fundamental physical constants G, h and c. Which of the following correctly gives the Planck length? [Online April 15, 2018]

Options:

A. G^2hc

B. $\left(\frac{Gh}{c^3}\right)^{\frac{1}{2}}$

C. $G^{\frac{1}{2}}h^2c$

D. Gh^2c^3

Answer: B

Solution:

Plank length is a unit of length, $l_p = 1.616229\times 10^{-35}$m

$$l_p = \sqrt{\frac{hG}{c^3}}$$

Question209

The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is:
[2018]

Options:

A. 2.5%

B. 3.5%

C. 4.5%

D. 6%

Answer: C

Solution:

1.5 % + 3 (1%) = 4.5%

Question210

The percentage errors in quantities P, Q, R and S are 0.5% 1%, 3% and 1.5% respectively in the measurement of aphysical quantity $A = \frac{P^3Q^2}{\sqrt{RS}}$.
The maximum percentage error in the value of A will be
[Online April 16, 2018]

Options:

A. 8.5%

B. 6.0%

C. 7.5%

D. 6.5%

Answer: D

Solution:

Maximum percentage error in A = + 3(% error in P) 2(% error in Q) + $\frac{1}{2}$(% error in R) + 1(% error in S)

$= 3 \times 0.5 + 2 \times 1 + \frac{1}{2} \times 3 + 1 \times 1.5$

$= 1.5 + 2 + 1.5 + 1.5 = 6.5\%$

Question211

The relative uncertainty in the period of a satellite orbiting around the earth is 10^{-2} . If the relative uncertainty in the radius of the orbit is negligible, the relative uncertainty in the mass of the earth is [Online April 16, 2018]

Options:

A. 3×10^{-2}

B. 10^{-2}

C. 2×10^{-2}

D. 6×10^{-2}

Answer: C

Solution:

From Kepler's law, time period of a satellite,

$T = 2\pi \sqrt{\frac{r^3}{Gm}} \Rightarrow T^2 = \frac{4\pi^2}{GM} r^3$

Relative uncertainty in the mass of the earth

$\left| \frac{\Delta M}{M} \right| = 2\frac{\Delta T}{T} = 2 \times 10^{-2}$

($\because 4\pi$ & G constant and relative uncertainty in radius $\frac{\Delta r}{r}$ negligible)

Question212

The relative error in the determination of the surface area of a sphere is a. Then the relative error in the determination of its volume is
[Online April 15, 2018]

Options:

A. $\frac{2}{3}\alpha$

B. 2α

C. $\frac{3}{2}\alpha$

D. α

Answer: C

Solution:

Relative error in Surface area, $\frac{\Delta s}{s} = 2 \times \frac{\Delta r}{r} = \alpha$ and relative error in volume, $\frac{\Delta v}{v} = 3 \times \frac{\Delta r}{r}$

$\therefore$ Relative error in volume w.r.t. relative error in area, $\frac{\Delta v}{v} = \frac{3}{2}\alpha$

Question213

In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm. There are 100 circular scale divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error, the thickness of the wire is: [Online April 15, 2018]

Options:

A. 0.0430 cm

B. 0.3150 cm

C. 0.4300 cm

D. 0.2150 cm

Answer: D

Solution:

Least count $= \dfrac{\text{Value of 1 part on main scale}}{\text{Number of parts on vernier scale}}$

$= \dfrac{0.25}{5 \times 100}\text{ cm} = 5 \times 10^{-4}\text{cm}$

Reading $= 4 \times 0.05\text{ cm} + 30 \times 5 \times 10^{-4}\text{ cm}$

$= (0.2 + 0.0150)\text{ cm} = 0.2150\text{ cm}$ (Thickness of wire)

Question214

Time (T), velocity (C) and angular momentum (h) are chosen as fundamental quantities instead of mass, length and time. In terms of these, the dimensions of mass would be : [Online April 8, 2017]

Options:

A. $[M] = [T^{-1}C^{-2}h]$

B. $[M] = [T^{-1}C^{2}h]$

C. $[M] = [T^{-1}C^{-2}h^{-1}]$

D. $[M] = [TC^{-2}h]$

Answer: A

Solution:

Let mass, related as $M \propto T^{x}C^{y}h^{z}$

$M^{1}L^{0}T^{0} = (T^{1})^{x}(L^{1}T^{-1})^{y}(M^{1}L^{2}T^{-1})^{z}$

$M^{1}L^{0}T^{0} = M^{z}L^{y+2z} + T^{x-y-z}$

$\therefore z = 1$

$\Rightarrow y + 2z = 0$

$\therefore y = -2$

$x - y - z = 0$

$x + 2 - 1 = 0$

$\therefore x = -1$
$M = [T^{-1}C^{-2}h^{1}]$

Question215

The following observations were taken for determining surface tensiton T of water by capillary method :
Diameter of capilary, $D = 1.25 \times 10^{-2}$m
rise of water, $h = 1.45 \times 10^{-2}$m
Using $g = 9.80 m/s^2$ and the simplified relation $T = \frac{rhg}{2} \times 10^3 N/m$, the possible error in surface tension is closest to :
[2017]

Options:

A. 2. 4 %

B. 10 %

C. 0.15%

D. 1.5%

Answer: D

Solution:

Surface tension, $T = \frac{rhg}{2} \times 10^3$
Relative error in surface tension,
$\frac{\Delta T}{T} = \frac{\Delta r}{r} + \frac{\Delta h}{h} + 0$ ($\because$ g, 2&10^3 are constant)
Percentage error
$100 \times \frac{\Delta T}{T} = \left(\frac{10^{-2} \times 0.01}{1.25 \times 10^{-2}} + \frac{10^{-2} \times 0.01}{1.45 \times 10^{-2}}\right) 100$
$= (0.8 + 0.689)$
$= (1.489) = 1.489\% \cong 1.5\%$

Question216

A physical quantity P is described by the relation $P = a^{\frac{1}{2}} b^2 c^3 d^{-4}$
If the relative errors in the measurement of a, b, c and d respectively, are 2%, 1%, 3% and 5%, then the relative error in P will be :
[Online April 9, 2017]

Options:

A. 8%

B. 12%

C. 32%

D. 25%

Answer: C

Solution:

Given, $P = a^{\frac{1}{2}} b^2 c^3 d^{-4}$
Maximum relative error,
$$\frac{\Delta P}{P} = \frac{1}{2}\frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 3\frac{\Delta c}{c} + 4\frac{\Delta d}{d}$$
$$= \frac{1}{2} \times 2 + 2 \times 1 + 3 \times 3 + 4 \times 5 = 32\%$$

Question217

A, B, C and D are four different physical quantities having different dimensions. None of them is dimensionless. But we know that the equation AD = C ln (BD) holds true. Then which of the combination is not a meaningful quantity ?
[Online April 10, 2016]

Options:

A. $\frac{C}{BD} - \frac{AD^2}{C}$

B. $A^2 - B^2C^2$

C. $AB - C$

D. $\frac{(A-C)}{D}$

Answer: D

Solution:

Dimension of A ≠ dimension of (C)
Hence A – C is not possible.

Question218

In the following 'I' refers to current and other symbols have their usual meaning, Choose the option that corresponds to the dimensions of electrical conductivity : [Online April 9, 2016]

Options:

A. $M^{-1}L^{-3}T^{3}I$

B. $M^{-1}L^{-3}T^{3}I^{2}$

C. $M^{-1}L^{3}T^{3}I$

D. $ML^{-3}T^{-3}I^{2}$

Answer: B

Solution:

We know that resistivity $\rho = \frac{RA}{l}$

Conductivity $= \frac{1}{\text{resistivity}} = \frac{l}{RA}$

$= \frac{lI}{VA} (\because V = RI)$

$= \frac{[L][I]}{\left[\frac{[ML^{2}T^{-2}]}{[I][T]}\right] \times [L^{2}]} \left[\because V = \frac{W}{q} = \frac{W}{it}\right]$

$= [M^{-1}L^{-3}T^{3}][I^{2}] = [M^{-1}L^{-3}T^{3}I^{2}]$

Question219

A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that wen the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line?
[2016]

Options:

A. 0.70 mm

B. 0.50 mm

C. 0.75 mm

D. 0.80 mm

Answer: D

Solution:

$L.C. = \frac{0.5}{50} = 0.01\ \text{mm}$
Zero error = 5 × 0.01 = 0.05 mm (Negative)
Reading = (0.5 + 25 × 0.01) + 0.05 = 0.80 mm

Question220

A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s, and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be:
[2016]

Options:

A. 92 ± 1.8 s

B. 92 ± 3s

C. 92 ± 1.5 s

D. 92 ± 5.0 s

Answer: C

Solution:

$$\Delta T = \frac{|\Delta T_1| + |\Delta T_2| + |\Delta T_3| + |\Delta T_4|}{4}$$
$$= \frac{2+1+3+0}{4} = 1.5$$

As the resolution of measuring clock is 1.5 therefore the mean time should be 92 ± 1.5

Question221

If electronic charge e, electron mass m, speed of light in vacuum c and Planck's constant h are taken as fundamental quantities, the permeability of vacuum μ_0 can be expressed in units of :

[Online April 11, 2015]

Options:

A. $\left(\frac{h}{me^2}\right)$

B. $\left(\frac{hc}{me^2}\right)$

C. $\left(\frac{h}{ce^2}\right)$

D. $\left(\frac{mc^2}{he^2}\right)$

Answer: C

Solution:

Let μ_0 related with e, m, c and h as follows.

$\mu_0 = ke^a m^b c^c h^d$

$[MLT^{-2}A^{-2}] = [AT]^a[M]^b[LT^{-1}]^c[ML^2T^{-1}]^d$

$= [M^{b+d}L^{c+2d}T^{a-c-d}A^a]$

On comparing both sides we get

a = – 2 ...(i)

b + d = 1 ...(ii)

c + 2d = 1 ...(iii)

a – c – d = –2 ...(iv)

By equation (i), (ii), (iii) & (iv) we get,

a = – 2, b = 0, c = – 1, d = 1

$\therefore [\mu_0] = \left[\frac{h}{ce^2}\right]$

Question222

If the capacitance of a nanocapacitor is measured in terms of a unit ‘u’ made by combining the electric charge ‘e’, Bohr radius ‘a_0’ , Planck’s constant ‘h’ and speed of light ‘c’ then: [Online April 10, 2015]

Options:

A. $u = \frac{e^2 h}{a_0}$

B. $u = \frac{hc}{e^2 a_0}$

C. $u = \frac{e^2 c}{h a_0}$

D. $u = \frac{e^2 a_0}{hc}$

Answer: D

Solution:

Let unit ' u ' related with e, a_0, h and c as follows.

$[u]=[e]^a[a_0]^b[h]^c[C]^d$

Using dimensional method,

$[M^{-1}L^{-2}T^{+4}A^{+2}]=[A^1T^1]^a[L]^b[ML2T^{-1}]^c[LT^{-1}]^d$

$[M^{-1}L^{-2}T^{+4}A^{+2}]=[M^cL^{b+2c+d}T^{a-c-d}A^a]$

$a=2, b=1, c=-1, d=-1$

$\therefore u=\dfrac{e^2a_0}{hc}$

Question223

The period of oscillation of a simple pendulum is $T=2\pi\sqrt{\frac{L}{g}}$.

Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1s resolution. The accuracy in the determination of g is : [2015]

Options:

A. 1%

B. 5%

C. 2%

D. 3%

Answer: D

Solution:

As, $g=4\pi^2\dfrac{L}{T^2}$

So, $\dfrac{\Delta g}{g}\times 100=\dfrac{\Delta L}{L}\times 100+2\dfrac{\Delta T}{T}\times 100$

$=\dfrac{0.1}{20}\times 100+2\times\dfrac{1}{90}\times 100=2.72\simeq 3\%$

Question224

Diameter of a steel ball is measured using a Vernier callipers which has divisions of 0.1 cm on its main scale (MS) and 10 divisions of its vernier scale (VS) match 9 divisions on the main scale. Three such measurements for a ball are given as:

S.No.	MS(cm)	VS divisions
1.	0.5	8
2.	0.5	4
3.	0.5	6

If the zero error is – 0.03 cm, then mean corrected diameter is: [Online April 10, 2015]

Options:

A. 0.52 cm

B. 0.59 cm

C. 0.56 cm

D. 0.53 cm

Answer: B

Solution:

Least count $= \frac{0.1}{10} = 0.01 \text{ cm}$

d_1 = 0.5 + 8 × 0.01 + 0.03 = 0.61 cm

d_2 = 0.5 + 4 × 0.01 + 0.03 = 0.57 cm

d_3 = 0.5 + 6 × 0.01 + 0.03 = 0.59 cm

Mean diameter $= \frac{0.61 + 0.57 + 0.59}{3}$ = 0.59 cm

Question225

From the following combinations of physical constants (expressed through their usual symbols) the only combination, that would have the same value in different systems of units, is:
[Online April 12, 2014]

Options:

A. $\frac{ch}{2\pi\varepsilon_o^{\,2}}$

B. $\frac{e^2}{2\pi\varepsilon_o G m_e^{\,2}}$ (m_e = mass of electron)

C. $\frac{\mu_0\varepsilon_0}{c^2}\frac{G}{he^2}$

D. $\frac{2\pi\sqrt{\mu_o\varepsilon_o}}{ce^2}\frac{h}{G}$

Answer: B

Solution:

The dimensional formulae of

$e = [M^0 L^0 T^1 A^1]$

$\varepsilon_0 = [M^{-1} L^3 T^4 A^2]$

$G = [M^{-1} L^3 T^{-2}]$ and $m_e = [M^1 L^0 T^0]$

Now, $\frac{e^2}{2\pi\varepsilon_0 G m_e^{\,2}}$

$$= \frac{[M^0 L^0 T^1 A^1]^2}{2\pi[M^{-1}L^{-3}T^4A^2][M^{-1}L^3T^{-2}][M^1L^0T^0]^2}$$

$$= \frac{[T^2A^2]}{2\pi[M^{-1-1+2}L^{-3+3}T^{4-2}A^2]}$$

$$= \frac{[T^2A^2]}{2\pi[M^0L^0T^2A^2]} = \frac{1}{2\pi}$$

$\because \frac{1}{2\pi}$ is dimensionless thus the combination $\frac{e^2}{2\pi\varepsilon_o G m_e^{\,2}}$

Question226

In terms of resistance R and time T, the dimensions of ratio $\frac{\mu}{\varepsilon}$ of the permeability m and permittivity ε is:
[Online April 11, 2014]

Options:

A. $[RT^{-2}]$

B. $[R^2T^{-1}]$

C. $[R^2]$

D. $[R^2T^2]$

Answer: C

Solution:

Dimensions of $\mu = [M LT^{-2}A^{-2}]$
Dimensions of $\varepsilon = [M^{-1}L^{-3}T^4A^2]$
Dimensions of $R = [M L^2T^{-3}A^{-2}]$
$\therefore \frac{\text{Dimensions of } \mu}{\text{Dimensions of } \varepsilon} = \frac{[M LT^{-2}A^{-2}]}{[M^{-1}L^{-3}T^4A^2]} = [M^2L^4T^{-6}A^{-4}] = [R^2]$

Question227

The current voltage relation of a diode is given by $I = (e^{1000V/T} - 1)$ mA, where the applied voltage V is in volts and the temperature T is in degree kelvin. If a student makes an error measuring ±0.01V while measuring the current of 5 mA at 300 K, what will be the error in the value of current in mA?
[2014]

Options:

A. 0.2 mA

B. 0.02 mA

C. 0.5 mA

D. 0.05 mA

Answer: A

Solution:

The current voltage relation of diode is
$I = (e^{1000V/T} - 1)\,mA$ (given)
When, $I = 5mA$, $e^{1000V/T} = 6\,mA$
Also, $dI = (e^{1000V/T}) \times \frac{1000}{T}$
Error $= \pm 0.01$ (By exponential function)
$= (6\,mA) \times \frac{1000}{300} \times (0.01) = 0.2\,mA$

Question228

A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it? [2014]

Options:

A. A meter scale.

B. A vernier calliper where the 10 divisions in vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 cm.

C. A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.

D. A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

Answer: B

Solution:

Measured length of rod = 3.50 cm
For Vernier Scale with 1 Main Scale Division = 1 mm
9 Main Scale Division = 10 Vernier Scale Division,
Least count = 1 MSD –1 VSD = 0.1 mm

Question229

Match List - I (Event) with List-II (Order of the time interval for happening of the event) and select the correct option from the options given below the lists:

List - I	List - II
(1) Rotation period of earth	(i) 10^5s
(2) Revolution period of earth	(ii) 10^7s
(3) Period of light wave	(iii) 10^{-15}s
(4) Period of sound wave	(iv) 10^{-3}s

[Online April 19, 2014]

Options:

A. (1)-(i), (2)-(ii), (3)-(iii), (4)-(iv)

B. (1)-(ii), (2)-(i), (3)-(iv), (4)-(iii)

C. (1)-(i), (2)-(ii), (3)-(iv), (4)-(iii)

D. (1)-(ii), (2)-(i), (3)-(iii), (4)-(iv)

Answer: A

Solution:

Rotation period of earth is about $24\,\text{hrs} \simeq 10^5\text{s}$
Revolution period of earth is about 365 days $\simeq 10^7\text{s}$
Speed of light wave $C = 3 \times 10^8 \text{m/s}$
Wavelength of visible light of spectrum $\lambda = 4000 - 7800\text{Å}$
$C = f\lambda \left(\text{and } T = \frac{1}{f}\right)$
Therefore period of light wave is 10^{-15}s (approx)

Question230

In the experiment of calibration of voltmeter, a standard cell of e.m.f. 1.1 volt is balanced against 440 cm of potential wire. The potential difference across the ends of resistance is found to balance against 220 cm of the wire. The corresponding reading of voltmeter is 0.5 volt. The error in the reading of voltmeter will be:
[Online April 12, 2014]

Options:

A. – 0. 15 volt

B. 0.15 volt

C. 0.5 volt

D. – 0.05 volt

Answer: D

Solution:

In a voltmeter
$V \propto l$
$V = kl$
Now, it is given $E = 1.1$ volt for $l_1 = 440\,\text{cm}$ and $V = 0.5$ volt for $l_2 = 220\,\text{cm}$
Let the error in reading of voltmeter be ΔV then,
$1.1 = 400K$ and $(0.5 - \Delta V) = 220K$
$\Rightarrow \frac{1.1}{440} = \frac{0.5 - \Delta V}{220}$
$\therefore \Delta V = -0.05\,\text{volt}$

Question231

An experiment is performed to obtain the value of acceleration due to gravity g by using a simple pendulum of length L. In this experiment time for 100 oscillations is measured by using a watch of 1 second least count and the value is 90.0 seconds. The length L is measured by using a meter scale of least count 1 mm and the value is 20.0 cm. The error in the determination of g would be:
[Online April 9, 2014]

Options:

A. 1.7%

B. 2.7%

C. 4.4%

D. 2.27%

Answer: B

Solution:

According to the question.
$t = (90 \pm 1)$ or, $\frac{\Delta t}{t} = \frac{1}{90}$
$l = (20 \pm 0.1)$ or, $\frac{\Delta l}{l} = \frac{0.1}{20}$
$\frac{\Delta g}{g}\% = ?$
As we know,
$t = 2\pi\sqrt{\frac{l}{g}} \Rightarrow g = \frac{4\pi^2 l}{t^2}$
or, $\frac{\Delta g}{g} = \pm\left(\frac{\Delta l}{l} + 2\frac{\Delta t}{t}\right) = \left(\frac{0.1}{20} + 2 \times \frac{1}{90}\right) = 0.027$
$\therefore \frac{\Delta g}{g}\% = 2.7\%$

Question232

A metal sample carrying a current along X -axis with density J_x is subjected to a magnetic field B_z (along z-axis). The electric field E_y developed along Y-axis is directly proportional to J_x as well as B_z. The constant of proportionality has SI unit [Online April 25, 2013]

Options:

A. $\frac{m^2}{A}$

B. $\frac{m^3}{As}$

C. $\frac{m^2}{As}$

D. $\frac{As}{m^3}$

Answer: B

Solution:

(b) According to question $E_y \propto J_x B_Z$ ∴ Constant of proportionality

$K = \frac{E_y}{B_Z J_x} = \frac{C}{J_x} = \frac{m^3}{As}$

[As $\frac{E}{B} = C($ speed of light $)$ and $J = I$ Area]

Question233

Let $\in_0$ denote the dimensional formula of the permittivity of vacuum. If M = mass, L = length, T = time and A = electric current, then:
[2013]

Options:

A. $\in_0 = [M^{-1}L^{-3}T^2A]$

B. $\in_0 = [M^{-1}L^{-3}T^4A^2]$

C. $\in_0 = [M^1L^2T^1A^2]$

D. $\in_0 = [M^1L^2T^1A]$

Answer: B

Solution:

As we know, $F = \frac{1}{4\pi\epsilon_0}\frac{q_1q_2}{R^2}$

$\Rightarrow \epsilon_0 = \frac{q_1q_2}{4\pi F R^2}$

Hence, $\epsilon_0 = \frac{C^2}{N.m^2} = \frac{[AT]^2}{[MLT^{-2}][L^2]}$

$\Rightarrow \epsilon_0 = [M^{-1}L^{-3}T^4A^2]$

Question234

If the time period t of the oscillation of a drop of liquid of density d , radius r, vibrating under surface tension s is given by the formula $t = \sqrt{r^{2b}s^c d^{a/2}}$. It is observed that the time period is directly proportional to $\sqrt{\frac{d}{s}}$. The value of b should therefore be:

[Online April 23, 2013]

Options:

A. $\frac{3}{4}$

B. $\sqrt{3}$

C. $\frac{3}{2}$

D. $\frac{2}{3}$

Answer: C

Solution:

Question235

The dimensions of angular momentum, latent heat and capacitance are, respectively.

[Online April 22, 2013]

Options:

A. $M L^{2}T^{1}A^{2}, L^{2}T^{-2}, M^{-1}L^{-2}T^{2}$

B. $M L^{2}T^{-2}, L^{2}T^{2}, M^{-1}L^{-2}T^{4}A^{2}$

C. $M L^{2}T^{-1}, L^{2}T^{-2}, M L^{2}T A^{2}$

D. $M L^{2}T^{-1}, L^{2}T^{-2}, M^{-1}L^{-2}T^{4}A^{2}$

Answer: D

Solution:

Angular momentum $= m \times v \times r = M L^{2}T^{-1}$

Latent heat $L = \frac{Q}{m} = \frac{M L^{2}T^{-2}}{M} = L^{2}T^{-2}$

Capacitance $C = \frac{\text{Charge}}{\text{P . d}} = M^{-1}L^{-2}T^{4}A^{2}$

Question236

Given that K = energy, V = velocity, T = time. If they are chosen as the fundamental units, then what is dimensional formula for surface tension?
[Online May 7, 2012]

Options:

A. $[K V^{-2}T^{-2}]$

B. $[K^{2}V^{2}T^{-2}]$

C. $[K^{2}V^{-2}T^{-2}]$

D. $[K V^{2}T^{2}]$

Answer: A

Solution:

Surface tension, $T = \frac{F}{l} = \frac{F}{l} \cdot \frac{l}{l} \cdot \frac{T^2}{T^2}$

(As, $F \cdot l = K$ (energy) ; $\frac{T^2}{l^2} = V^{-2}$)

Therefore, surface tension $= [K V^{-2} T^{-2}]$

Question237

Resistance of a given wire is obtained by measuring the current flowing in it and the voltage difference applied across it. If the percentage errors in the measurement of the current and the voltage difference are 3% each, then error in the value of resistance of the wire is
[2012]

Options:

A. 6%

B. zero

C. 1%

D. 3%

Answer: A

Solution:

According to ohm's law, $V = IR$

$R = \frac{V}{I}$

$\therefore$ Percentage error $= \frac{\text{Absolute error}}{\text{Measurement}} \times 10^2$

where, $\frac{\Delta V}{V} \times 100 = \frac{\Delta I}{I} \times 100 = 3\%$

then, $\frac{\Delta R}{R} \times 100 = \frac{\Delta V}{V} \times 10^2 + \frac{\Delta I}{I} \times 10^2$

$= 3\% + 3\% = 6\%$

Question238

A spectrometer gives the following reading when used to measure the angle of a prism.
Main scale reading : 58.5 degree
Vernier scale reading : 09 divisions
Given that 1 division on main scale corresponds to 0.5 degree. Total divisions on the Vernier scale is 30 and match with 29 divisions of the main scale. The angle of the prism from the above data is
[2012]

Options:

A. 58.59 degree

B. 58.77 degree

C. 58.65 degree

D. 59 degree

Answer: C

Solution:

$\because$ Reading of Vernier = Main scale reading + Vernier scale reading × least count.
Main scale reading = 58.5
Vernier scale reading = 09 division
least count of Vernier = $\frac{0.5°}{30}$
Thus, $R = 58.5° + 9 \times \frac{0.5°}{30} = 58.65°$

Question239

N divisions on the main scale of a vernier calliper coincide with (N + 1) divisions of the vernier scale. If each division of main scale is 'a' units, then the least count of the instrument is [Online May 19, 2012]

Options:

A. a

B. $\frac{a}{N}$

C. $\frac{N}{N+1} \times a$

D. $\frac{a}{N+1}$

Answer: D

Solution:

No. of divisions on main scale = N
No. of divisions on vernier scale = N + 1
size of main scale division = a
Let size of vernier scale division be b
then we have

$aN = b(N+1) \Rightarrow b = \frac{aN}{N+1}$

Least count is $a - b = a - \frac{aN}{N+1}$

$= a\left[\frac{N+1-N}{N+1}\right] = \frac{a}{N+1}$

Question240

A student measured the diameter of a wire using a screw gauge with the least count 0.001 cm and listed the measurements. The measured value should be recorded as
[Online May 12, 2012]

Options:

A. 5.3200 cm

B. 5.3 cm

C. 5.32 cm

D. 5.320 cm

Answer: D

Solution:

The least count (L.C.) of a screw guage is the smallest length which can be measured accurately with it.

As least count is 0.001 cm $= \frac{1}{1000}$ cm

Hence measured value should be recorded upto 3 decimal places i.e., 5.320 cm

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Question241

A screw gauge gives the following reading when used to measure the diameter of a wire.
Main scale reading : 0 mm
Circular scale reading : 52 divisions
Given that 1mm on main scale corresponds to 100 divisions of the circular scale. The diameter of wire from the above data is
[2011]

Options:

A. 0.052 cm

B. 0.026 cm

C. 0.005 cm

D. 0.52 cm

Answer: A

Solution:

Least count, L.C. $= \frac{1}{100}$ mm

Diameter of wire = MSR + CSR × L.C.

∵ 1 mm = 0.1 cm $= 0 + \frac{1}{100} \times 52$ = 0.52 mm = 0.052 cm

----

Question242

The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are

[2010]

Options:

A. 5, 1, 2

B. 5, 1, 5

C. 5, 5, 2

D. 4, 4, 2

Answer: A

Solution:

Number of significant figures in 23.023 = 5
Number of significant figures in 0.0003 = 1
Number of significant figures in $2.1 \times 10^{-3} = 2$
So, the radiation belongs to X-rays part of the spectrum

Question243

In an experiment the angles are required to be measured using an instrument, 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half- a degree (= 0.5°), then the least count of the instrument is:
[2009]

Options:

A. half minute

B. one degree

C. half degree

D. one minute

Answer: D

Solution:

30 Divisions of V.S. coincide with 29 divisions of M.S.

∴ 1 V.S.D $= \frac{29}{30}$ MSD

L.C. = 1 MSD – 1VSD = 1 MSD - $\frac{29}{30}$ MSD

$= \frac{1}{30}\text{MSD} = \frac{1}{30} \times 0.5° = 1$ minute

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Question244

The dimensions of magnetic field in M, L, T and C (coulomb) is given as
[2008]

Options:

A. $[M LT^{-1}C^{-1}]$

B. $[M T^{2}C^{-2}]$

C. $[M T^{-1}C^{-1}]$

D. $[M T^{-2}C^{-1}]$

Answer: C

Solution:

Magnitude of Lorentz formula $F = qvB\sin\theta$

$B = \frac{F}{qv} = \frac{M LT^{-2}}{C \times LT^{-1}} = [M T^{-1}C^{-1}]$

Question245

A body of mass m = 3.513 kg is moving along the x-axis with a speed of 5.00 ms^{-1}. The magnitude of its momentum is recorded as
[2008]

Options:

A. 17.6 kg ms^{-1}

B. 17.565 kg ms^{-1}

C. 17.56 kg ms^{-1}

D. 17.57 kg ms^{-1}

Answer: A

Solution:

Momentum, p = m × v
Given, mass of a body = 3.513 kg speed of body
= (3.513) × (5.00) = 17.565 kg m/s
= 17.6 (Rounding off to get three significant figures)

Question246

Two full turns of the circular scale of a screw gauge cover a distance of 1mm on its main scale. The total number of divisions on the circular scale is 50. Further, it is found that the screw gauge has a zero error of – 0.03 mm. While measuring the diameter of a thin wire, a student notes the main scale reading of 3 mm and the number of circular scale divisions in line with the main scale as 35. The diameter of the wire is [2008]

Options:

A. 3.32 mm

B. 3.73 mm

C. 3.67 mm

D. 3.38 mm

Answer: D

Solution:

Least count of screw gauge = 0.01 mm
$\because \frac{0.5}{50}$ mm
Reading = [M.S.R. + C.S.R. × L.C.] – (zero error)
= [3 + 35 × 0.01] – (–0.03) = 3.38 mm

Question247

Which of the following units denotes the dimension $\frac{ML^2}{Q^2}$, where Q denotes the electric charge?
[2006]

Options:

A. $\frac{Wb}{m^2}$

B. Henry (H)

C. $\frac{H}{m^2}$

D. Weber (Wb)

Answer: B

Solution:

Mutual inductance $= \frac{\phi}{I} = \frac{BA}{I}$

[Henry] $= \frac{[MT^{-1}Q^{-1}L^2]}{[QT^{-1}]} = ML^2Q^{-2}$

Question248

Out of the following pair, which one does NOT have identical dimensions ?
[2005]

Options:

A. Impulse and momentum

B. Angular momentum and planck's constant

C. Work and torque

D. Moment of inertia and moment of a force

Answer: D

Solution:

Moment of Inertia, $I = MR^2$
$[I] = [ML^2]$
Moment of force, $\vec{\tau} = \vec{r} \times \vec{F}$
$\vec{\tau} = [L][MLT^{-2}] = [ML^2T^{-2}]$

Question249

Which one of the following represents the correct dimensions of the coefficient of viscosity? [2004]

Options:

A. $[ML^{-1}T^{-1}]$

B. $[MLT^{-1}]$

C. $[ML^{-1}T^{-2}]$

D. $[ML^{-2}T^{-2}]$

Answer: A

Solution:

According to, Stokes law,

$$F = 6\pi\eta rv \Rightarrow \eta = \frac{F}{6\pi rv}$$

$$\eta = \frac{[MLT^{-2}]}{[L][LT^{-1}]}$$

$$\Rightarrow \eta = [ML^{-1}T^{-1}]$$

Question250

Dimensions of $\frac{1}{\mu_o \varepsilon_o}$, where symbols have their usual meaning, are [2003]

Options:

A. $[L^{-1}T]$

B. $[L^{-2}T^{2}]$

C. $[L^{2}T^{-2}]$

D. $[LT^{-1}]$

Answer: C

Solution:

As we know, the velocity of light in free space is given by

$$c = \frac{1}{\sqrt{\mu_o \varepsilon_o}}$$

$$\therefore \frac{1}{\mu_o \varepsilon_o} = e^2 = Z_1{}^2T^2$$

$$\frac{1}{\sqrt{\mu_o \varepsilon_o}} = C^2[m/s]^2 = [LT^{-1}]^2 = [M^0L^2T^{-2}]$$

Question251

The physical quantities not having same dimensions are [2003]

Options:

A. torque and work

B. momentum and planck's constant

C. stress and young's modulus

D. speed and $(\mu_o\varepsilon_o)^{-1/2}$

Answer: B

Solution:

Momentum, $= mv = [MLT^{-1}]$
Planck's constant,
$$h = \frac{E}{v} = \frac{[ML^2T^{-2}]}{[T^{-1}]} = [ML^2T^{-1}]$$

Question252

Identify the pair whose dimensions are equal [2002]

Options:

A. torque and work

B. stress and energy

C. force and stress

D. force and work

Answer: A

Solution:

Work $W = \vec{F} \cdot \vec{s} = F s \cos\theta$

$\because \vec{A} \cdot \vec{B} = AB\cos\theta$

$= [MLT^{-2}][L] = [ML^2T^{-2}]$

Torque, $\vec{\tau} = \vec{r} \times \vec{F} \Rightarrow \tau = rF \sin\theta$

$\because \vec{A} \times \vec{B} = AB\sin\theta = [L][MLT^{-2}] = [ML^2T^{-2}]$

Question253

A spherical body of radius r and density σ falls freely through a viscous liquid having density ρ and viscosity η and attains a terminal velocity v_0. Estimated maximum error in the quantity η is : (Ignore errors associated with σ, ρ and g, gravitational acceleration)

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Options:

A.

$2\left[\frac{\Delta r}{r} - \frac{\Delta v_0}{v_0}\right]$

B.

$2\left[\frac{\Delta r}{r} + \frac{\Delta v_0}{v_0}\right]$

C.

$\frac{2\Delta r}{r}+\frac{\Delta v_0}{v_0}$

D.

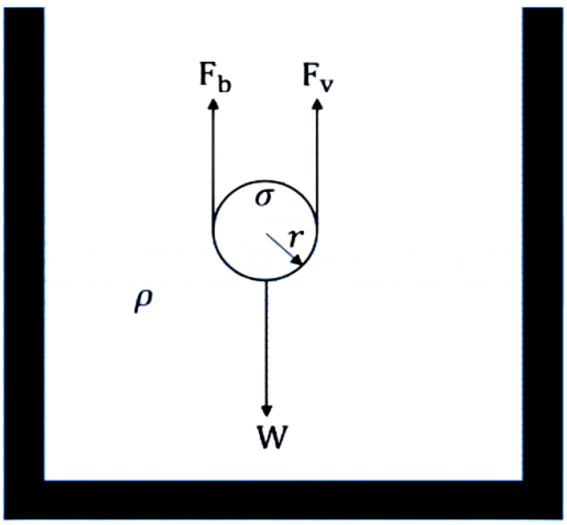
$2\frac{\Delta r}{r}-\frac{\Delta v_0}{v_0}$

Answer: C

Solution:

When a spherical body falls through a viscous liquid, the forces acting on it are,

- Weight (W) which is acting downwards.
- Up thrust (F_b) is acting upwards (due to buoyancy).
- Viscous Drag (F_v) is acting upwards (opposing the motion).



The radius of the spherical body is r.

The density of the body is σ

So, the weight of the spherical body is,

$W=mg=(\sigma V)g=\sigma\left[\frac{4}{3}\pi r^3\right]g$

The density of the liquid is ρ, and when body is completely inside the liquid, the up thrust is,

$F_B = \rho V g = \frac{4}{3}\pi r^3 \rho\, g$

The viscous drag acting on the body is

$F_v = 6\pi\eta r v$

Terminal velocity is reached when the net force on the body becomes zero (acceleration is zero).

$W = F_B + F_v$

$\Rightarrow \frac{4}{3}\pi r^3 \sigma g = \frac{4}{3}\pi r^3 \rho g + 6\pi\eta r v_0$

$\Rightarrow 6\pi\eta r v_0 = \frac{4}{3}\pi r^3 g(\sigma - \rho)$

$\Rightarrow \eta = \frac{\frac{4}{3}\pi r^3 g(\sigma-\rho)}{6\pi r v_0}$

$\Rightarrow \eta = \frac{2r^2 g(\sigma-\rho)}{9v_0}$

$\Rightarrow \eta = \frac{\left[\frac{2g(\sigma-\rho)}{9}\right] r^2}{v_0}$

As it is given to ignore errors associated with σ, ρ, and g . Therefore, $\left[\frac{2\text{ g}(\sigma-\rho)}{9}\right]$ is a constant (k).

$\eta = k \cdot \frac{r^2}{v_0}$

Taking natural logarithm on both sides:

$\ln(\eta) = \ln(k) + 2\ln(r) - \ln(v_0)$

On differentiating both sides:

$\frac{d\eta}{\eta} = 0 + \frac{2dr}{r} - \frac{dv_0}{v_0}$

For maximum error, we sum the absolute values of the individual errors because errors always propagate in a way that maximizes the uncertainty:

$\frac{\Delta\eta}{\eta} = \frac{2\Delta r}{r} + \frac{\Delta v_0}{v_0}$ or $\frac{\Delta\eta}{\eta} = -\frac{2\Delta r}{r} - \frac{\Delta v_0}{v_0}$

Therefore, the correct option is (C).
